



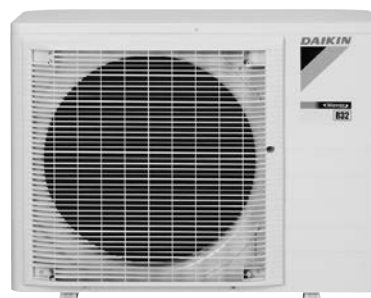
EDTNZ041928

R-32

Engineering Data

Split Type Air Conditioners
- Heat Pump -

FTXM-U Series



INVERTER

Split Type Air Conditioners FTXM-U Series

Heat Pump	FTXM20UVMZ	RXM20UVMZ
	FTXM25UVMZ	RXM25UVMZ
	FTXM35UVMZ	RXM35UVMZ
	FTXM46UVMZ	RXM46UVMZ
	FTXM50UVMZ	RXM50UVMZ
	FTXM60UVMZ	RXM60UVMZ
	FTXM71UVMZ	RXM71UVMZ

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Cautions

1. Air conditioners should not be installed in areas where corrosive gases, such as acid gas or alkaline gas, are produced.
2. If the outdoor unit is to be installed close to the sea shore, direct exposure to the sea breeze should be avoided and choose an outdoor unit with anti-corrosion treatment.

1. Power Supply

Indoor Unit	Outdoor Unit	Power Supply
FTXM20UVMZ	RXM20UVMZ	1 phase, 220 - 230 - 240 V, 50 Hz 1 phase, 220 - 230 V, 60 Hz
FTXM25UVMZ	RXM25UVMZ	
FTXM35UVMZ	RXM35UVMZ	
FTXM46UVMZ	RXM46UVMZ	
FTXM50UVMZ	RXM50UVMZ	
FTXM60UVMZ	RXM60UVMZ	
FTXM71UVMZ	RXM71UVMZ	

Note: Power Supply Intake ; Outdoor Unit

2. Functions

Category	Functions	FTXM20/25/35/46UVMZ RXM20/25/35/46UVMZ	FTXM50/60/71UVMZ RXM50/60/71UVMZ	Category	Functions	FTXM20/25/35/46UVMZ RXM20/25/35/46UVMZ	FTXM50/60/71UVMZ RXM50/60/71UVMZ	
Basic Functions	Inverter (with inverter power control)	●	●	Health & Cleanliness	Titanium apatite deodorizing filter	●	●	
	Operation limit for cooling	Refer to P. 27			Dust collection filter (PM2.5) (option)	●	●	
	Operation limit for heating				Mould-proof air filter (prefilter)	●	●	
	PAM control	●	●		Wipe-clean flat panel	●	●	
	Standby electricity saving	●	●		Washable panel	—	—	
	Demand controller (option)	●	●		Removable drain pan	●	●	
Compressor	Oval scroll compressor	—	—	Timer	Mould-proof operation	—	—	
	Swing compressor	●	●		Heating dry operation	—	—	
	Rotary compressor	—	—		Good-sleep cooling operation	—	—	
	Reluctance DC motor	●	●		Odour removing	●	●	
Comfortable Airflow	Power-airflow flap	●	—		Worry Free (Reliability & Durability)	Weekly timer	●	●
	Power-airflow dual flaps	—	●			24-hour ON/OFF timer	●	●
	Wide-angle louvres	●	●	Count up-down ON/OFF timer		●	●	
	Auto-swing (up and down)	●	●	Night set mode		●	●	
	Auto-swing (left and right)	●	●	Auto-restart after power failure	●	●		
	3D airflow	●	●	Self-diagnosis with remo-con	●	●		
	Comfort airflow mode	●	●	Wiring error check	—	—		
Comfort Control	Auto fan speed	●	●	Flexibility	Anti-corrosion treatment of outdoor heat exchanger fins	●	●	
	Indoor unit quiet operation	●	●		Protection net of outdoor unit	●	●	
	Outdoor unit quiet operation	●	●		Low/high voltage shield	—	—	
	Intelligent eye (auto energy saving)	●	●		Multi-split/split compatible indoor unit	—	—	
	2-area intelligent eye (focus and comfort)	●	—	Chargeless	10 m	10 m		
	Hot-start function	●	●	Either side drain (right/left)	●	●		
	Automatic defrost	●	●	Power selection	—	—		
Operation	Automatic operation (cooling and heating)	●	●	Remote Control	Low temperature cooling operation (–15°C)	●	●	
	Programme dry function	●	●		Remote control adaptor (normal open pulse contact) (option)	●	●	
	Fan operation	●	●		Remote control adaptor (normal open contact) (option)	●	●	
Lifestyle Convenience	Powerful operation (non-inverter)	—	—	Remote Controller	HA and DIII-NET application (option)	●	●	
	Inverter powerful operation	●	●		Wireless LAN adaptor (option) ★	●	●	
	Cooling/heating mode lock	—	—		Wireless	●	●	
	Home leave operation	—	—	Wired (option)	●	●		
	Econo mode	●	●					
	Indoor unit ON/OFF switch	●	●					
	Signal reception indicator	●	●					
	0.5°C step room temperature setting	●	●					
	16°C setting for cooling mode	—	—					
	Wireless remote controller with backlight	●	●					
	Wireless remote controller with luminous button	—	—					

Note: ● : Available
— : Not Available

★ : Can be installed inside indoor unit

3. Specifications

50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V

Model	Indoor Unit		FTXM20UVMZ		FTXM25UVMZ	
	Outdoor Unit		RXM20UVMZ		RXM25UVMZ	
			Cooling	Heating	Cooling	Heating
Capacity Rated (Min. ~ Max.) / Low Temperature	kW		2.0 (0.9 ~ 2.8) / —	2.8 (0.9 ~ 4.4) / 3.1	2.5 (0.9 ~ 3.7) / —	3.3 (0.9 ~ 5.4) / 3.6
	Btu/h		6,800 (3,100 ~ 9,600) / —	9,600 (3,100 ~ 15,000) / 10,600	8,500 (3,100 ~ 12,600) / —	11,300 (3,100 ~ 18,400) / 12,300
	kcal/h		1,720 (770 ~ 2,410) / —	2,410 (770 ~ 3,780) / 2,670	2,150 (770 ~ 3,180) / —	2,840 (770 ~ 4,640) / 3,100
Moisture Removal	L/h		0.9	—	1.2	—
Running Current (Rated)	A		1.9 - 1.8 - 1.7 / 1.9 - 1.8	2.8 - 2.7 - 2.6 / 2.8 - 2.7	2.3 - 2.2 - 2.1 / 2.3 - 2.2	3.2 - 3.1 - 2.9 / 3.2 - 3.1
Power Consumption Rated (Min. ~ Max.) / Low Temperature	W		400 (220 ~ 640) / —	580 (190 ~ 1,200) / 960	490 (230 ~ 1,050) / —	680 (190 ~ 1,480) / 1,100
Power Factor (Rated)	%		95.7 - 96.6 - 98.0 / 95.7 - 96.6	94.2 - 93.4 - 92.9 / 94.2 - 93.4	96.8 - 96.8 - 97.2 / 96.8 - 96.8	96.6 - 95.4 - 97.7 / 96.6 - 95.4
COP (Rated)	W/W		5.00	4.83	5.10	4.85
AEER / ACOP			4.92	4.77	5.03	4.81
Piping Connections	Liquid	mm	φ 6.4		φ 6.4	
	Gas	mm	φ 9.5		φ 9.5	
	Drain	mm	φ 16.0		φ 16.0	
Heat Insulation			Both Liquid and Gas Pipes		Both Liquid and Gas Pipes	
Max. Interunit Piping Length	m		20		20	
Max. Interunit Height Difference	m		15		15	
Chargeless	m		10		10	
Amount of Additional Charge of Refrigerant	g/m		20		20	
Indoor Unit			FTXM20UVMZ		FTXM25UVMZ	
Front Panel Colour			Pure White		Pure White	
Airflow Rate	H	m³/min (cfm)	9.1 (321)	9.4 (332)	10.4 (367)	10.1 (357)
	M		7.1 (251)	7.7 (272)	7.1 (251)	7.7 (272)
	L		5.0 (177)	5.5 (194)	5.0 (177)	5.5 (194)
	SL		3.6 (127)	3.5 (124)	3.6 (127)	3.5 (124)
Fan	Type		Cross Flow Fan		Cross Flow Fan	
	Motor Output	W	27		27	
	Speed	Steps	5 Steps, Quiet, Auto		5 Steps, Quiet, Auto	
Air Direction Control			Right, Left, Horizontal, Downwards		Right, Left, Horizontal, Downwards	
Air Filter			Removable, Washable, Mildew Proof		Removable, Washable, Mildew Proof	
Running Current (Rated)	A		0.15 - 0.15 - 0.14 / 0.15 - 0.15	0.19 - 0.19 - 0.18 / 0.19 - 0.19	0.23 - 0.23 - 0.22 / 0.23 - 0.23	0.23 - 0.23 - 0.22 / 0.23 - 0.23
Power Consumption (Rated)	W		18	24	36	36
Power Factor (Rated)	%		54.5 - 52.2 - 53.6 / 54.5 - 52.2	57.4 - 54.9 - 55.6 / 57.4 - 54.9	71.1 - 68.1 - 68.2 / 71.1 - 68.1	71.1 - 68.1 - 68.2 / 71.1 - 68.1
Temperature Control			Microcomputer Control		Microcomputer Control	
Dimensions (H × W × D)	mm		285 × 770 × 226		285 × 770 × 226	
Packaged Dimensions (H × W × D)	mm		320 × 830 × 360		320 × 830 × 360	
Weight (Mass)	kg		9		9	
Gross Weight (Gross Mass)	kg		12		12	
Sound Pressure Level	H / M / L / SL	dB(A)	38 / 32 / 25 / 19	39 / 34 / 28 / 20	42 / 32 / 25 / 19	40 / 34 / 28 / 20
Sound Power Level		dB	54	55	58	56
Outdoor Unit			RXM20UVMZ		RXM25UVMZ	
Casing Colour			Ivory White		Ivory White	
Heat Exchanger	Fin / Spec. Tube		Waffle Fin (PE) / φ 7 Hi-XD Tube		Waffle Fin / φ 7 Hi-XD Tube	
Compressor	Type		Hermetically Sealed Swing Type		Hermetically Sealed Swing Type	
	Model		1YC25FXD		1YC25FXD	
	Motor Output	W	800		800	
Refrigerant Oil	Type		FW68DA		FW68DA	
	Charge	L	0.375		0.375	
Refrigerant	Type		R-32		R-32	
	Charge	kg	0.73		0.85	
Airflow Rate	H	m³/min (cfm)	26.2 (925)	23.6 (833)	25.3 (893)	21.8 (770)
	SL		23.2 (819)	20.2 (713)	21.2 (749)	15.7 (554)
Fan	Type		Propeller		Propeller	
	Motor Output	W	50		50	
Running Current (Rated)	A		1.8 - 1.7 - 1.6 / 1.8 - 1.7	2.6 - 2.5 - 2.4 / 2.6 - 2.5	2.1 - 2.0 - 1.9 / 2.1 - 2.0	3.0 - 2.9 - 2.7 / 3.0 - 2.9
Power Consumption (Rated)	W		382	556	454	644
Power Factor (Rated)	%		96.5 - 97.7 - 99.5 / 96.5 - 97.7	97.2 - 96.7 - 96.5 / 97.2 - 96.7	98.3 - 98.7 - 99.6 / 98.3 - 98.7	97.6 - 96.6 - 99.4 / 97.6 - 96.6
Starting Current	A		2.8		3.2	
Dimensions (H × W × D)	mm		550 × 675 × 284		550 × 675 × 284	
Packaged Dimensions (H × W × D)	mm		605 × 798 × 373		605 × 798 × 373	
Weight (Mass)	kg		28		30	
Gross Weight (Gross Mass)	kg		32		35	
Sound Pressure Level	H / SL	dB(A)	46 / 43	47 / 44	47 / 43	48 / 44
Sound Power Level	H	dB	58	59	59	60
Drawing No.			3D123949A		3D123950A	

Notes:

1. SL: The Quiet fan level of the airflow rate setting.
2. The data are based on the conditions shown in the table below.

Cooling	Heating	Piping Length
Indoor ; 27°CDB / 19°CWB Outdoor ; 35°CDB / 24°CWB	Indoor ; 20°CDB / 15°CWB Outdoor ; 7°CDB / 6°CWB	5.0 m

Conversion Formulae

kcal/h = kW × 860
 Btu/h = kW × 3412
 cfm = m³/min × 35.3

50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V

Model	Indoor Unit		FTXM35UVMZ		FTXM46UVMZ	
	Outdoor Unit		RXM35UVMZ		RXM46UVMZ	
			Cooling	Heating	Cooling	Heating
Capacity Rated (Min. ~ Max.) / Low Temperature	kW		3.5 (0.9 ~ 4.2) / —	4.0 (0.9 ~ 5.5) / 3.7	4.6 (0.9 ~ 5.5) / —	5.4 (1.0 ~ 6.7) / 4.9
	Btu/h		11,900 (3,100 ~ 14,300) / —	13,600 (3,100 ~ 18,800) / 12,600	15,700 (3,100 ~ 18,800) / —	18,400 (3,400 ~ 22,900) / 16,700
	kcal/h		3,010 (770 ~ 3,610) / —	3,440 (770 ~ 4,730) / 3,180	3,960 (770 ~ 4,730) / —	4,640 (860 ~ 5,760) / 4,210
Moisture Removal	L/h		1.9	—	2.9	—
Running Current (Rated)	A		3.9 - 3.7 - 3.5 / 3.9 - 3.7	4.2 - 4.0 - 3.7 / 4.2 - 4.0	5.8 - 5.5 - 5.2 / 5.8 - 5.5	6.6 - 6.4 - 6.1 / 6.6 - 6.4
Power Consumption Rated (Min. ~ Max.) / Low Temperature	W		810 (230 ~ 1,300) / —	870 (190 ~ 1,470) / 1,130	1,200 (230 ~ 1,730) / —	1,390 (220 ~ 2,350) / 1,630
Power Factor (Rated)	%		94.4 - 95.2 - 96.4 / 94.4 - 95.2	94.2 - 94.6 - 98.0 / 94.2 - 94.6	94.0 - 94.9 - 96.2 / 94.0 - 94.9	95.7 - 94.4 - 94.9 / 95.7 - 94.4
COP (Rated)	W/W		4.32	4.60	3.83	3.89
AEER / ACOP			4.20	4.56	3.80	3.86
Piping Connections	Liquid	mm	φ 6.4		φ 6.4	
	Gas	mm	φ 9.5		φ 12.7	
	Drain	mm	φ 16.0		φ 16.0	
Heat Insulation			Both Liquid and Gas Pipes		Both Liquid and Gas Pipes	
Max. Interunit Piping Length	m		20		30	
Max. Interunit Height Difference	m		15		20	
Chargeless	m		10		10	
Amount of Additional Charge of Refrigerant	g/m		20		20	
Indoor Unit			FTXM35UVMZ		FTXM46UVMZ	
Front Panel Colour			Pure White		Pure White	
Airflow Rate	H	m³/min (cfm)	10.5 (371)	11.0 (388)	12.2 (431)	12.1 (427)
	M		7.6 (268)	8.4 (297)	10.0 (353)	9.3 (328)
	L		5.4 (191)	5.8 (205)	8.4 (297)	7.5 (265)
	SL		3.6 (127)	3.5 (124)	6.1 (215)	5.2 (184)
Fan	Type		Cross Flow Fan		Cross Flow Fan	
	Motor Output	W	27		27	
	Speed	Steps	5 Steps, Quiet, Auto		5 Steps, Quiet, Auto	
Air Direction Control			Right, Left, Horizontal, Downwards		Right, Left, Horizontal, Downwards	
Air Filter			Removable, Washable, Mildew Proof		Removable, Washable, Mildew Proof	
Running Current (Rated)	A		0.26 - 0.26 - 0.25 / 0.26 - 0.26	0.26 - 0.26 - 0.25 / 0.26 - 0.26	0.30 - 0.29 - 0.28 / 0.30 - 0.29	0.29 - 0.29 - 0.28 / 0.29 - 0.29
Power Consumption (Rated)	W		40	40	44	42
Power Factor (Rated)	%		69.9 - 66.9 - 66.7 / 69.9 - 66.9	69.9 - 66.9 - 66.7 / 69.9 - 66.9	66.7 - 66.0 - 65.5 / 66.7 - 66.0	65.8 - 63.0 - 62.5 / 65.8 - 63.0
Temperature Control			Microcomputer Control		Microcomputer Control	
Dimensions (H × W × D)	mm		285 × 770 × 226		285 × 770 × 226	
Packaged Dimensions (H × W × D)	mm		320 × 830 × 360		320 × 830 × 360	
Weight (Mass)	kg		9		9	
Gross Weight (Gross Mass)	kg		12		12	
Sound Pressure Level	H / M / L / SL	dB(A)	42 / 34 / 26 / 19	42 / 36 / 29 / 20	44 / 40 / 35 / 26	43 / 38 / 33 / 26
Sound Power Level		dB	58	58	60	59
Outdoor Unit			RXM35UVMZ		RXM46UVMZ	
Casing Colour			Ivory White		Ivory White	
Heat Exchanger	Fin / Spec. Tube		Waffle Fin / φ 7 Hi-XD Tube		Waffle Fin / φ 7 Hi-XD Tube	
Compressor	Type		Hermetically Sealed Swing Type		Hermetically Sealed Swing Type	
	Model		1YC25FXD		2YC40GXD	
	Motor Output	W	800		1,300	
Refrigerant Oil	Type		FW68DA		FW68DA	
	Charge	L	0.375		0.650	
Refrigerant	Type		R-32		R-32	
	Charge	kg	0.85		1.20	
Airflow Rate	H	m³/min (cfm)	28.4 (1,003)	23.5 (830)	55.8 (1,970)	45.8 (1,617)
	SL		22.7 (802)	16.3 (576)	46.1 (1,628)	35.3 (1,246)
Fan	Type		Propeller		Propeller	
	Motor Output	W	50		128	
Running Current (Rated)	A		3.6 - 3.4 - 3.3 / 3.6 - 3.4	3.9 - 3.7 - 3.5 / 3.9 - 3.7	5.5 - 5.2 - 4.9 / 5.5 - 5.2	6.3 - 6.1 - 5.8 / 6.3 - 6.1
Power Consumption (Rated)	W		770	830	1,156	1,348
Power Factor (Rated)	%		97.2 - 98.5 - 97.2 / 97.2 - 98.5	96.7 - 97.5 - 98.8 / 96.7 - 97.5	95.5 - 96.7 - 98.3 / 95.5 - 96.7	97.3 - 96.1 - 96.8 / 97.3 - 96.1
Starting Current	A		4.2		6.6	
Dimensions (H × W × D)	mm		550 × 675 × 284		695 × 930 × 350	
Packaged Dimensions (H × W × D)	mm		605 × 798 × 373		762 × 1,004 × 475	
Weight (Mass)	kg		30		51	
Gross Weight (Gross Mass)	kg		35		58	
Sound Pressure Level	H / SL	dB(A)	49 / 44	50 / 45	47 / 44	49 / 45
Sound Power Level	H	dB	61	62	59	61
Drawing No.			3D123951A		3D123952A	

Notes:

1. SL: The Quiet fan level of the airflow rate setting.
2. The data are based on the conditions shown in the table below.

Cooling	Heating	Piping Length
Indoor ; 27°CDB / 19°CWB Outdoor ; 35°CDB / 24°CWB	Indoor ; 20°CDB / 15°CWB Outdoor ; 7°CDB / 6°CWB	5.0 m

Conversion Formulae
kcal/h = kW × 860 Btu/h = kW × 3412 cfm = m³/min × 35.3

50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V

Model	Indoor Unit		FTXM50UVMZ		FTXM60UVMZ	
	Outdoor Unit		RXM50UVMZ		RXM60UVMZ	
			Cooling	Heating	Cooling	Heating
Capacity Rated (Min. ~ Max.) / Low Temperature	kW		5.0 (1.1 ~ 6.7) / —	6.1 (1.0 ~ 8.0) / 6.1	6.0 (1.1 ~ 7.5) / —	7.3 (1.0 ~ 9.8) / 6.5
	Btu/h		17,100 (3,800 ~ 22,900) / —	20,800 (3,400 ~ 27,300) / 20,800	20,500 (3,800 ~ 25,600) / —	24,900 (3,400 ~ 33,400) / 22,200
	kcal/h		4,300 (950 ~ 5,760) / —	5,250 (860 ~ 6,880) / 5,250	5,160 (950 ~ 6,450) / —	6,280 (860 ~ 8,430) / 5,590
Moisture Removal	L/h		2.9	—	3.9	—
Running Current (Rated)	A		5.5 - 5.3 - 5.1 / 5.5 - 5.3	6.4 - 6.1 - 5.8 / 6.4 - 6.1	7.2 - 6.9 - 6.7 / 7.2 - 6.9	7.9 - 7.5 - 7.2 / 7.9 - 7.5
Power Consumption Rated (Min. ~ Max.) / Low Temperature	W		1,140 (280 ~ 2,000) / —	1,340 (220 ~ 2,500) / 1,890	1,520 (280 ~ 2,380) / —	1,660 (220 ~ 2,750) / 2,060
Power Factor (Rated)	%		94.2 - 93.5 - 93.1 / 94.2 - 93.5	95.2 - 95.5 - 96.3 / 95.2 - 95.5	96.0 - 95.8 - 94.5 / 96.0 - 95.8	95.5 - 96.2 - 96.1 / 95.5 - 96.2
COP (Rated)	W/W		4.39	4.55	3.95	4.40
AEER / ACOP			4.35	4.52	3.92	4.37
Piping Connections	Liquid	mm	φ 6.4		φ 6.4	
	Gas	mm	φ 12.7		φ 12.7	
	Drain	mm	φ 16.0		φ 16.0	
Heat Insulation			Both Liquid and Gas Pipes		Both Liquid and Gas Pipes	
Max. Interunit Piping Length	m		30		30	
Max. Interunit Height Difference	m		20		20	
Chargeless	m		10		10	
Amount of Additional Charge of Refrigerant	g/m		20		20	
Indoor Unit			FTXM50UVMZ		FTXM60UVMZ	
Front Panel Colour			Pure White		Pure White	
Airflow Rate	H	m³/min (cfm)	16.0 (565)	17.2 (607)	18.6 (657)	20.0 (706)
	M		13.6 (480)	14.3 (505)	15.0 (530)	15.9 (561)
	L		11.1 (392)	11.4 (403)	12.0 (424)	11.4 (403)
	SL		8.9 (314)	9.2 (325)	9.1 (321)	9.7 (343)
Fan	Type		Cross Flow Fan		Cross Flow Fan	
	Motor Output	W	58		58	
	Speed	Steps	5 Steps, Quiet, Auto		5 Steps, Quiet, Auto	
Air Direction Control			Right, Left, Horizontal, Downwards		Right, Left, Horizontal, Downwards	
Air Filter			Removable, Washable, Mildew Proof		Removable, Washable, Mildew Proof	
Running Current (Rated)	A		0.35 - 0.34 - 0.33 / 0.35 - 0.34	0.37 - 0.36 - 0.35 / 0.37 - 0.36	0.43 - 0.39 - 0.37 / 0.43 - 0.39	0.46 - 0.44 - 0.43 / 0.46 - 0.44
Power Consumption (Rated)	W		42	44	58	60
Power Factor (Rated)	%		54.5 - 53.7 - 53.0 / 54.5 - 53.7	54.1 - 53.1 - 52.4 / 54.1 - 53.1	61.3 - 64.7 - 65.3 / 61.3 - 64.7	59.3 - 59.3 - 58.1 / 59.3 - 59.3
Temperature Control			Microcomputer Control		Microcomputer Control	
Dimensions (H × W × D)	mm		295 × 990 × 266		295 × 990 × 266	
Packaged Dimensions (H × W × D)	mm		386 × 1,102 × 389		386 × 1,102 × 389	
Weight (Mass)	kg		13		13	
Gross Weight (Gross Mass)	kg		18		18	
Sound Pressure Level	H / M / L / SL	dB(A)	45 / 40 / 35 / 28	45 / 39 / 33 / 28	48 / 42 / 36 / 29	48 / 41 / 33 / 29
Sound Power Level		dB	61	61	64	64
Outdoor Unit			RXM50UVMZ		RXM60UVMZ	
Casing Colour			Ivory White		Ivory White	
Heat Exchanger	Fin / Spec. Tube		Waffle Fin / φ 7 Hi-XD Tube		Waffle Fin / φ 7 Hi-XD Tube	
Compressor	Type		Hermetically Sealed Swing Type		Hermetically Sealed Swing Type	
	Model		2YC40GXD		2YC40GXD	
	Motor Output	W	1,300		1,300	
Refrigerant Oil	Type		FW68DA		FW68DA	
	Charge	L	0.650		0.650	
Refrigerant	Type		R-32		R-32	
	Charge	kg	1.50		1.50	
Airflow Rate	H	m³/min (cfm)	50.6 (1,787)	41.2 (1,455)	54.6 (1,928)	50.6 (1,787)
	SL		44.1 (1,557)	36.1 (1,275)	45.7 (1,614)	36.1 (1,275)
Fan	Type		Propeller		Propeller	
	Motor Output	W	128		128	
Running Current (Rated)	A		5.2 - 5.0 - 4.8 / 5.2 - 5.0	6.0 - 5.7 - 5.5 / 6.0 - 5.7	6.8 - 6.5 - 6.3 / 6.8 - 6.5	7.4 - 7.1 - 6.8 / 7.4 - 7.1
Power Consumption (Rated)	W		1,098	1,296	1,462	1,600
Power Factor (Rated)	%		96.0 - 95.5 - 95.3 / 96.0 - 95.5	98.2 - 98.9 - 98.2 / 98.2 - 98.9	97.7 - 97.8 - 96.7 / 97.7 - 97.8	98.3 - 98.0 - 98.0 / 98.3 - 98.0
Starting Current	A		6.4		7.9	
Dimensions (H × W × D)	mm		695 × 930 × 350		695 × 930 × 350	
Packaged Dimensions (H × W × D)	mm		762 × 1,004 × 475		762 × 1,004 × 475	
Weight (Mass)	kg		54		54	
Gross Weight (Gross Mass)	kg		61		61	
Sound Pressure Level	H / SL	dB(A)	47 / 44	48 / 45	49 / 45	52 / 45
Sound Power Level	H	dB	59	60	61	64
Drawing No.			3D123953A		3D123954A	

Notes:

- SL: The Quiet fan level of the airflow rate setting.
- The data are based on the conditions shown in the table below.

Cooling	Heating	Piping Length
Indoor ; 27°CDB / 19°CWB Outdoor ; 35°CDB / 24°CWB	Indoor ; 20°CDB / 15°CWB Outdoor ; 7°CDB / 6°CWB	5.0 m

Conversion Formulae

kcal/h = kW × 860
Btu/h = kW × 3412
cfm = m³/min × 35.3

50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V

Model			Indoor Unit		FTXM71UVMZ	
			Outdoor Unit		RXM71UVMZ	
			Cooling		Heating	
Capacity Rated (Min. ~ Max.) / Low Temperature			kW	7.1 (2.0 ~ 8.9) / —		8.1 (2.0 ~ 10.8) / 7.7
			Btu/h	24,200 (6,800 ~ 30,400) / —		27,600 (6,800 ~ 36,900) / 26,300
			kcal/h	6,110 (1,720 ~ 7,650) / —		6,970 (1,720 ~ 9,290) / 6,620
Moisture Removal			L/h	4.5		—
Running Current (Rated)			A	9.2 - 8.8 - 8.4 / 9.2 - 8.8		9.0 - 8.7 - 8.4 / 9.0 - 8.7
Power Consumption Rated (Min. ~ Max.) / Low Temperature			W	1,930 (450 ~ 3,350) / —		1,920 (390 ~ 3,680) / 2,690
Power Factor (Rated)			%	95.4 - 95.4 - 95.7 / 95.4 - 95.4		97.0 - 96.0 - 95.2 / 97.0 - 96.0
COP (Rated)			W/W	3.68		4.22
AEER / ACOP				3.59		4.13
Piping Connections	Liquid	mm	φ 6.4			
	Gas	mm	φ 15.9			
	Drain	mm	φ 16.0			
Heat Insulation			Both Liquid and Gas Pipes			
Max. Interunit Piping Length			m	30		
Max. Interunit Height Difference			m	20		
Chargeless			m	10		
Amount of Additional Charge of Refrigerant			g/m	20		
Indoor Unit			FTXM71UVMZ			
Front Panel Colour			Pure White			
Airflow Rate	H	m³/min (cfm)	18.6 (657)		19.7 (696)	
	M		15.6 (551)		16.8 (593)	
	L		12.3 (434)		12.8 (452)	
	SL		9.1 (321)		10.7 (378)	
Fan	Type	Cross Flow Fan				
	Motor Output	W	58			
	Speed	Steps	5 Steps, Quiet, Auto			
Air Direction Control			Right, Left, Horizontal, Downwards			
Air Filter			Removable, Washable, Mildew Proof			
Running Current (Rated)			A	0.51 - 0.50 - 0.49 / 0.51 - 0.50		0.46 - 0.44 - 0.43 / 0.46 - 0.44
Power Consumption (Rated)			W	64		60
Power Factor (Rated)			%	57.0 - 55.7 - 54.4 / 57.0 - 55.7		59.3 - 59.3 - 58.1 / 59.3 - 59.3
Temperature Control			Microcomputer Control			
Dimensions (H × W × D)			mm	295 × 990 × 266		
Packaged Dimensions (H × W × D)			mm	386 × 1,102 × 389		
Weight (Mass)			kg	13		
Gross Weight (Gross Mass)			kg	18		
Sound Pressure Level			H / M / L / SL	dB(A)		49 / 45 / 37 / 30
Sound Power Level			dB	65		65
Outdoor Unit			RXM71UVMZ			
Casing Colour			Ivory White			
Heat Exchanger			Fin / Spec. Tube		Waffle Fin / φ 7 Hi-XD Tube	
Compressor	Type		Hermetically Sealed Swing Type			
	Model		2YC71BXD			
	Motor Output	W	2,400			
Refrigerant Oil	Type		FW68DA			
	Charge	L	0.900			
Refrigerant	Type		R-32			
	Charge	kg	1.60			
Airflow Rate	H	m³/min (cfm)	64.3 (2,270)		52.0 (1,836)	
	SL		49.8 (1,758)		43.3 (1,529)	
Fan	Type		Propeller			
	Motor Output	W	128			
Running Current (Rated)			A	8.7 - 8.3 - 7.9 / 8.7 - 8.3		8.5 - 8.3 - 8.0 / 8.5 - 8.3
Power Consumption (Rated)			W	1,866		1,860
Power Factor (Rated)			%	97.5 - 97.7 - 98.4 / 97.5 - 97.7		99.5 - 97.4 - 96.9 / 99.5 - 97.4
Starting Current			A	9.2		
Dimensions (H × W × D)			mm	695 × 930 × 350		
Packaged Dimensions (H × W × D)			mm	762 × 1,004 × 475		
Weight (Mass)			kg	60		
Gross Weight (Gross Mass)			kg	67		
Sound Pressure Level			H / SL	dB(A)		53 / 49
Sound Power Level			H	dB		65
Drawing No.			3D123955A			

Notes:

1. SL: The Quiet fan level of the airflow rate setting.
2. The data are based on the conditions shown in the table below.

Cooling	Heating	Piping Length
Indoor ; 27°CDB / 19°CWB Outdoor ; 35°CDB / 24°CWB	Indoor ; 20°CDB / 15°CWB Outdoor ; 7°CDB / 6°CWB	5.0 m

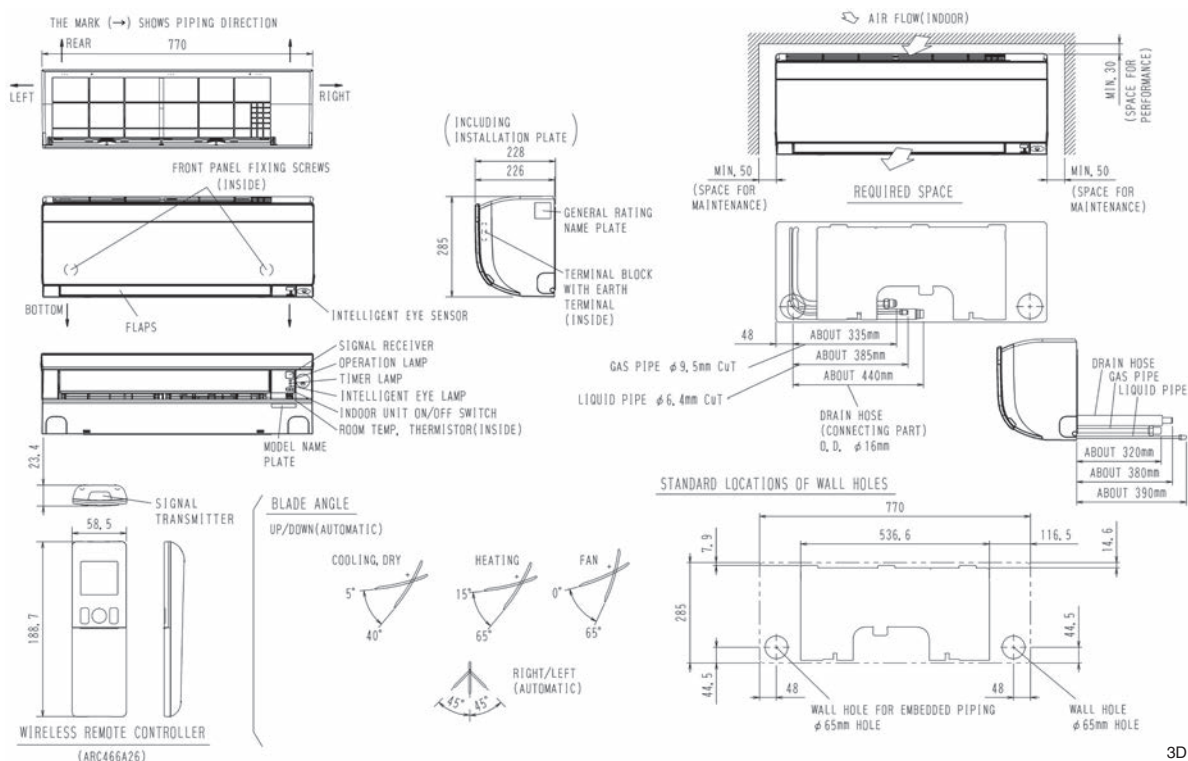
Conversion Formulae

kcal/h = kW × 860
 Btu/h = kW × 3412
 cfm = m³/min × 35.3

4. Dimensions

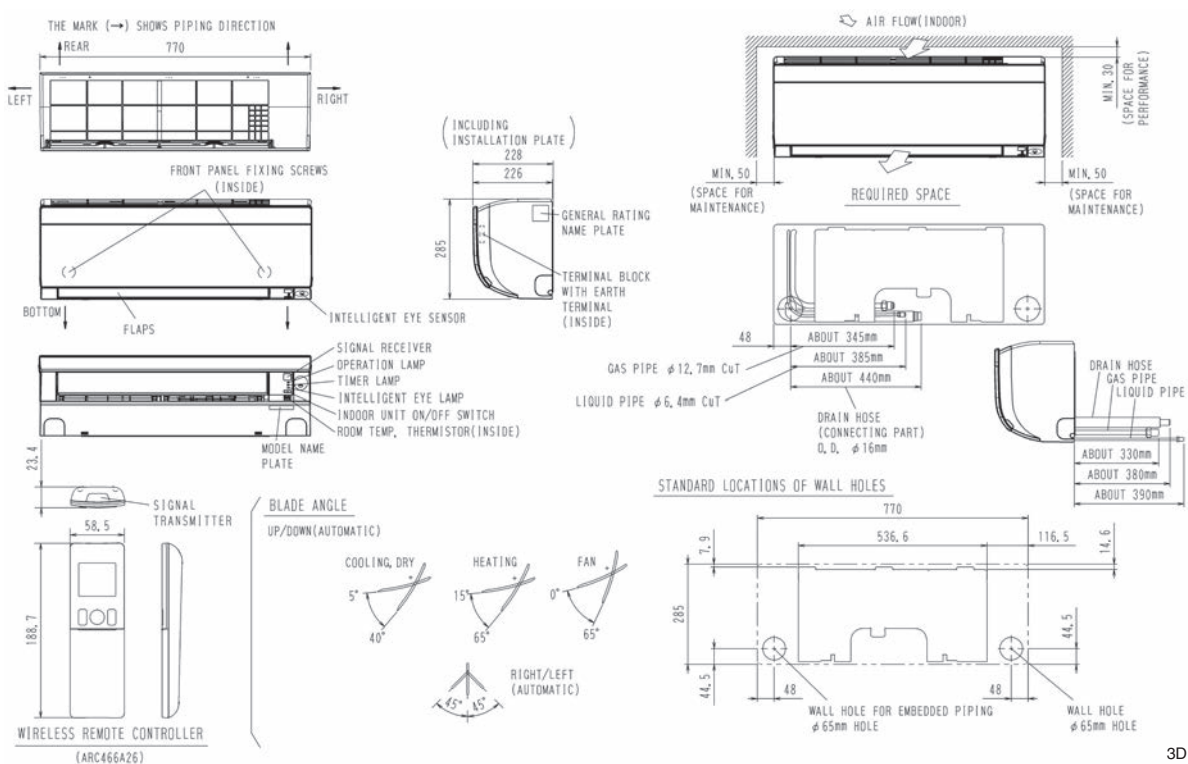
4.1 Indoor Unit

FTXM20/25/35UVMZ



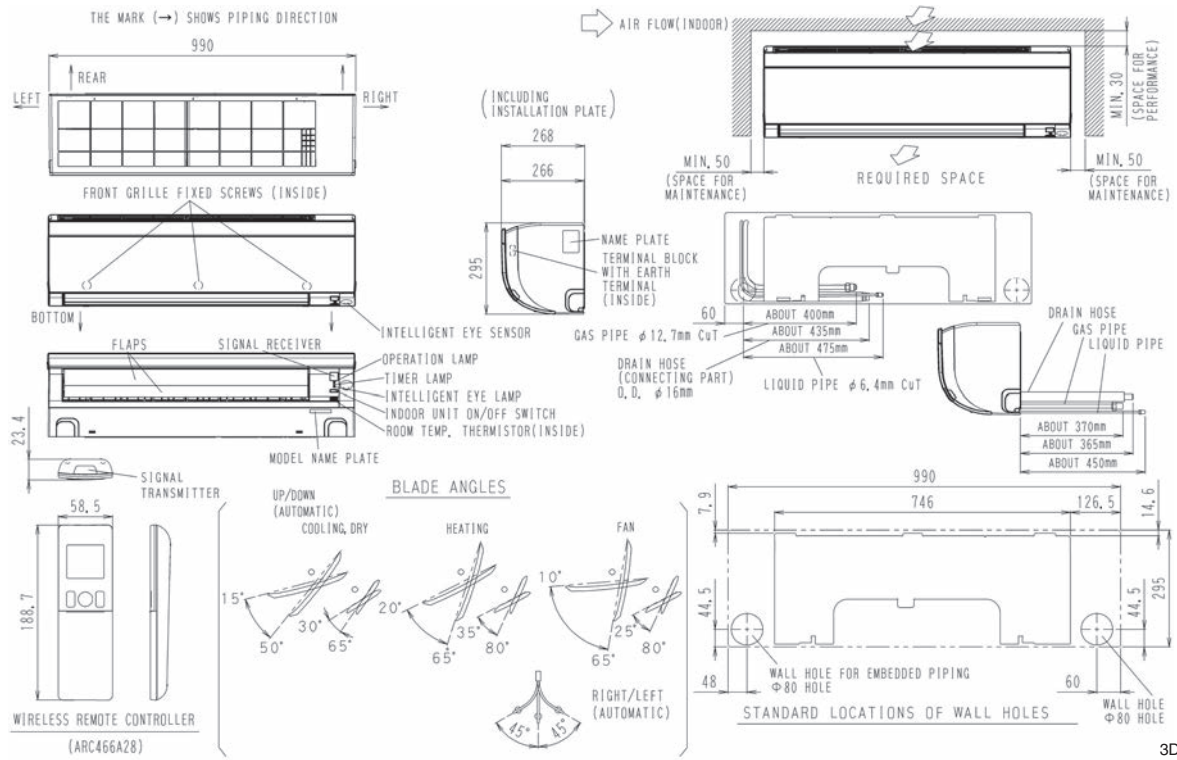
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FTXM46UVMZ

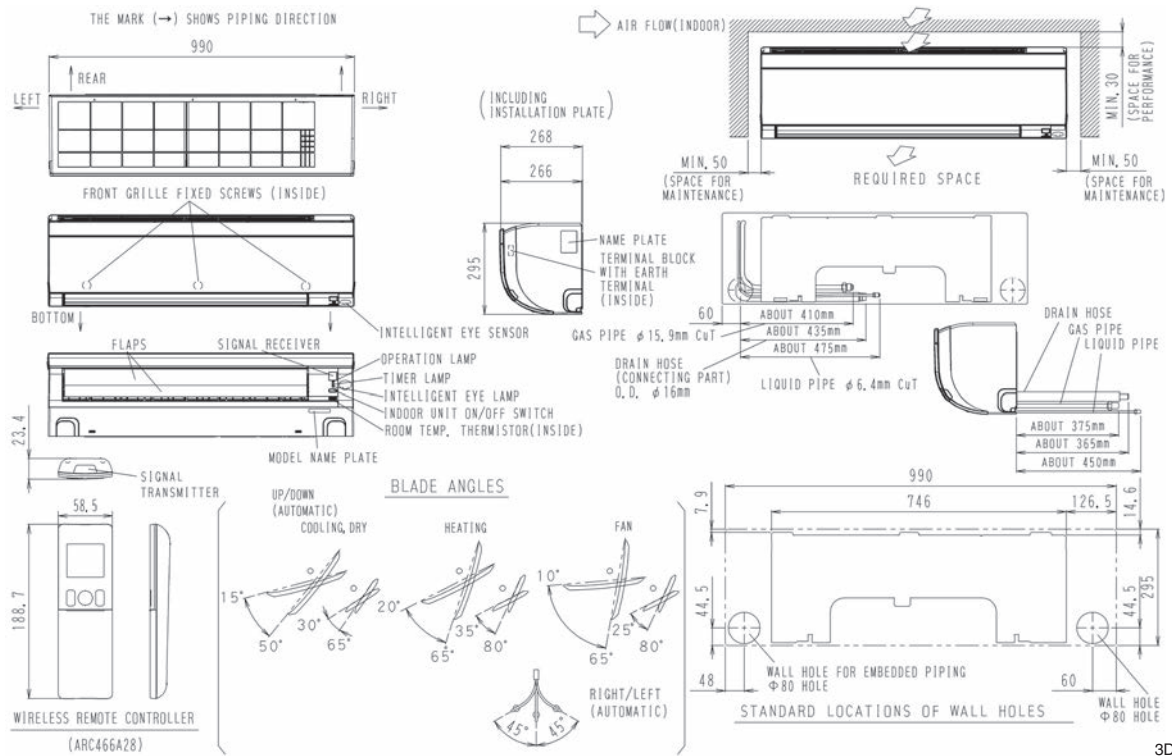


3D123289

FTXM50/60UVMZ

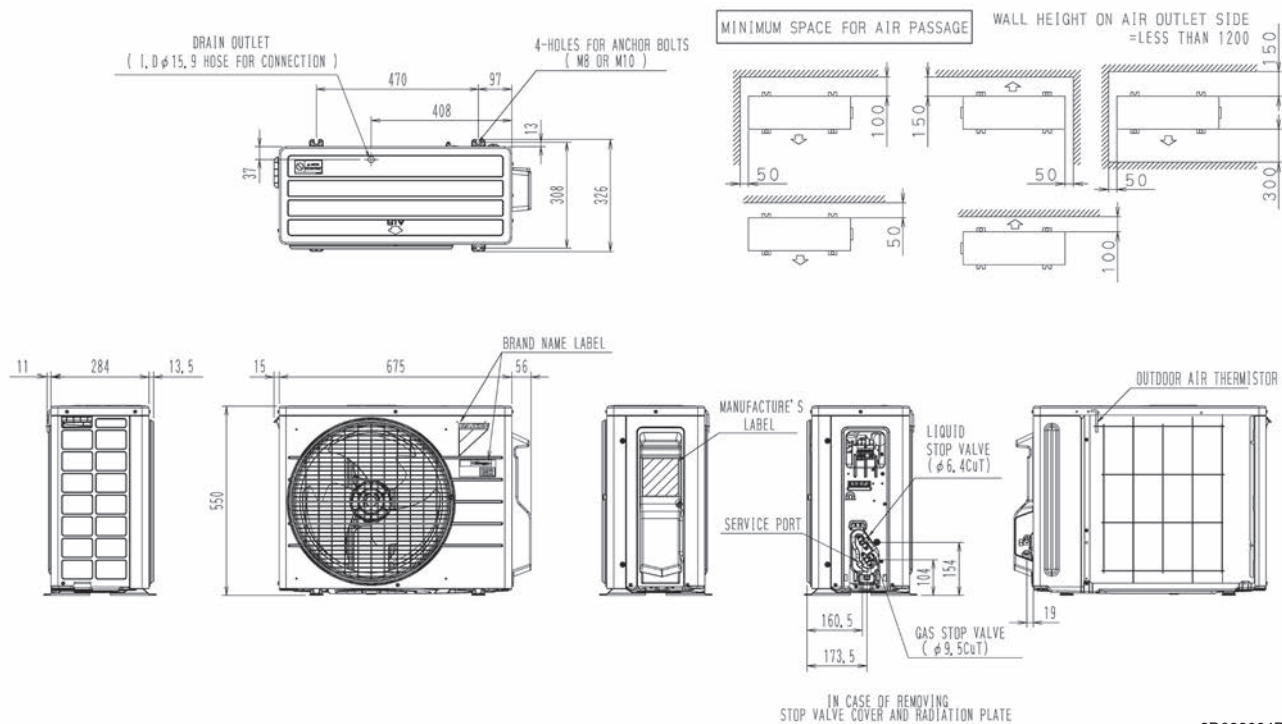


FTXM71UVMZ



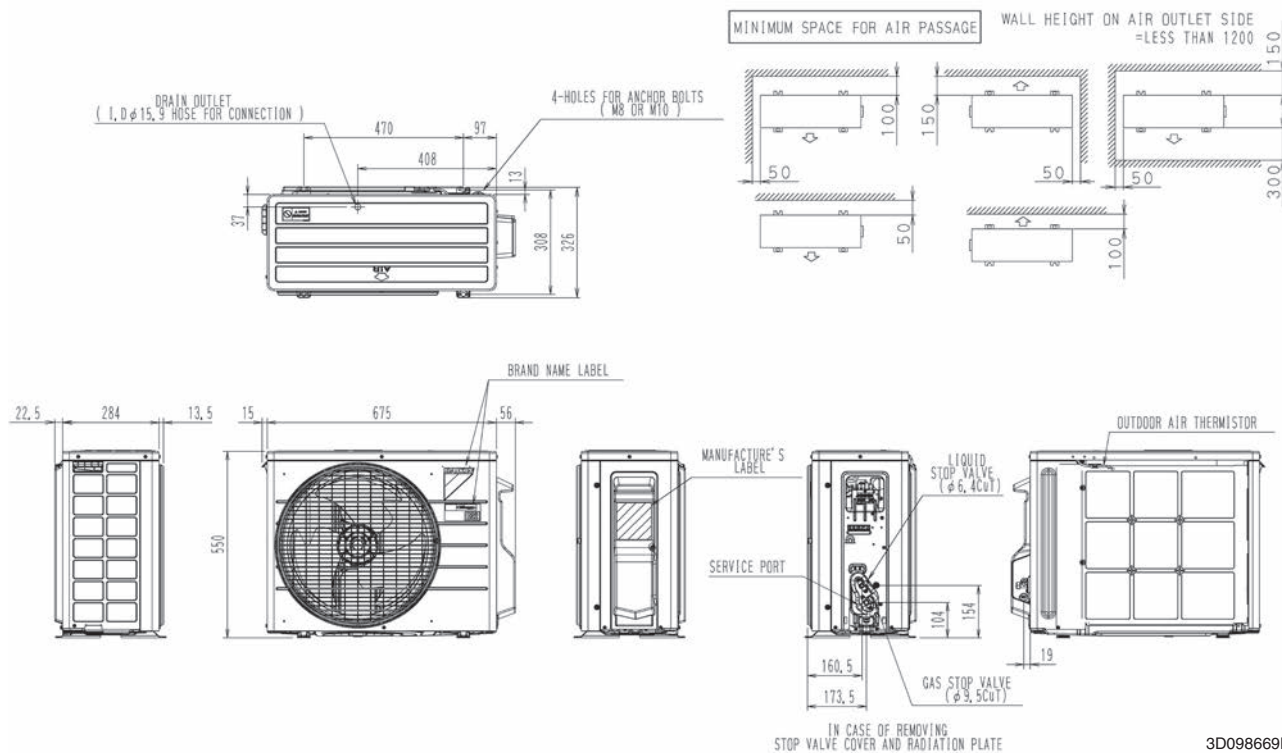
4.2 Outdoor Unit

RXM20UVMZ



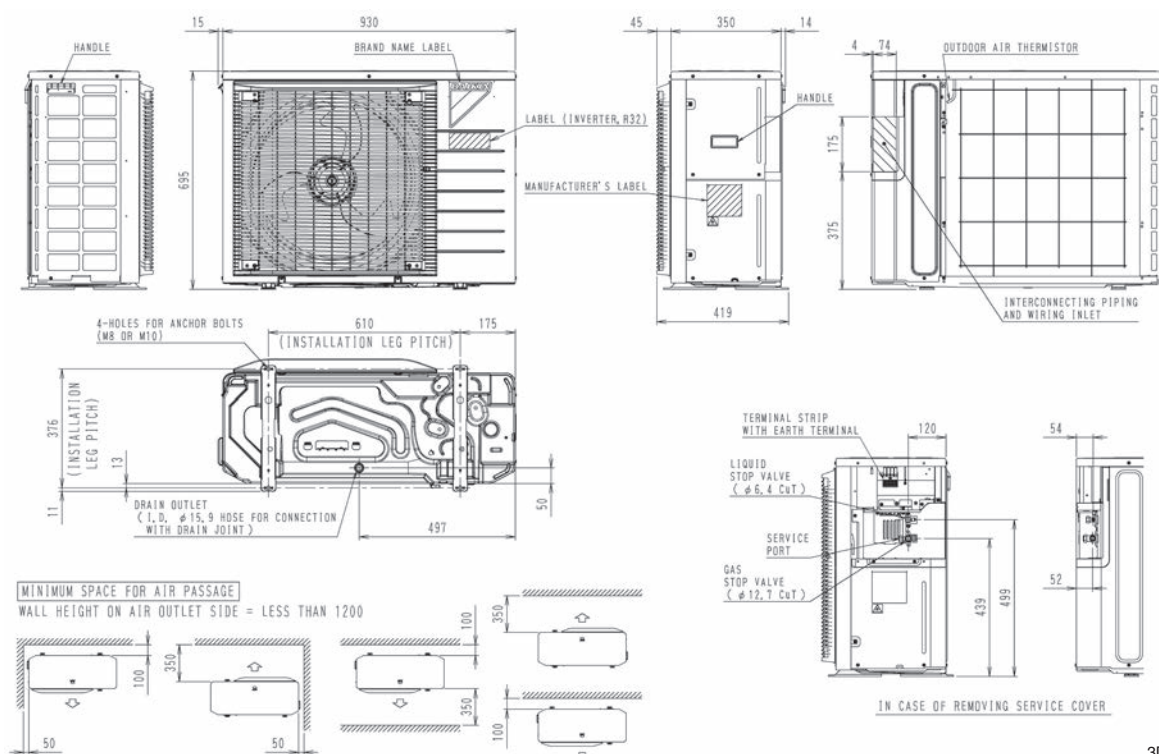
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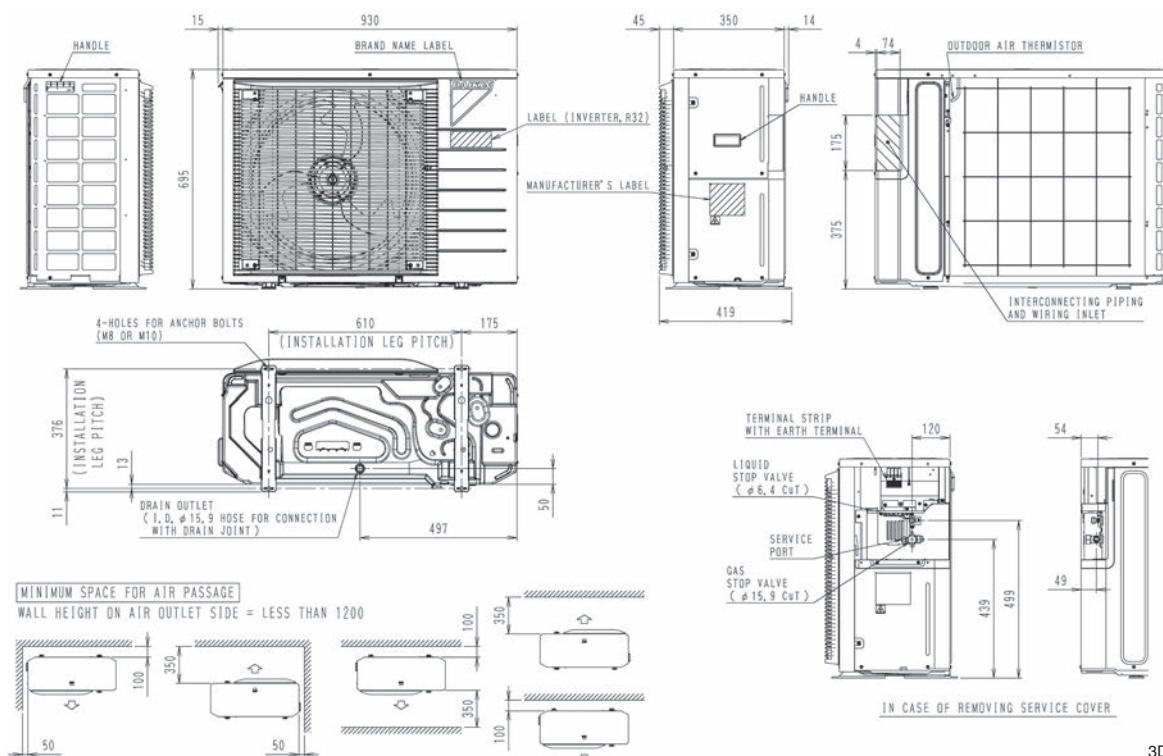
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RXM46/50/60UVMZ



3D091732E

RXM71UVMZ

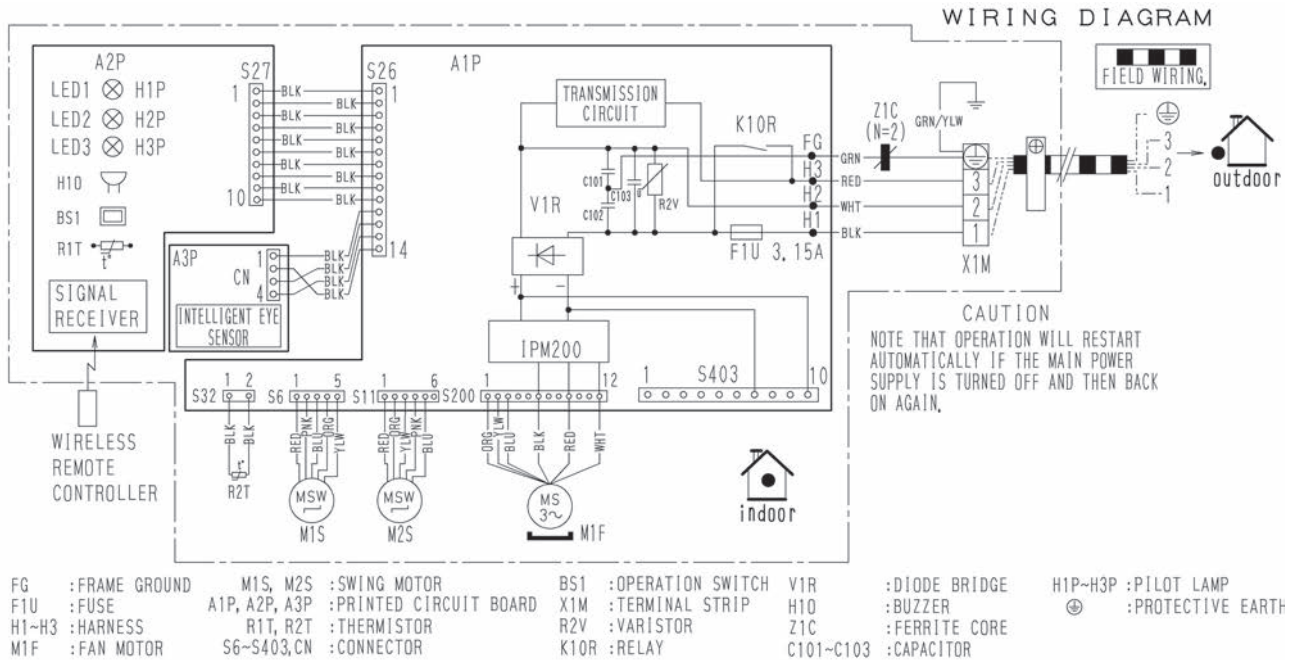


3D091733C

5. Wiring Diagrams

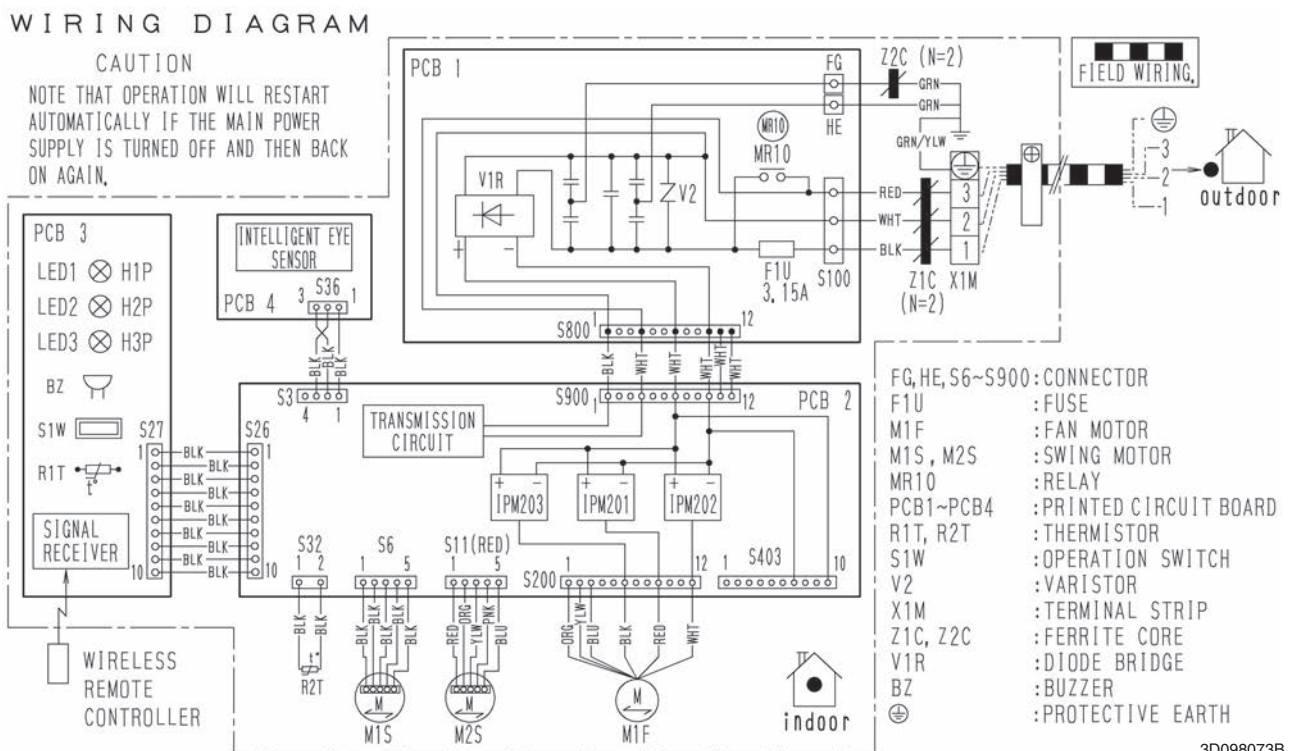
5.1 Indoor Unit

FTXM20/25/35/46UVMZ



3D097333B

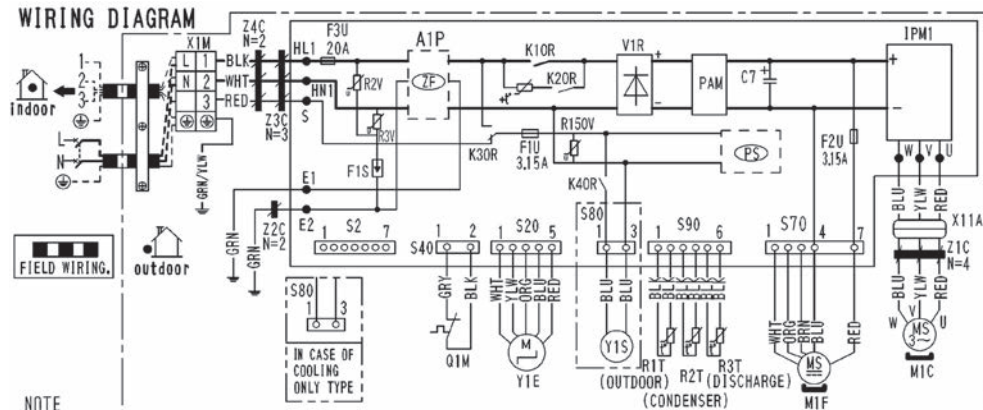
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3D098073B

5.2 Outdoor Unit

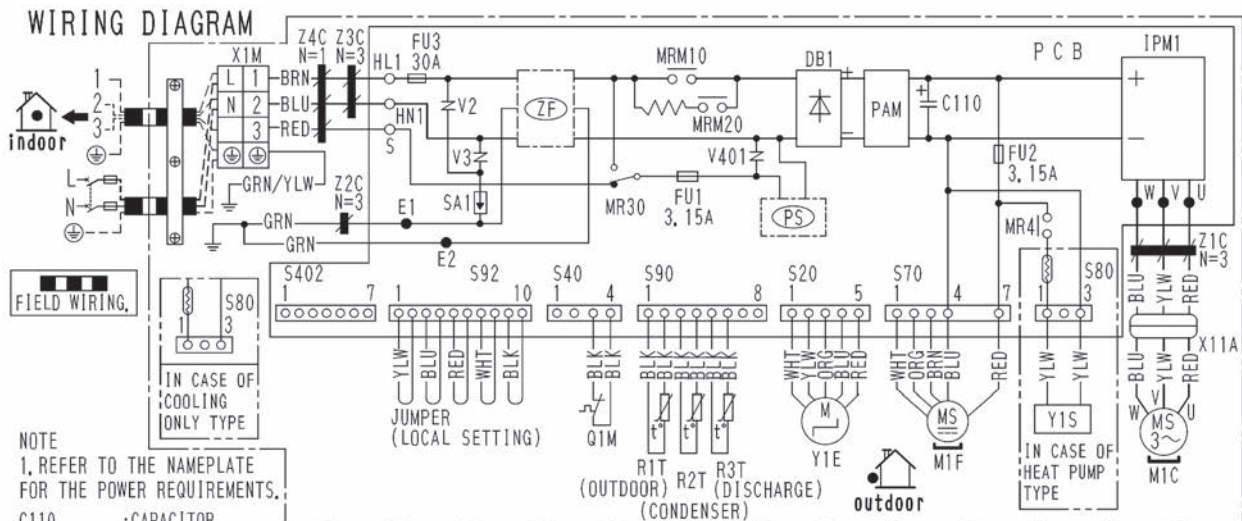
RXM20/25/35UVMZ



C7	:CAPACITOR	Q1M	:OVERLOAD PROTECTOR	⊕	:PROTECTIVE EARTH
V1R	:DIODE BRIDGE	R1T, R2T, R3T	:THERMISTOR	⊥	:EARTH
F1U, F2U, F3U	:FUSE	F1S	:SURGE ARRESTER		
IPM1	:INTELLIGENT POWER MODULE	S20, S40, S70, S80,			
L	:LIVE	S90, X11A, S2	:CONNECTOR		
M1C	:COMPRESSOR MOTOR	R2V, R3V, R150V	:VARISTOR		
M1F	:FAN MOTOR	X1M	:TERMINAL STRIP		
K10R, K20R,		Y1E	:ELECTRONIC EXPANSION		
K30R, K40R	:MAGNETIC RELAY	Y1S	:REVERSING SOLENOID		
N	:NEUTRAL		VALVE COIL		
PAM	:PULSE AMPLITUDE MODULATION		VALVE COIL		
A1P	:PRINTED CIRCUIT BOARD	Z1C~Z4C	:FERRITE CORE		
PS	:SWITCHING POWER SUPPLY	ZF	:NOISE FILTER		

3D098279C

RXM46/50/60/71UVMZ



C110	:CAPACITOR	PAM	:PULSE AMPLITUDE MODULATION	SA1	:SURGE ARRESTER
DB1	:DIODE BRIDGE	PCB	:PRINTED CIRCUIT BOARD	V2, V3, V401	:VARISTOR
FU1, FU2, FU3	:FUSE	PS	:SWITCHING POWER SUPPLY	X1M	:TERMINAL STRIP
IPM1	:INTELLIGENT POWER MODULE	Q1M	:OVERLOAD PROTECTOR	Y1E	:ELECTRONIC EXPANSION VALVE COIL
L	:LIVE	R1T, R2T, R3T	:THERMISTOR	Y1S	:REVERSING SOLENOID VALVE COIL
M1C	:COMPRESSOR MOTOR	S20, S40, S70,		Z1C, Z2C, Z3C, Z4C	:FERRITE CORE
M1F	:FAN MOTOR	S80, S90, S92,		ZF	:NOISE FILTER
MR4, MR30,		S402, X11A	:CONNECTOR	⊕	:PROTECTIVE EARTH
MRM10, MRM20	:MAGNETIC RELAY			⊥	:EARTH
N	:NEUTRAL				

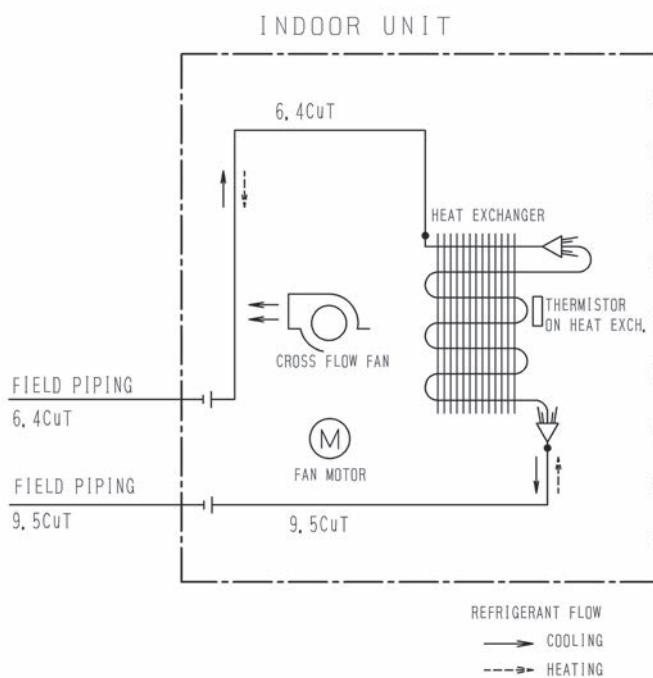
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6. Piping Diagrams

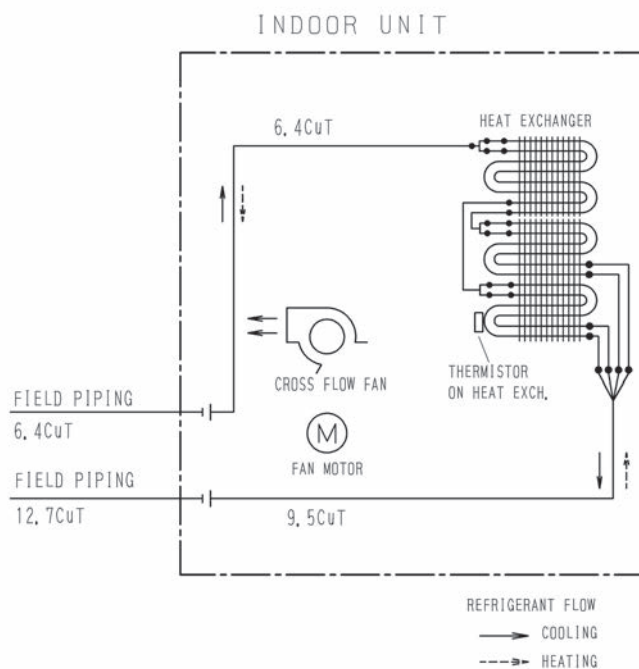
6.1 Indoor Unit

FTXM20/25/35UVMZ

FTXM46UVMZ



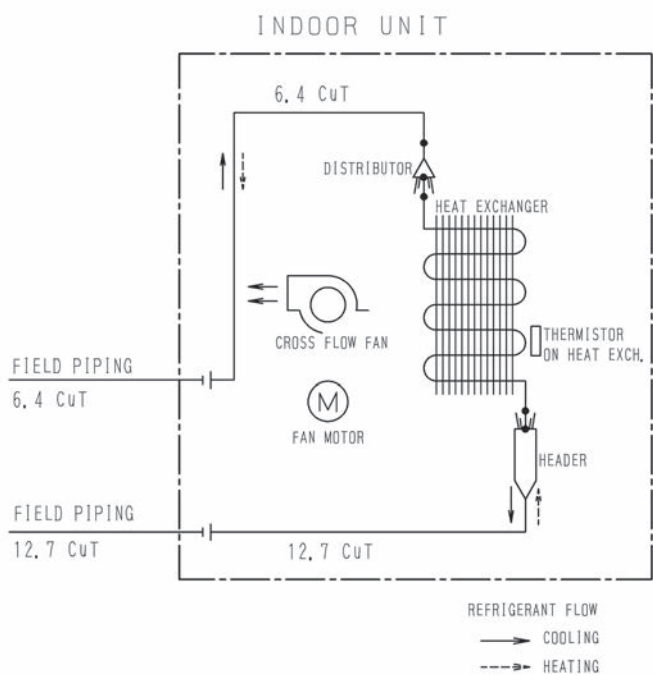
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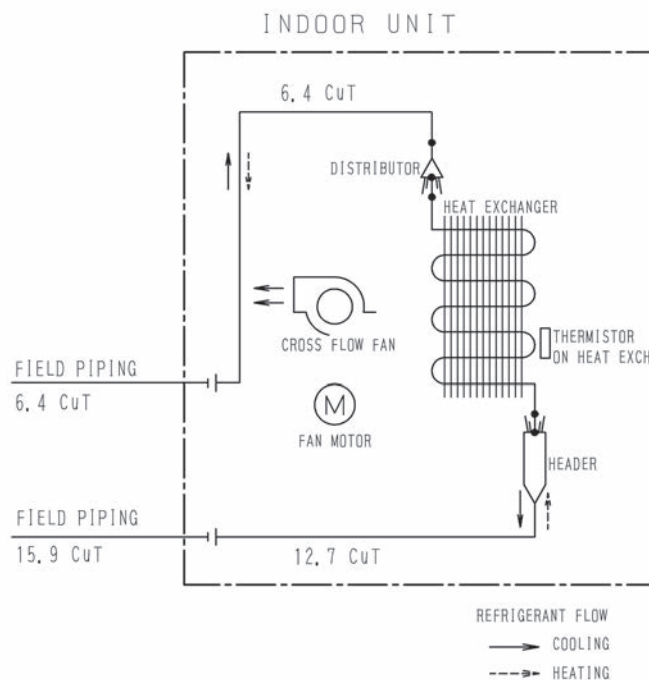
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FTXM50/60UVMZ

FTXM71UVMZ



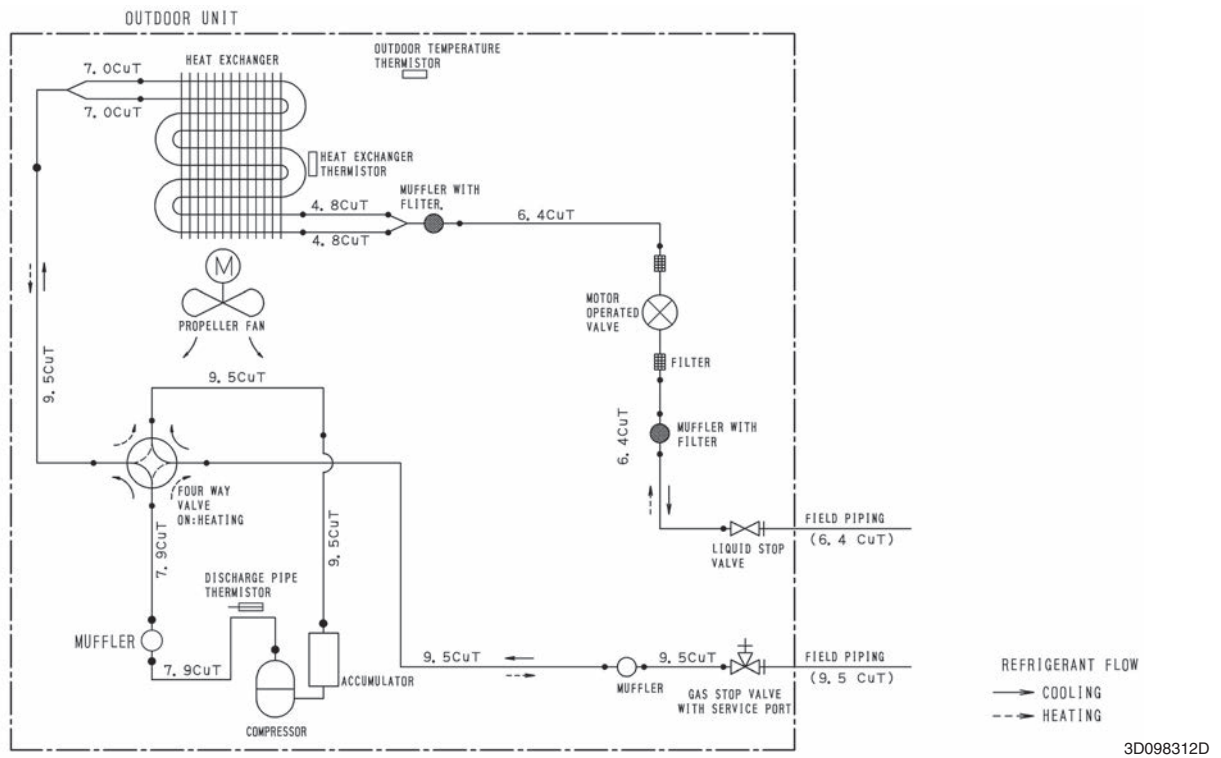
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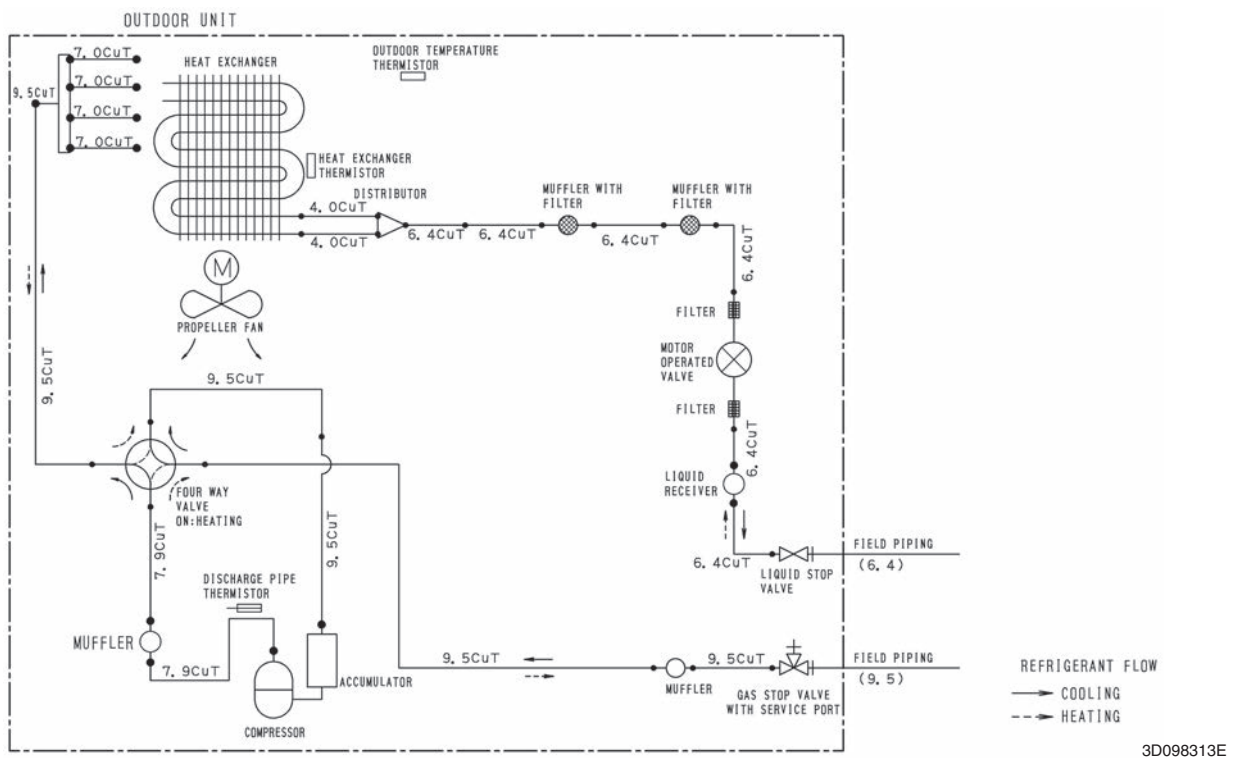
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6.2 Outdoor Unit

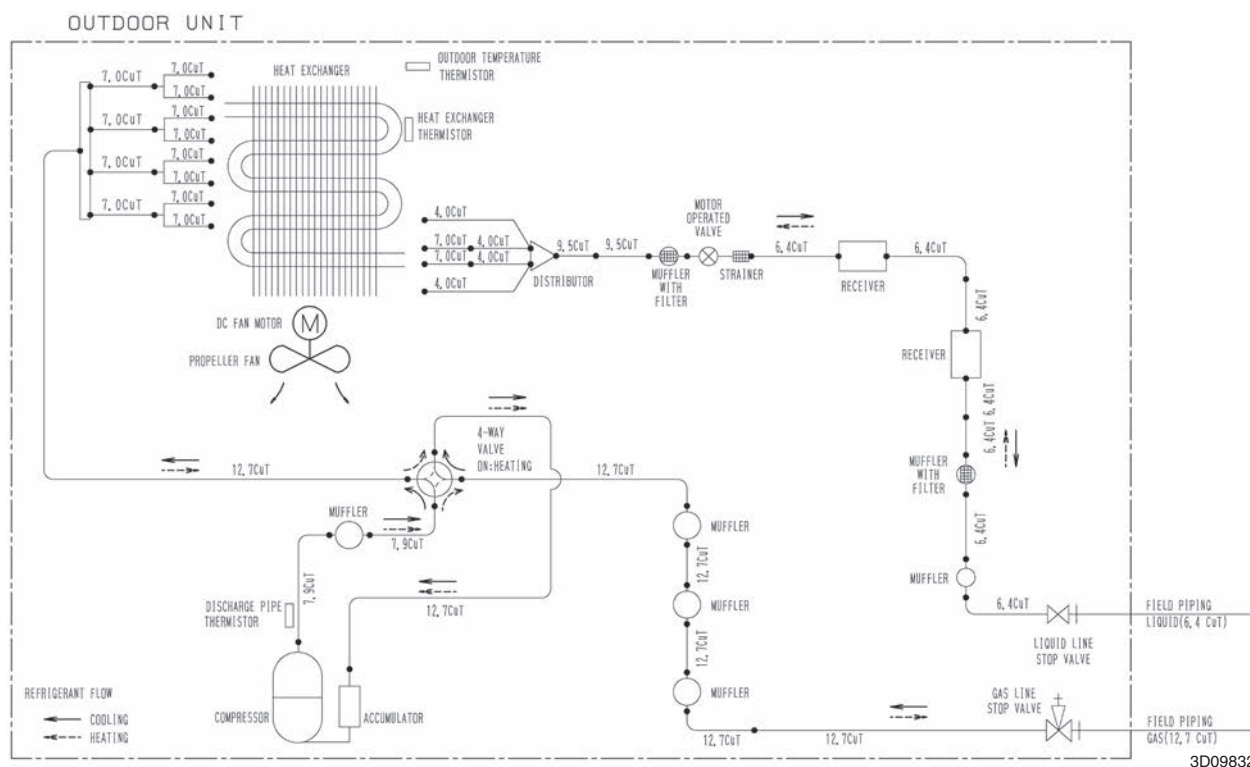
RXM20UVMZ



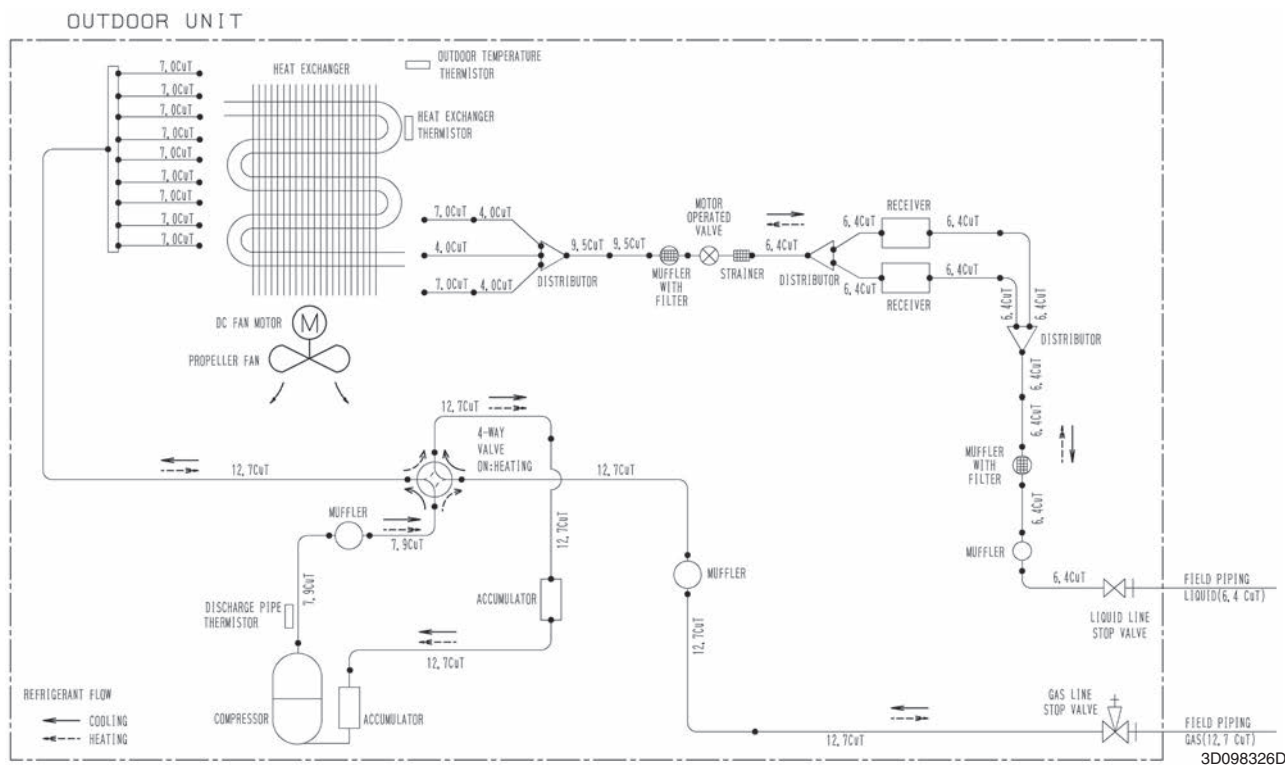
RXM25/35UVMZ



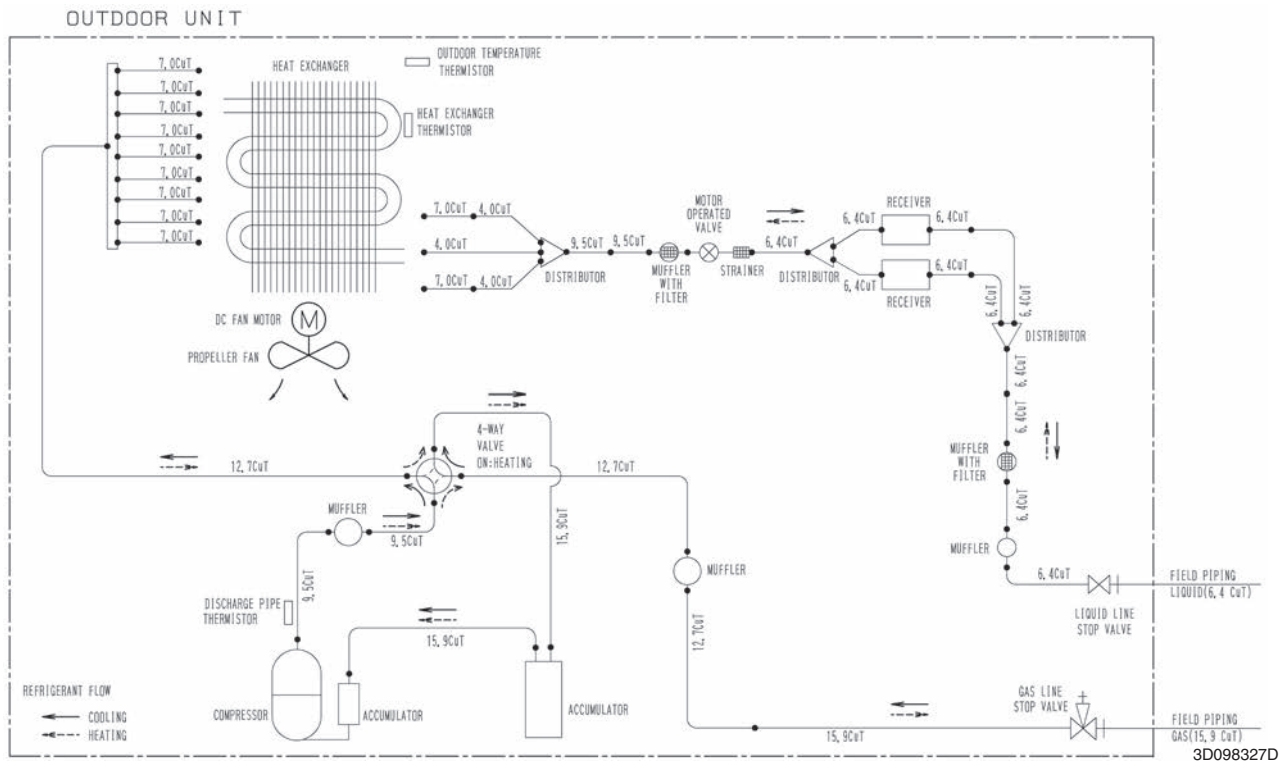
RXM46UVMZ



RXM50/60UVMZ



RXM71UVMZ



7. Capacity Tables

7.1 Heat Pump

FTXM20UVMZ + RXM20UVMZ (50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V)

Cooling

AFR	9.1
BF	0.15

INDOOR		OUTDOOR TEMPERATURE (°CDB)																	
EWB	EDB	20			25			30			32			35			40		
°C	°C	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
14.0	20	1.97	1.76	0.31	1.88	1.72	0.34	1.79	1.68	0.37	1.83	1.70	0.38	1.77	1.67	0.40	1.68	1.63	0.43
16.0	22	2.05	1.74	0.31	1.97	1.70	0.34	1.88	1.66	0.37	1.92	1.68	0.38	1.86	1.65	0.40	1.77	1.62	0.43
18.0	25	2.14	1.86	0.31	2.05	1.82	0.34	1.96	1.79	0.37	2.01	1.81	0.38	1.95	1.79	0.40	1.86	1.75	0.43
19.0	27	2.19	2.00	0.31	2.10	1.97	0.34	2.01	1.93	0.37	2.06	1.95	0.38	2.00	1.93	0.40	1.91	1.89	0.43
22.0	30	2.32	1.94	0.32	2.23	1.91	0.35	2.14	1.88	0.38	2.19	1.90	0.38	2.14	1.88	0.40	2.05	1.85	0.43
24.0	32	2.41	1.91	0.32	2.32	1.88	0.35	2.23	1.85	0.38	2.29	1.87	0.39	2.23	1.85	0.40	2.14	1.82	0.43

Heating

AFR	9.4
-----	-----

INDOOR		OUTDOOR TEMPERATURE (°CWB)									
EDB		-10		-5		0		6		10	
°C		TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
15.0		1.88	0.49	2.20	0.51	2.52	0.54	2.90	0.57	3.15	0.59
20.0		1.79	0.50	2.10	0.53	2.42	0.55	2.80	0.58	3.05	0.60
22.0		1.75	0.51	2.07	0.53	2.38	0.56	2.76	0.59	3.01	0.60
24.0		1.71	0.51	2.03	0.54	2.34	0.56	2.72	0.59	2.98	0.61
25.0		1.69	0.52	2.01	0.54	2.32	0.56	2.70	0.59	2.96	0.61
27.0		1.65	0.52	1.97	0.55	2.29	0.57	2.66	0.60	2.92	0.62

Symbols:

AFR	: Airflow rate	(m ³ /min.)
BF	: Bypass factor	
EWB	: Entering wet bulb temp.	(°C)
EDB	: Entering dry bulb temp.	(°C)
TC	: Total capacity	(kW)
SHC	: Sensible heat capacity	(kW)
PI	: Power input	(kW)

Notes:

1. ■ shows nominal (rated) capacities and power input.
2. TC, PI and SHC must be calculated by interpolation using the figures in the above tables. (Figures out of the tables should not be used for calculation.)
3. Capacities are based on the following conditions.
Corresponding refrigerant piping length : 5 m
Level difference : 0 m

3D123934

FTXM25UVMZ + RXM25UVMZ (50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V)

Cooling

AFR	10.4
BF	0.07

INDOOR		OUTDOOR TEMPERATURE (°CDB)																	
EWB	EDB	20			25			30			32			35			40		
°C	°C	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
14.0	20	2.46	2.21	0.38	2.35	2.16	0.41	2.23	2.11	0.45	2.28	2.13	0.46	2.21	2.10	0.48	2.10	2.04	0.52
16.0	22	2.57	2.17	0.38	2.46	2.12	0.42	2.34	2.08	0.45	2.40	2.10	0.47	2.33	2.07	0.49	2.21	2.02	0.52
18.0	25	2.68	2.33	0.38	2.57	2.28	0.42	2.46	2.24	0.46	2.51	2.26	0.47	2.44	2.24	0.49	2.33	2.19	0.53
19.0	27	2.73	2.50	0.38	2.62	2.46	0.42	2.51	2.42	0.46	2.57	2.44	0.47	2.50	2.42	0.49	2.38	2.37	0.53
22.0	30	2.90	2.43	0.39	2.79	2.40	0.42	2.68	2.36	0.46	2.74	2.38	0.47	2.67	2.36	0.49	2.56	2.32	0.53
24.0	32	3.01	2.39	0.39	2.90	2.35	0.43	2.79	2.32	0.46	2.86	2.34	0.47	2.79	2.32	0.50	2.67	2.28	0.53

Heating

AFR	10.1
-----	------

INDOOR		OUTDOOR TEMPERATURE (°CWB)									
EDB		-10		-5		0		6		10	
°C		TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
15.0		2.22	0.58	2.59	0.60	2.97	0.63	3.41	0.66	3.71	0.69
20.0		2.11	0.59	2.48	0.62	2.85	0.65	3.30	0.68	3.60	0.70
22.0		2.06	0.60	2.43	0.62	2.81	0.65	3.25	0.69	3.55	0.71
24.0		2.02	0.60	2.39	0.63	2.76	0.66	3.21	0.69	3.51	0.71
25.0		1.99	0.61	2.37	0.63	2.74	0.66	3.19	0.70	3.48	0.72
27.0		1.95	0.61	2.32	0.64	2.69	0.67	3.14	0.70	3.44	0.72

Symbols:

AFR	: Airflow rate	(m ³ /min.)
BF	: Bypass factor	
EWB	: Entering wet bulb temp.	(°C)
EDB	: Entering dry bulb temp.	(°C)
TC	: Total capacity	(kW)
SHC	: Sensible heat capacity	(kW)
PI	: Power input	(kW)

Notes:

1. ■ shows nominal (rated) capacities and power input.
2. TC, PI and SHC must be calculated by interpolation using the figures in the above tables. (Figures out of the tables should not be used for calculation.)
3. Capacities are based on the following conditions.
Corresponding refrigerant piping length : 5 m
Level difference : 0 m

3D123942

FTXM35UVMZ + RXM35UVMZ (50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V)**Cooling**

AFR	10.5
BF	0.09

INDOOR		OUTDOOR TEMPERATURE (°CDB)																	
EWB	EDB	20			25			30			32			35			40		
°C	°C	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
14.0	20	3.44	2.66	0.62	3.28	2.58	0.69	3.13	2.50	0.75	3.19	2.54	0.77	3.10	2.49	0.80	2.93	2.41	0.86
16.0	22	3.60	2.61	0.63	3.44	2.54	0.69	3.28	2.47	0.75	3.36	2.50	0.77	3.26	2.46	0.80	3.10	2.38	0.86
18.0	25	3.75	2.75	0.63	3.59	2.68	0.69	3.44	2.62	0.75	3.52	2.65	0.77	3.42	2.61	0.81	3.26	2.54	0.87
19.0	27	3.83	2.91	0.63	3.67	2.85	0.69	3.51	2.79	0.75	3.60	2.82	0.77	3.50	2.78	0.81	3.34	2.72	0.87
22.0	30	4.06	2.82	0.64	3.90	2.76	0.70	3.75	2.70	0.76	3.84	2.74	0.78	3.74	2.70	0.82	3.58	2.64	0.88
24.0	32	4.21	2.75	0.64	4.06	2.70	0.70	3.90	2.64	0.76	4.00	2.68	0.78	3.90	2.64	0.82	3.74	2.59	0.88

Heating

AFR	11.0
-----	------

INDOOR		OUTDOOR TEMPERATURE (°CWB)									
EDB		-10		-5		0		6		10	
°C		TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
15.0		2.69	0.74	3.14	0.77	3.60	0.81	4.14	0.85	4.50	0.88
20.0		2.55	0.76	3.01	0.79	3.46	0.83	4.00	0.87	4.36	0.90
22.0		2.50	0.76	2.95	0.80	3.40	0.83	3.94	0.88	4.31	0.91
24.0		2.44	0.77	2.90	0.81	3.35	0.84	3.89	0.89	4.25	0.91
25.0		2.42	0.77	2.87	0.81	3.32	0.85	3.86	0.89	4.22	0.92
27.0		2.36	0.78	2.81	0.82	3.26	0.85	3.81	0.90	4.17	0.93

Symbols:

AFR	: Airflow rate	(m ³ /min.)
BF	: Bypass factor	
EWB	: Entering wet bulb temp.	(°C)
EDB	: Entering dry bulb temp.	(°C)
TC	: Total capacity	(kW)
SHC	: Sensible heat capacity	(kW)
PI	: Power input	(kW)

Notes:

1. ■ shows nominal (rated) capacities and power input.
2. TC, PI and SHC must be calculated by interpolation using the figures in the above tables. (Figures out of the tables should not be used for calculation.)
3. Capacities are based on the following conditions.
Corresponding refrigerant piping length : 5 m
Level difference : 0 m

3D123943

FTXM46UVMZ + RXM46UVMZ (50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V)

Cooling

AFR	12.2
BF	0.20

INDOOR		OUTDOOR TEMPERATURE (°CDB)																	
EWB	EDB	20			25			30			32			35			40		
°C	°C	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
14.0	20	4.52	3.22	0.93	4.32	3.12	1.01	4.11	3.01	1.10	4.20	3.06	1.13	4.07	2.99	1.19	3.86	2.88	1.28
16.0	22	4.73	3.17	0.93	4.52	3.07	1.02	4.31	2.97	1.11	4.41	3.01	1.14	4.28	2.95	1.19	4.07	2.85	1.28
18.0	25	4.93	3.30	0.94	4.72	3.20	1.03	4.52	3.11	1.11	4.62	3.16	1.14	4.49	3.10	1.20	4.28	3.00	1.29
19.0	27	5.03	3.46	0.94	4.83	3.37	1.03	4.62	3.28	1.12	4.73	3.32	1.15	4.60	3.27	1.20	4.39	3.18	1.29
22.0	30	5.34	3.33	0.95	5.13	3.24	1.04	4.92	3.17	1.13	5.05	3.21	1.15	4.92	3.16	1.21	4.70	3.08	1.30
24.0	32	5.54	3.23	0.95	5.33	3.16	1.04	5.13	3.09	1.13	5.26	3.13	1.16	5.13	3.09	1.21	4.92	3.01	1.30

Heating

AFR	12.1
-----	------

INDOOR		OUTDOOR TEMPERATURE (°CWB)									
EDB		-10		-5		0		6		10	
°C		TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
15.0		3.64	1.18	4.24	1.23	4.85	1.29	5.59	1.36	6.07	1.40
20.0		3.45	1.21	4.06	1.26	4.67	1.32	5.40	1.39	5.89	1.44
22.0		3.37	1.22	3.98	1.28	4.59	1.33	5.33	1.40	5.81	1.45
24.0		3.30	1.23	3.91	1.29	4.52	1.35	5.25	1.42	5.74	1.46
25.0		3.26	1.24	3.87	1.30	4.48	1.35	5.21	1.42	5.70	1.47
27.0		3.19	1.25	3.80	1.31	4.41	1.37	5.14	1.43	5.63	1.48

Symbols:

AFR	: Airflow rate	(m ³ /min.)
BF	: Bypass factor	
EWB	: Entering wet bulb temp.	(°C)
EDB	: Entering dry bulb temp.	(°C)
TC	: Total capacity	(kW)
SHC	: Sensible heat capacity	(kW)
PI	: Power input	(kW)

Notes:

1. ■ shows nominal (rated) capacities and power input.
2. TC, PI and SHC must be calculated by interpolation using the figures in the above tables. (Figures out of the tables should not be used for calculation.)
3. Capacities are based on the following conditions.
Corresponding refrigerant piping length : 5 m
Level difference : 0 m

3D123944

FTXM50UVMZ + RXM50UVMZ (50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V)

Cooling

AFR	16.0
BF	0.07

INDOOR		OUTDOOR TEMPERATURE (°CDB)																	
EWB	EDB	20			25			30			32			35			40		
°C	°C	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
14.0	20	4.92	3.93	0.88	4.69	3.82	0.96	4.47	3.72	1.05	4.56	3.76	1.08	4.42	3.70	1.13	4.19	3.59	1.21
16.0	22	5.14	3.86	0.88	4.91	3.76	0.97	4.69	3.67	1.05	4.79	3.71	1.08	4.65	3.65	1.13	4.42	3.55	1.22
18.0	25	5.36	4.09	0.89	5.13	4.00	0.97	4.91	3.90	1.06	5.02	3.95	1.09	4.88	3.89	1.14	4.65	3.80	1.22
19.0	27	5.47	4.35	0.89	5.24	4.26	0.98	5.02	4.17	1.06	5.14	4.22	1.09	5.00	4.16	1.14	4.77	4.07	1.22
22.0	30	5.80	4.21	0.90	5.58	4.13	0.98	5.35	4.05	1.07	5.49	4.10	1.10	5.35	4.05	1.15	5.11	3.97	1.23
24.0	32	6.02	4.11	0.90	5.80	4.04	0.99	5.57	3.97	1.07	5.72	4.01	1.10	5.58	3.97	1.15	5.34	3.90	1.24

Heating

AFR	17.2
-----	------

INDOOR		OUTDOOR TEMPERATURE (°CWB)									
EDB		-10		-5		0		6		10	
°C		TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
15.0		4.11	1.13	4.80	1.19	5.48	1.24	6.31	1.31	6.86	1.35
20.0		3.90	1.16	4.58	1.22	5.27	1.27	6.10	1.34	6.65	1.38
22.0		3.81	1.18	4.50	1.23	5.19	1.29	6.02	1.35	6.57	1.40
24.0		3.73	1.19	4.42	1.24	5.10	1.30	5.93	1.36	6.48	1.41
25.0		3.69	1.19	4.37	1.25	5.06	1.30	5.89	1.37	6.44	1.41
27.0		3.60	1.21	4.29	1.26	4.98	1.32	5.81	1.38	6.36	1.43

Symbols:

AFR	: Airflow rate	(m ³ /min.)
BF	: Bypass factor	
EWB	: Entering wet bulb temp.	(°C)
EDB	: Entering dry bulb temp.	(°C)
TC	: Total capacity	(kW)
SHC	: Sensible heat capacity	(kW)
PI	: Power input	(kW)

Notes:

1. ■ shows nominal (rated) capacities and power input.
2. TC, PI and SHC must be calculated by interpolation using the figures in the above tables. (Figures out of the tables should not be used for calculation.)
3. Capacities are based on the following conditions.
Corresponding refrigerant piping length : 5 m
Level difference : 0 m

3D123945

FTXM60UVMZ + RXM60UVMZ (50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V)

Cooling

AFR	18.6
BF	0.08

INDOOR		OUTDOOR TEMPERATURE (°CDB)																	
EWB	EDB	20			25			30			32			35			40		
°C	°C	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
14.0	20	5.90	4.63	1.17	5.63	4.50	1.29	5.36	4.37	1.40	5.48	4.43	1.44	5.31	4.35	1.50	5.03	4.21	1.62
16.0	22	6.16	4.55	1.18	5.90	4.43	1.29	5.63	4.31	1.41	5.75	4.36	1.44	5.59	4.29	1.51	5.31	4.17	1.62
18.0	25	6.43	4.80	1.19	6.16	4.69	1.30	5.89	4.58	1.41	6.03	4.64	1.45	5.86	4.57	1.52	5.58	4.45	1.63
19.0	27	6.56	5.10	1.19	6.29	4.99	1.30	6.03	4.88	1.42	6.17	4.94	1.45	6.00	4.87	1.52	5.72	4.76	1.63
22.0	30	6.96	4.93	1.20	6.69	4.84	1.31	6.42	4.74	1.43	6.58	4.80	1.46	6.41	4.74	1.53	6.14	4.64	1.64
24.0	32	7.23	4.81	1.21	6.96	4.73	1.32	6.69	4.64	1.43	6.86	4.69	1.47	6.69	4.64	1.54	6.41	4.55	1.65

Heating

AFR	20.0
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INDOOR		OUTDOOR TEMPERATURE (°CWB)									
EDB		-10		-5		0		6		10	
°C		TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
15.0		4.91	1.40	5.74	1.47	6.56	1.54	7.55	1.62	8.21	1.68
20.0		4.66	1.44	5.49	1.51	6.31	1.58	7.30	1.66	7.96	1.71
22.0		4.56	1.46	5.39	1.52	6.21	1.59	7.20	1.67	7.86	1.73
24.0		4.46	1.47	5.28	1.54	6.11	1.61	7.10	1.69	7.76	1.74
25.0		4.41	1.48	5.23	1.55	6.06	1.62	7.05	1.70	7.71	1.75
27.0		4.31	1.49	5.13	1.56	5.96	1.63	6.95	1.71	7.61	1.77

Symbols:

AFR	: Airflow rate	(m ³ /min.)
BF	: Bypass factor	
EWB	: Entering wet bulb temp.	(°C)
EDB	: Entering dry bulb temp.	(°C)
TC	: Total capacity	(kW)
SHC	: Sensible heat capacity	(kW)
PI	: Power input	(kW)

Notes:

1. ■ shows nominal (rated) capacities and power input.
2. TC, PI and SHC must be calculated by interpolation using the figures in the above tables. (Figures out of the tables should not be used for calculation.)
3. Capacities are based on the following conditions.
Corresponding refrigerant piping length : 5 m
Level difference : 0 m

3D123946

FTXM71UVMZ + RXM71UVMZ (50 Hz, 220 - 230 - 240 V / 60 Hz, 220 - 230 V)

Cooling

AFR	18.6
BF	0.08

INDOOR		OUTDOOR TEMPERATURE (°CDB)																	
EWB	EDB	20			25			30			32			35			40		
°C	°C	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
14.0	20	6.98	5.16	1.49	6.66	5.00	1.63	6.35	4.85	1.78	6.48	4.91	1.82	6.28	4.82	1.91	5.95	4.65	2.05
16.0	22	7.29	5.07	1.50	6.98	4.92	1.64	6.66	4.78	1.78	6.81	4.84	1.83	6.61	4.75	1.92	6.28	4.60	2.06
18.0	25	7.61	5.31	1.51	7.29	5.17	1.65	6.97	5.04	1.79	7.13	5.10	1.84	6.94	5.02	1.93	6.61	4.88	2.07
19.0	27	7.76	5.60	1.51	7.45	5.47	1.65	7.13	5.33	1.80	7.30	5.40	1.84	7.10	5.32	1.93	6.77	5.18	2.07
22.0	30	8.24	5.40	1.52	7.92	5.28	1.67	7.60	5.16	1.81	7.79	5.23	1.86	7.59	5.16	1.94	7.26	5.04	2.09
24.0	32	8.55	5.26	1.53	8.23	5.15	1.68	7.92	5.04	1.82	8.12	5.11	1.87	7.92	5.04	1.95	7.59	4.93	2.09

Heating

AFR	19.7
-----	------

INDOOR		OUTDOOR TEMPERATURE (°CWB)									
EDB		-10		-5		0		6		10	
°C		TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
15.0		5.45	1.62	6.37	1.70	7.28	1.78	8.38	1.88	9.11	1.94
20.0		5.17	1.67	6.09	1.75	7.00	1.83	8.10	1.92	8.83	1.98
22.0		5.06	1.68	5.98	1.76	6.89	1.84	7.99	1.94	8.72	2.00
24.0		4.95	1.70	5.86	1.78	6.78	1.86	7.88	1.95	8.61	2.02
25.0		4.89	1.71	5.81	1.79	6.72	1.87	7.82	1.96	8.55	2.03
27.0		4.78	1.73	5.70	1.81	6.61	1.89	7.71	1.98	8.44	2.04

Symbols:

AFR	: Airflow rate	(m ³ /min.)
BF	: Bypass factor	
EWB	: Entering wet bulb temp.	(°C)
EDB	: Entering dry bulb temp.	(°C)
TC	: Total capacity	(kW)
SHC	: Sensible heat capacity	(kW)
PI	: Power input	(kW)

Notes:

1. ■ shows nominal (rated) capacities and power input.
2. TC, PI and SHC must be calculated by interpolation using the figures in the above tables. (Figures out of the tables should not be used for calculation.)
3. Capacities are based on the following conditions.
Corresponding refrigerant piping length : 5 m
Level difference : 0 m

3D123947

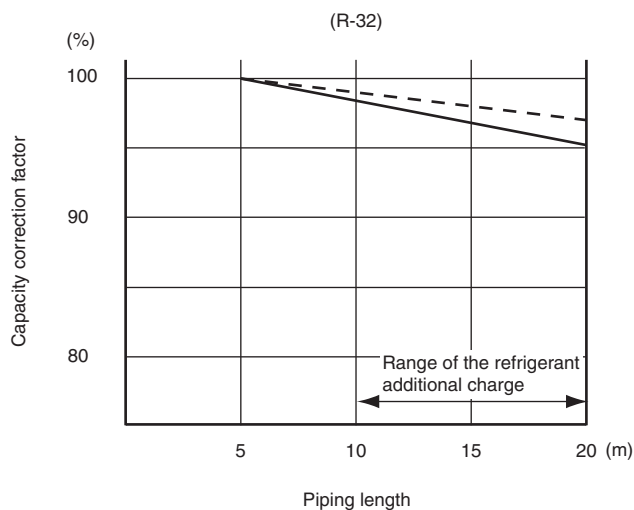
7.2 Capacity Correction Factor by the Length of Refrigerant Piping (Reference)

The cooling capacity and the heating capacity of the unit have to be corrected in accordance with the length of refrigerant piping — the distance between the indoor unit and the outdoor unit.

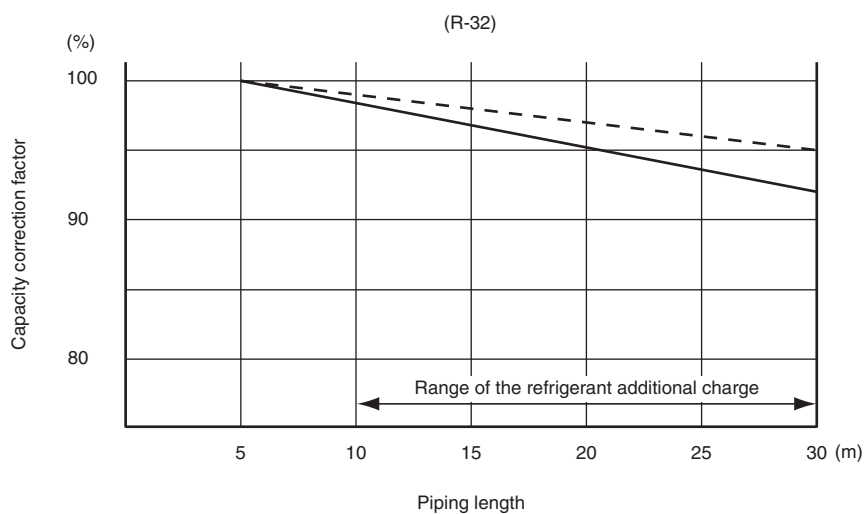
<— line : cooling capacity>

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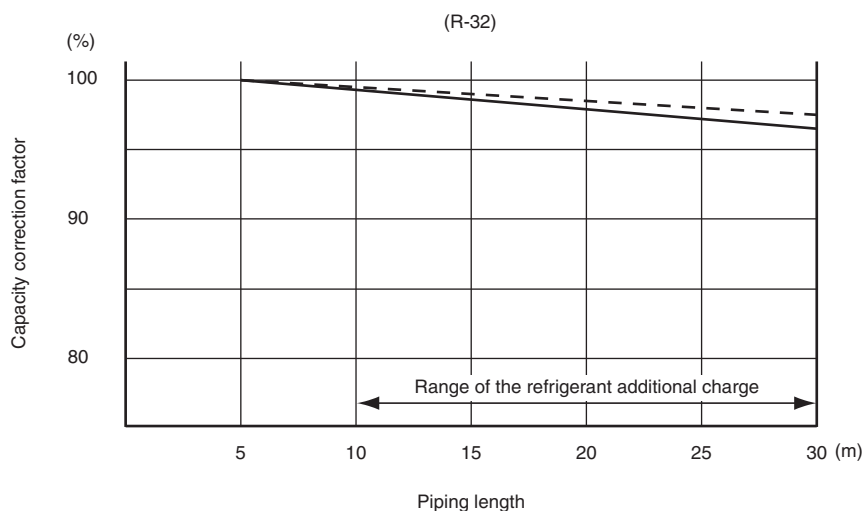
7.2.1 20/25/35 Class



7.2.2 46/50/60 Class



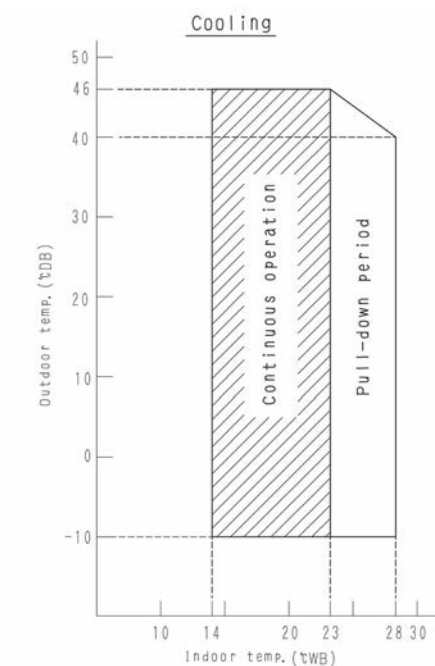
7.2.3 71 Class



Note: The graphs show the factor when additional refrigerant of the proper quantity is charged.

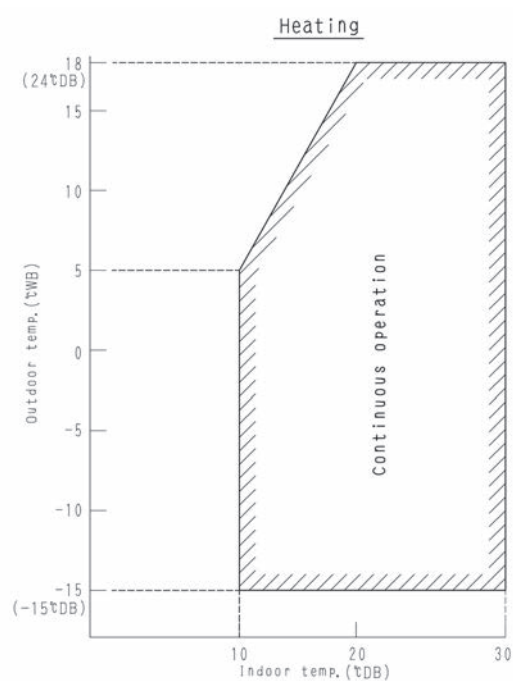
8. Operation Limit

RXM20/25/35/46/50/60/71UVMZ



Notes:
The graphs are based on the following conditions.

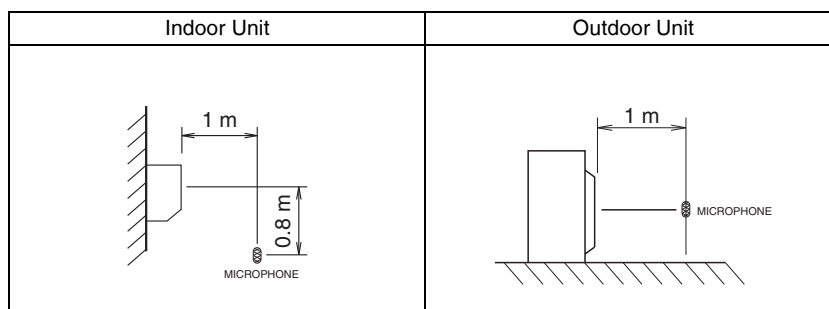
- Equivalent piping length 5m
- Level difference 0m
- Air flow rate High



3D098659C

9. Sound Level

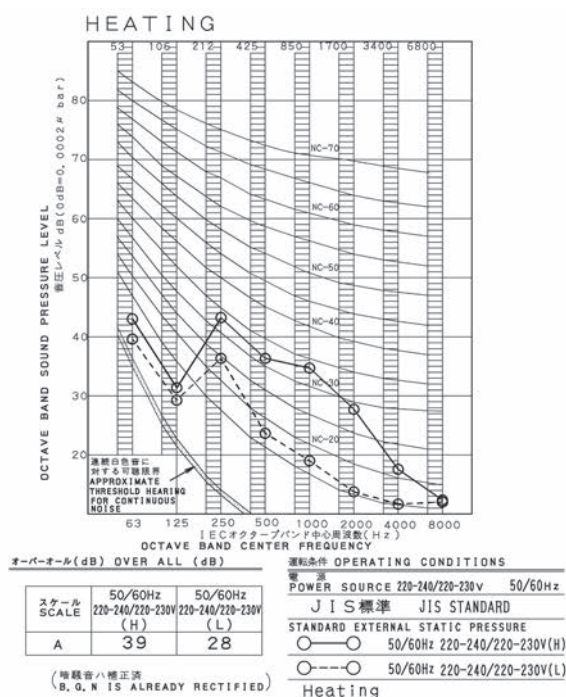
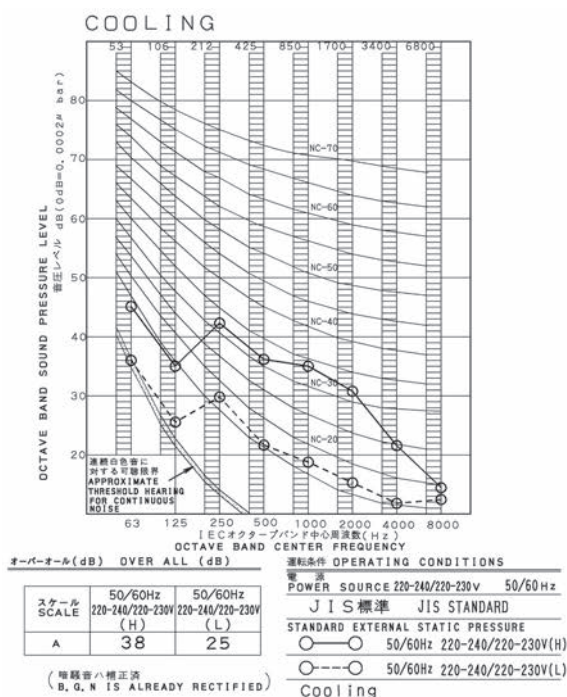
9.1 Measuring Location



- Notes:**
1. Operation sound is measured in an anechoic chamber.
 2. The operation sound measuring method is based on JIS standard.

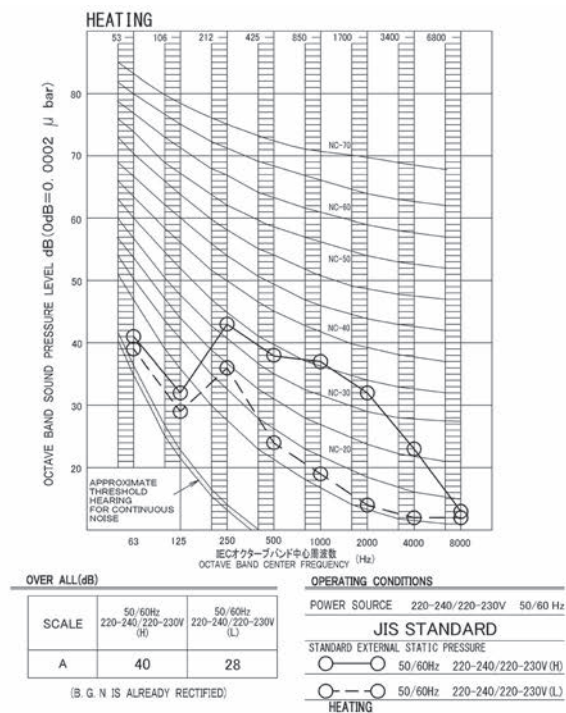
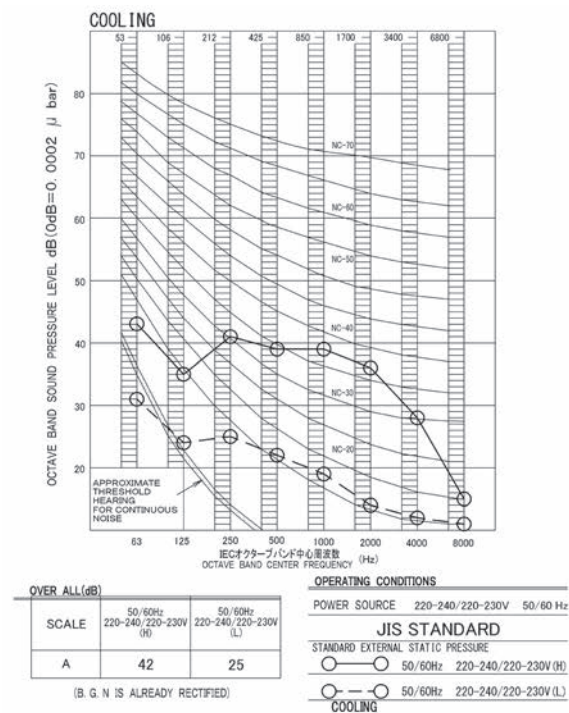
9.2 Indoor Unit

FTXM20UVMZ



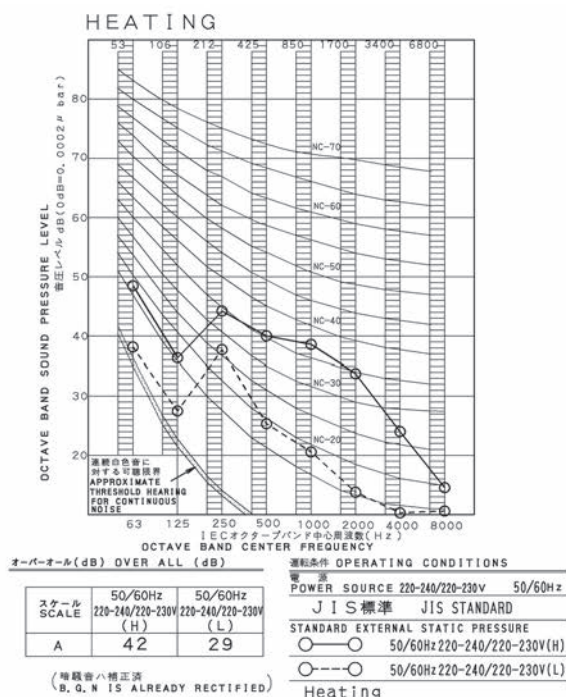
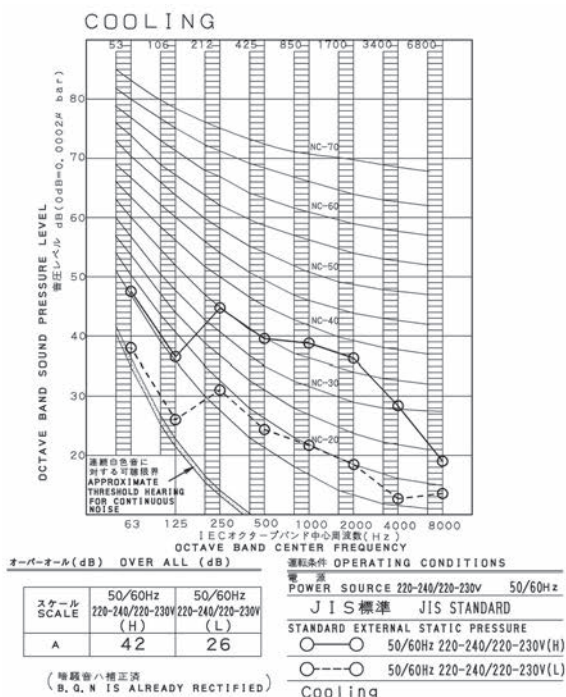
3D098930A

FTXM25UVMZ



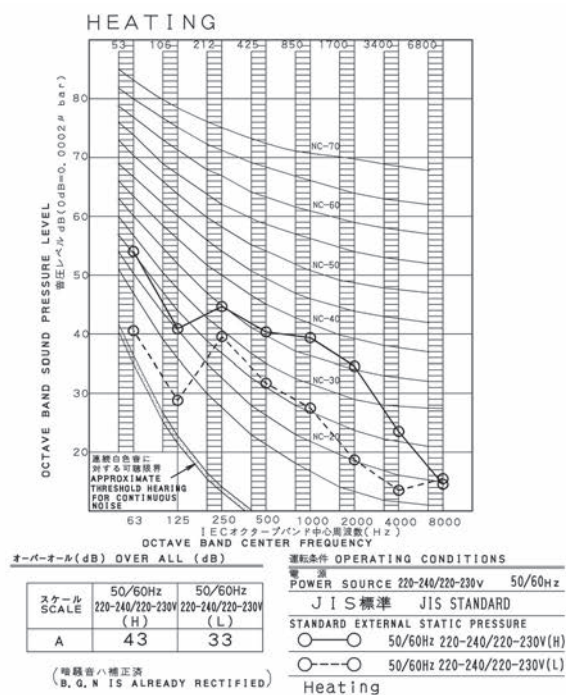
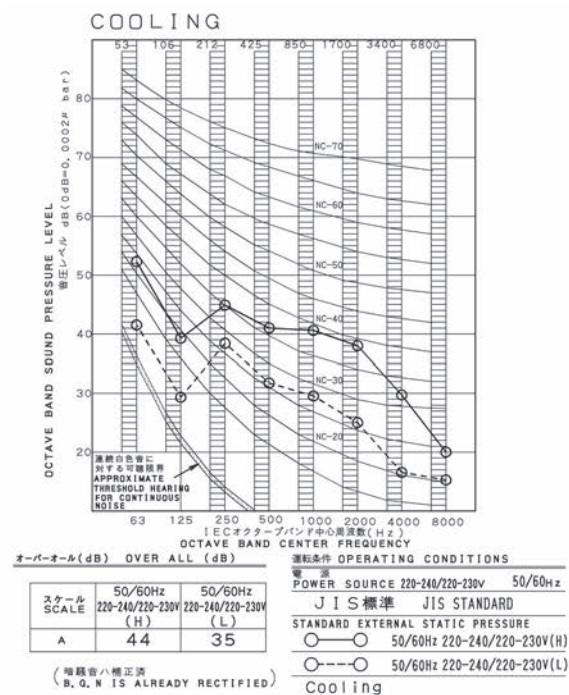
3D121305

FTXM35UVMZ



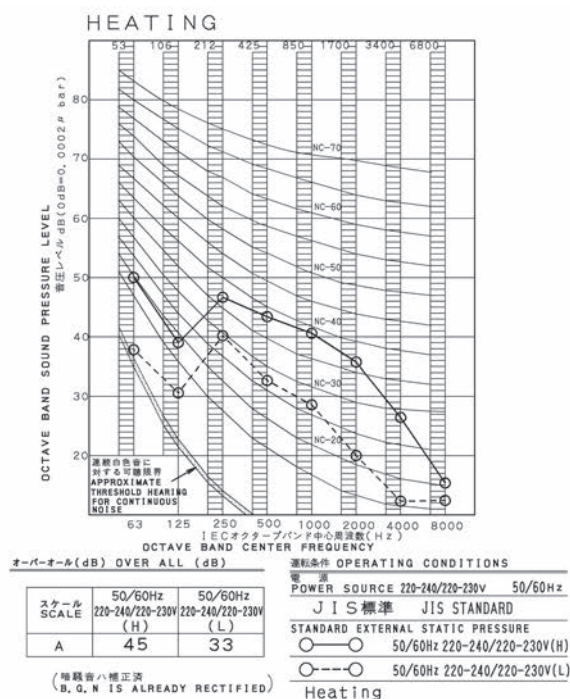
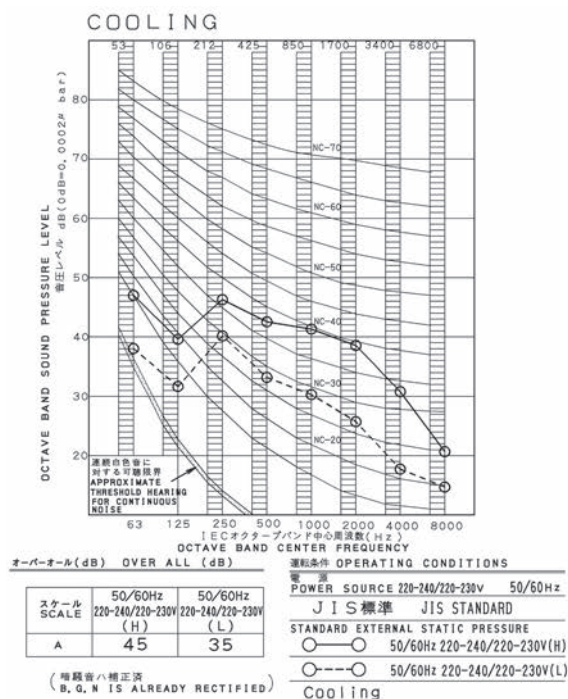
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FTXM46UVMZ



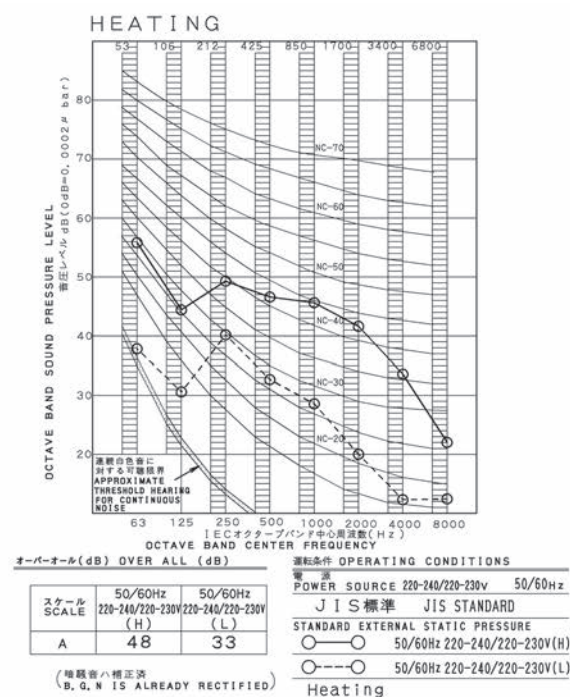
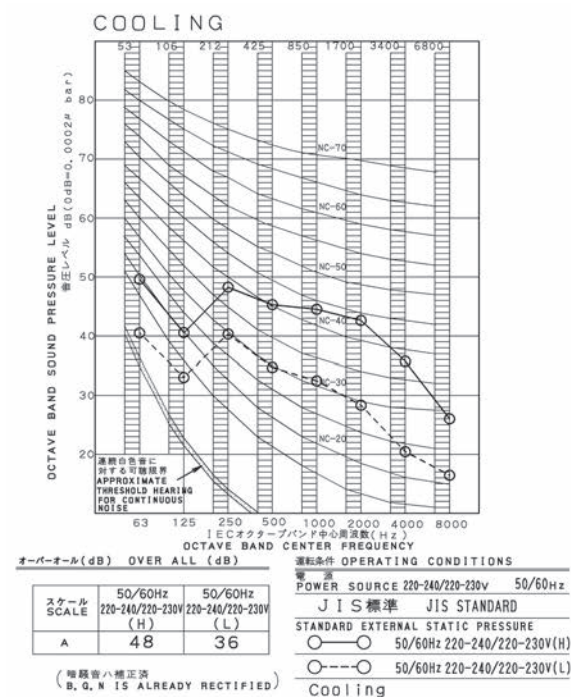
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FTXM50UVMZ



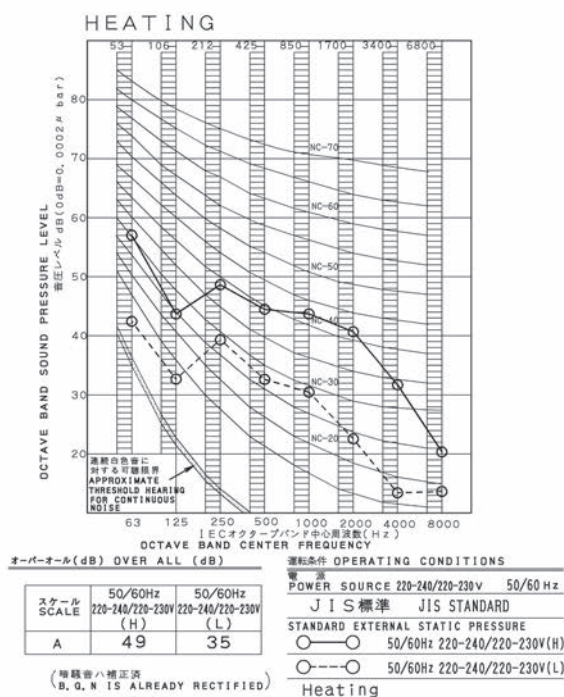
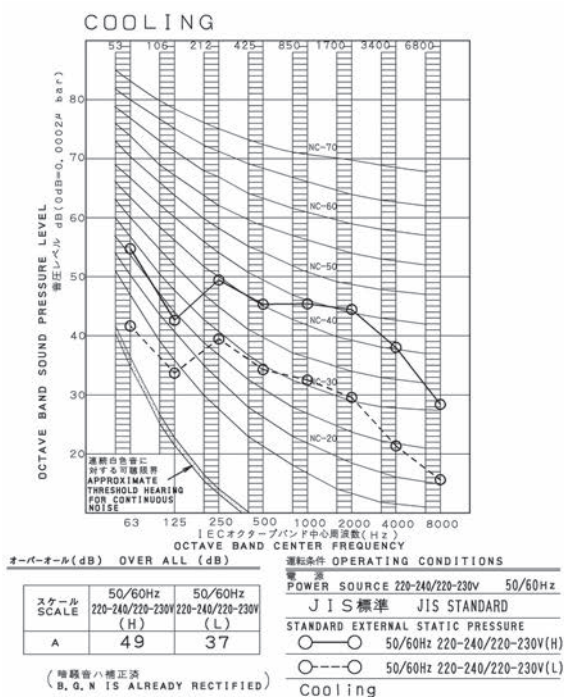
3D098954D

FTXM60UVMZ



3D098956D

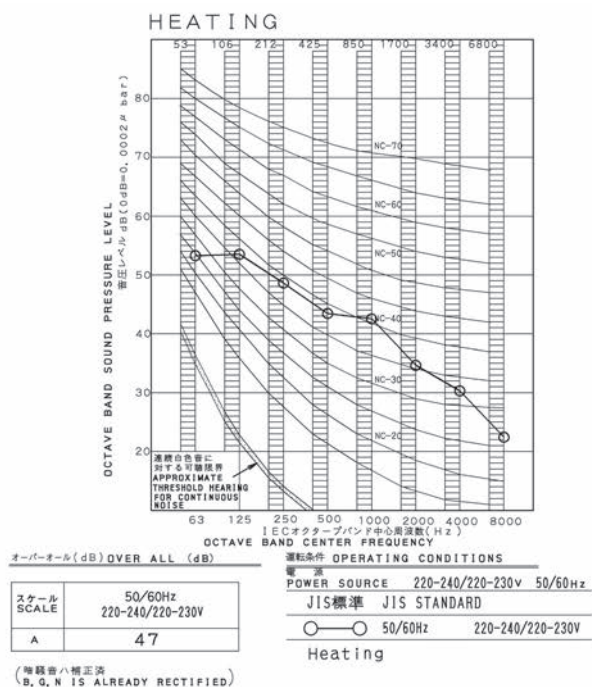
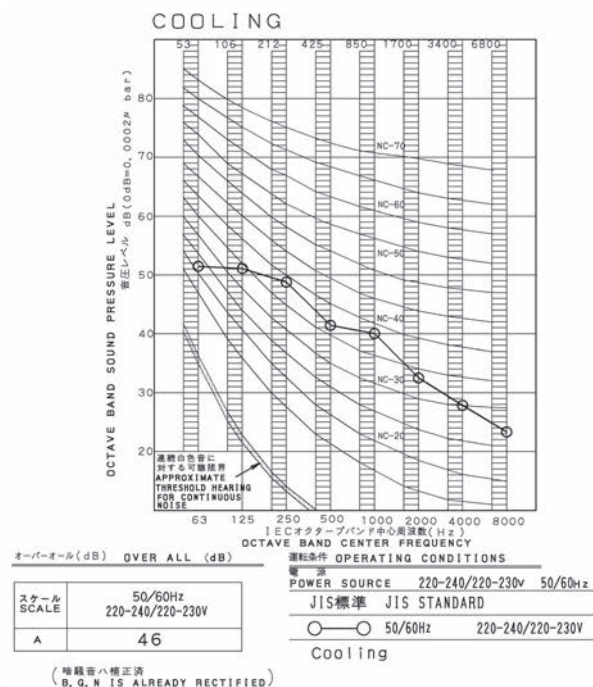
FTXM71UVMZ



3D098975D

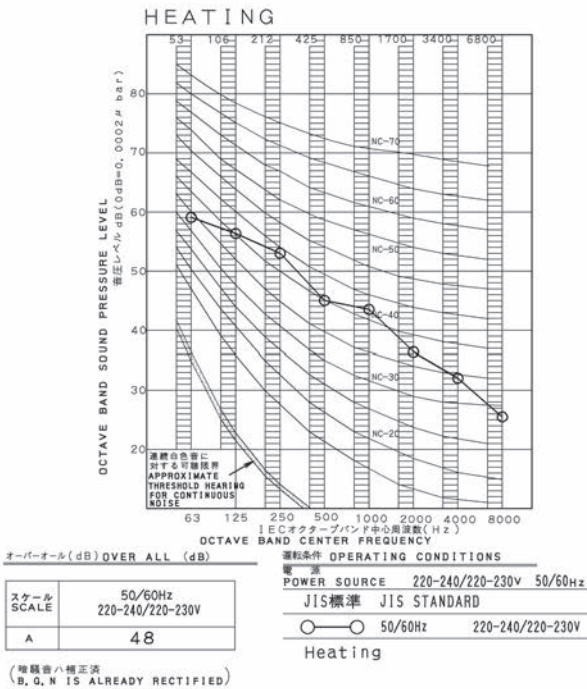
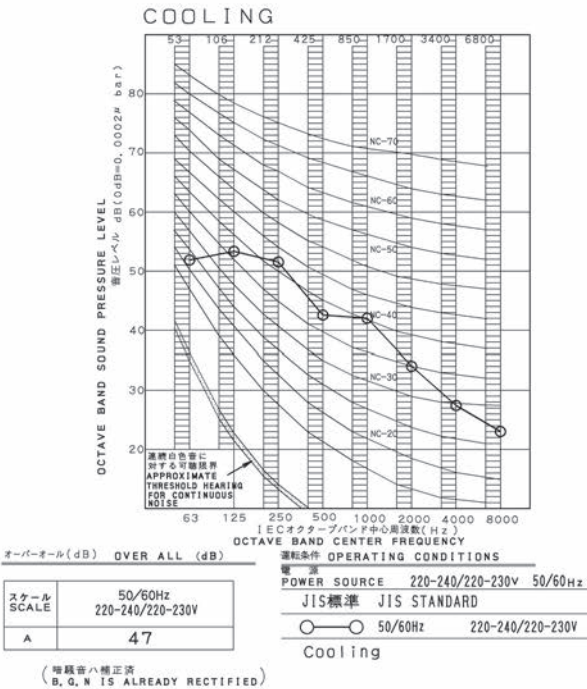
9.3 Outdoor Unit

RXM20UVMZ



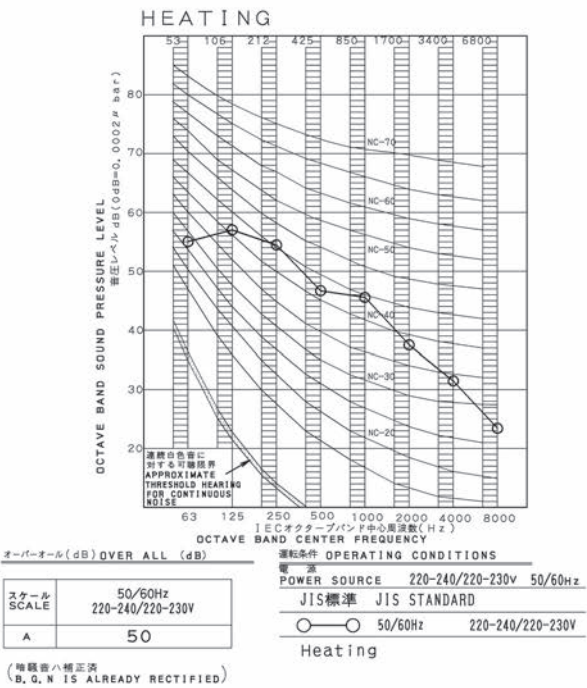
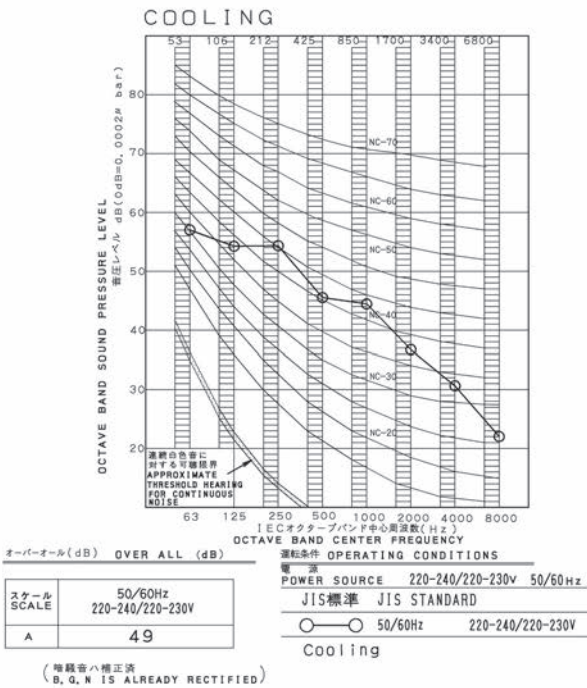
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RXM25UVMZ



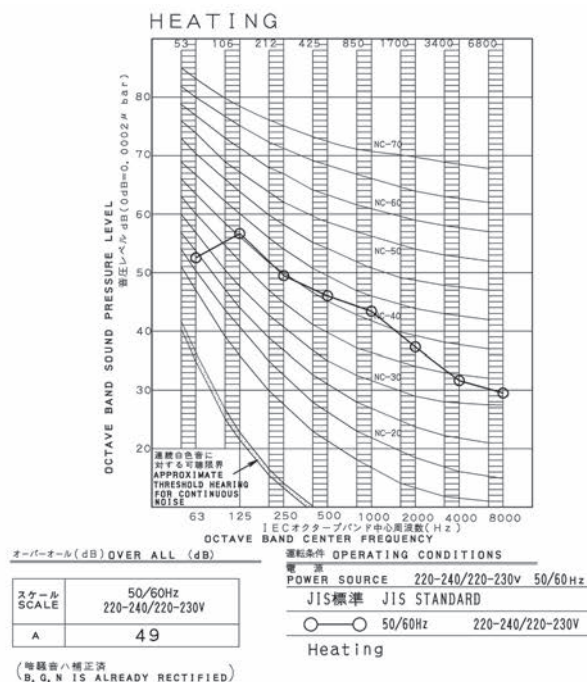
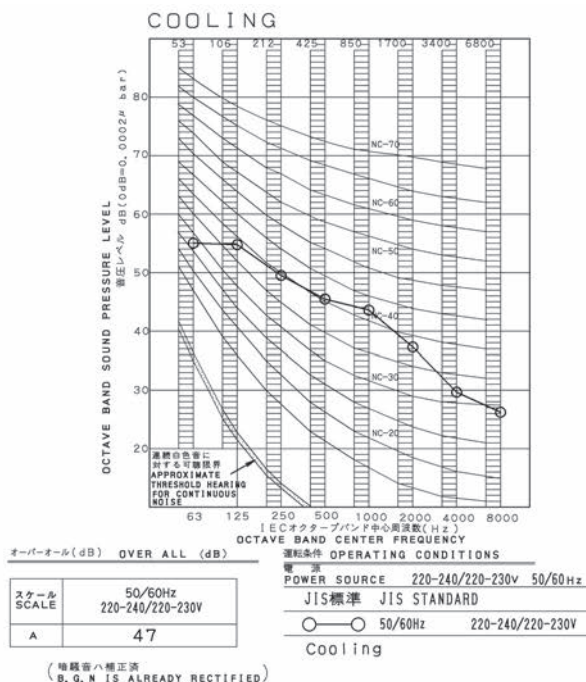
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RXM35UVMZ



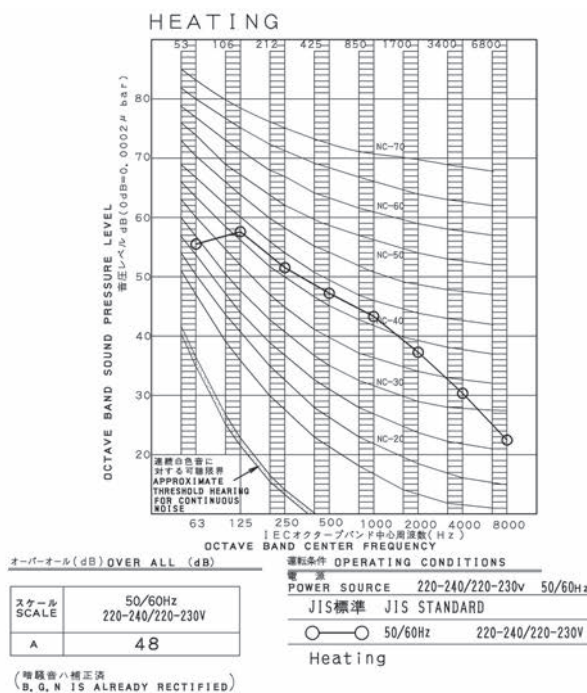
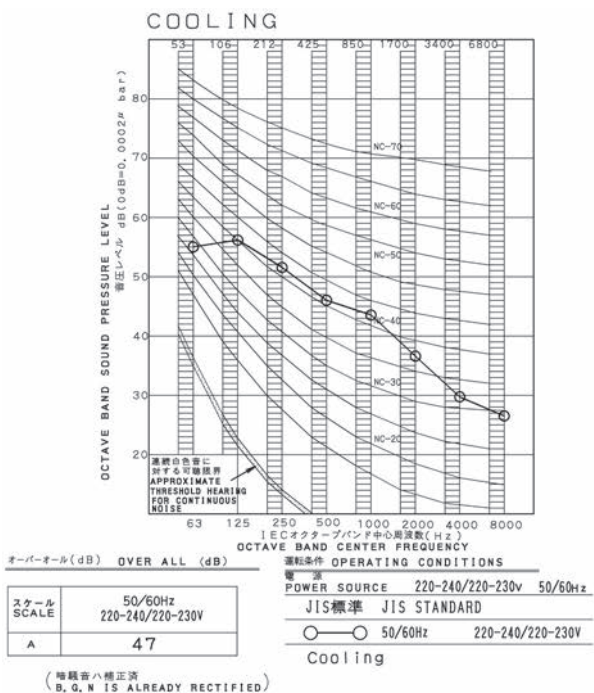
3D108697

RXM46UVMZ



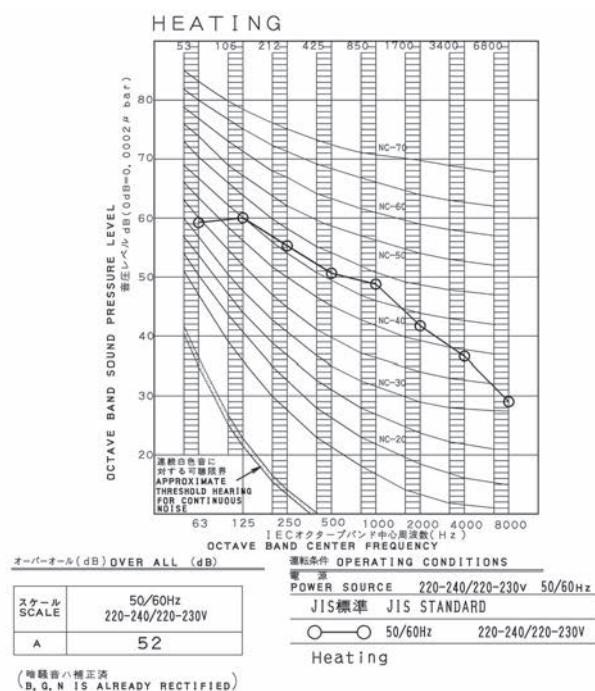
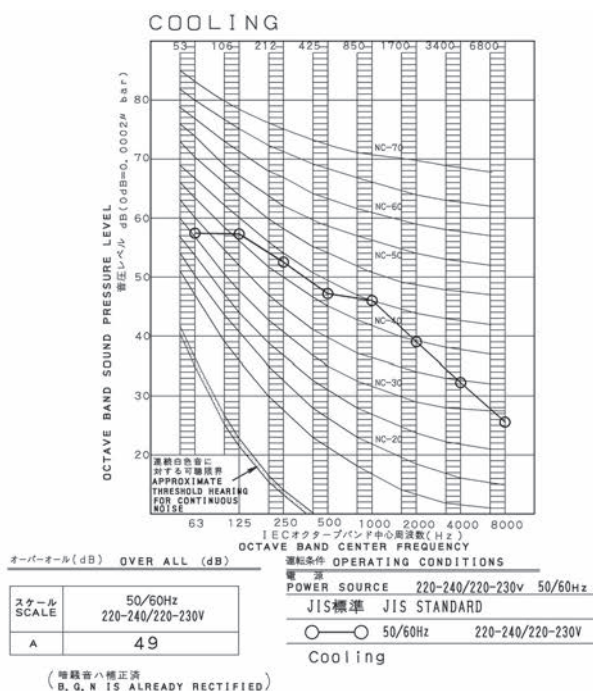
3D108698

RXM50UVMZ



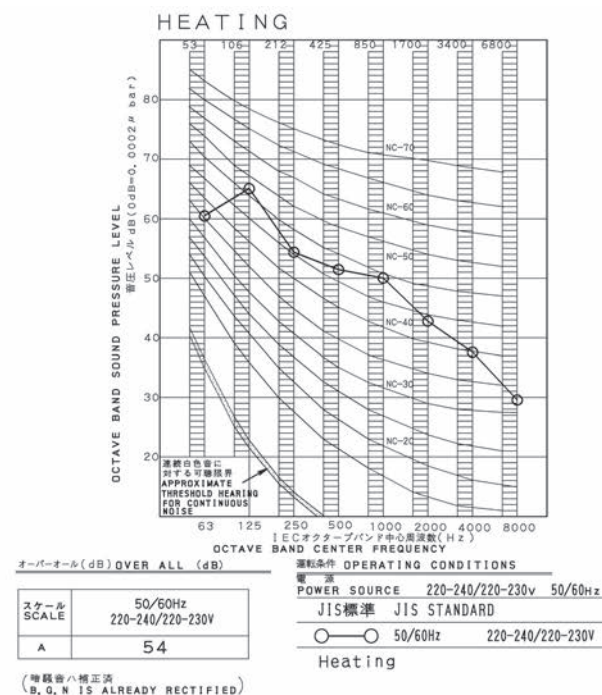
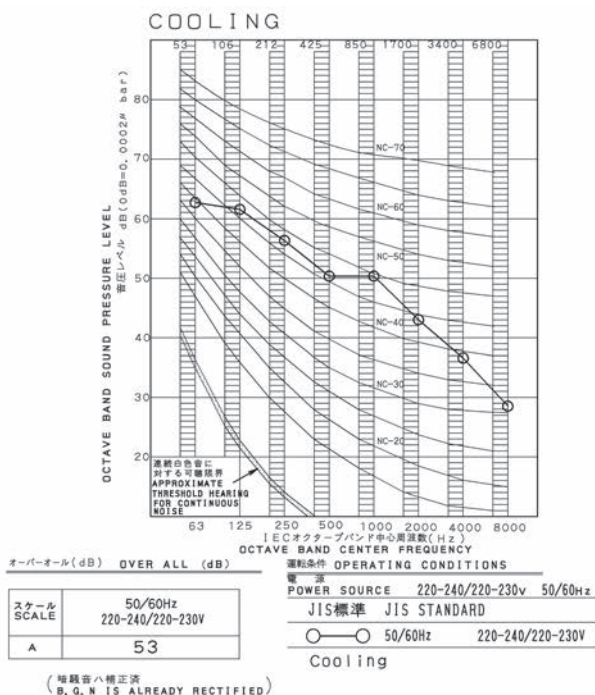
3D098928B

RXM60UVMZ



3D098929B

RXM71UVMZ



3D098977B

10. Electric Characteristics

Indoor Unit	Outdoor Unit	Power Supply				Compressor		OFM		IFM	
		Hz - Volts	Voltage Range	MCA	MFA	RHz	RLA	W	FLA	W	FLA
FTXM20UVMZ	RXM20UVMZ	50 - 220	Max. 50 Hz, 264 V Min. 50 Hz, 198 V	8.5	10.0	32	1.50	50	0.32	27.0	0.15
		50 - 230					1.40		0.31		0.15
		50 - 240					1.30		0.30		0.14
		60 - 220	Max. 60 Hz, 253 V Min. 60 Hz, 198 V				1.50		0.32		0.15
		60 - 230					1.40		0.31		0.15
FTXM25UVMZ	RXM25UVMZ	50 - 220	Max. 50 Hz, 264 V Min. 50 Hz, 198 V	8.5	10.0	39	1.80	50	0.34	27.0	0.23
		50 - 230					1.70		0.33		0.23
		50 - 240					1.60		0.32		0.22
		60 - 220	Max. 60 Hz, 253 V Min. 60 Hz, 198 V				1.80		0.34		0.23
		60 - 230					1.70		0.33		0.23
FTXM35UVMZ	RXM35UVMZ	50 - 220	Max. 50 Hz, 264 V Min. 50 Hz, 198 V	8.5	10.0	60	3.20	50	0.40	27.0	0.26
		50 - 230					3.00		0.39		0.26
		50 - 240					2.90		0.38		0.25
		60 - 220	Max. 60 Hz, 253 V Min. 60 Hz, 198 V				3.20		0.40		0.26
		60 - 230					3.00		0.39		0.26
FTXM46UVMZ	RXM46UVMZ	50 - 220	Max. 50 Hz, 264 V Min. 50 Hz, 198 V	9.3	15.0	55	4.70	128	0.85	27.0	0.30
		50 - 230					4.40		0.84		0.29
		50 - 240					4.10		0.83		0.28
		60 - 220	Max. 60 Hz, 253 V Min. 60 Hz, 198 V				4.70		0.85		0.30
		60 - 230					4.40		0.84		0.29
FTXM50UVMZ	RXM50UVMZ	50 - 220	Max. 50 Hz, 264 V Min. 50 Hz, 198 V	13.5	15.0	53.5	4.40	128	0.81	58.0	0.35
		50 - 230					4.20		0.80		0.34
		50 - 240					4.00		0.79		0.33
		60 - 220	Max. 60 Hz, 253 V Min. 60 Hz, 198 V				4.40		0.81		0.35
		60 - 230					4.20		0.80		0.34
FTXM60UVMZ	RXM60UVMZ	50 - 220	Max. 50 Hz, 264 V Min. 50 Hz, 198 V	13.5	15.0	66	5.90	128	0.91	58.0	0.43
		50 - 230					5.60		0.89		0.39
		50 - 240					5.40		0.88		0.37
		60 - 220	Max. 60 Hz, 253 V Min. 60 Hz, 198 V				5.90		0.91		0.43
		60 - 230					5.60		0.89		0.39
FTXM71UVMZ	RXM71UVMZ	50 - 220	Max. 50 Hz, 264 V Min. 50 Hz, 198 V	17.3	20.0	49	7.60	128	1.08	58.0	0.51
		50 - 230					7.20		1.06		0.50
		50 - 240					6.90		1.04		0.49
		60 - 220	Max. 60 Hz, 253 V Min. 60 Hz, 198 V				7.60		1.08		0.51
		60 - 230					7.20		1.06		0.50

Symbols:

MCA : Min. circuit amps (A)
 MFA : Max. fuse amps (A)
 RHz : Rated operating frequency (Hz)
 RLA : Rated load amps (A)
 OFM : Outdoor fan motor
 IFM : Indoor fan motor
 W : Fan motor rated output (W)
 FLA : Full load amps (A)

Notes:

1. RLA is based on the following conditions.
Indoor temp. 27°CDB / 19°CWB
Outdoor temp. 35°CDB
2. Maximum allowable voltage variation between phases is 2%.
3. Select wire size based on the larger value of MCA.
4. Instead of fuse, use circuit breaker.
5. Be sure to install an earth leak detector.
(This unit uses an inverter, which means that an earth leak detector capable of handling high harmonics must be used in order to prevent malfunctioning of the earth leak detector.)

3D123932

11. Installation Manual

11.1 Indoor Unit

SAFETY PRECAUTIONS (1)



Read the precautions in this manual carefully before operating the unit.



This appliance is filled with R32.

- The precautions described herein are classified as WARNING and CAUTION. They both contain important information regarding safety. Be sure to observe all precautions without fail.
- Meaning of WARNING and CAUTION notices

	WARNING	Failure to follow these instructions properly may result in personal injury or loss of life.
	CAUTION	Failure to follow these instructions properly may result in property damage or personal injury, which may be serious depending on the circumstances.

- The safety marks shown in this manual have the following meanings:

	Be sure to follow the instructions.		Be sure to establish an earth connection.		Never attempt.
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- After completing installation, conduct a trial operation to check for faults and explain to the user how to operate the air conditioner and take care of it with the aid of the operation manual.



WARNING

- Ask your dealer or qualified personnel to carry out installation work.
Do not attempt to install the air conditioner yourself. Improper installation may result in water leakage, electric shock or fire.
- Install the air conditioner in accordance with the instructions in this installation manual. Improper installation may result in water leakage, electric shock or fire.
- Be sure to use only the specified accessories and parts for installation work. Failure to use the specified parts may result in the unit falling, water leakage, electric shock or fire.
- Install the air conditioner on a foundation strong enough to hold the weight of the unit.
A foundation of insufficient strength may result in the equipment falling and causing injury.
- Electrical work must be performed in accordance with relevant local and national regulations and with the instructions in this installation manual.
Be sure to use a dedicated power supply circuit only. Insufficient power supply and improper workmanship may result in electric shock or fire.
- Use a cable of suitable length. Do not use tapped wires or an extension lead, as this may cause overheating, electric shock or fire.
- Make sure that all wiring is secured, the specified wires are used, and that there is no strain on the terminal connections or wires. Improper connections or securing of wires may result in abnormal heat build-up or fire.
- When wiring the power supply and connecting the wiring between the indoor and outdoor units, position the wires so that the electrical wiring box cover can be securely fastened. Improper positioning of the electrical wiring box cover may result in electric shock, fire or overheating terminals.
- If refrigerant gas leaks during installation, ventilate the area immediately. Toxic gas may be produced if the refrigerant comes into contact with fire.
- After completing installation, check for refrigerant gas leakage. Toxic gas may be produced if the refrigerant gas leaks into the room and comes into contact with a source of fire, such as a fan heater, stove or cooker.
- When installing or relocating the air conditioner, do not let any other substances besides R32, such as air, enter the refrigerant circuit.
The presence of air or foreign matter in the refrigerant circuit causes an abnormal pressure rise, which may result in equipment damage and even injury.
- During installation, attach the refrigerant piping securely before operating the compressor. If the refrigerant pipes are not attached and the stop valve is open when the compressor is operated, air will be sucked in, causing abnormal pressure in the refrigeration cycle, which may result in equipment damage and even injury.
- During pump down, stop the compressor before removing the refrigerant piping. If the compressor is still operating and the stop valve is open during pump down, air will be sucked in when the refrigerant piping is removed, causing abnormal pressure in the refrigeration cycle, which may result in equipment damage and even injury.
- Be sure to earth the air conditioner.
Do not earth the unit to a utility pipe, lightning conductor or telephone earth lead. Imperfect earthing may result in electric shock.
- Be sure to install an earth leakage circuit breaker. Failure to install an earth leakage circuit breaker may result in electric shock or fire.
- Do not use means to accelerate the defrosting process (if possible) or to clean, other than those recommended by the manufacturer.
- The appliance must be stored in a room without continuously operating ignition sources (for example: open flames, an operating gas appliance or an operating electric heater).
- Do not pierce or burn.
- Be aware that refrigerants may not contain an odour.
- The appliance must be installed, operated and stored in a room with floor area larger than –*m² for 20/25/35 class, 2.5m² for 46 class, 3.5m² for 50/60 class, 3.9m² for 71 class.
* Since the maximum refrigerant charge of the models is below 1.22kg, it does not have any limitation for the minimum floor area.
Maintain an installation height of 1.8m or more from the floor surface to the bottom of the appliance. If the size of the room where the appliance is to be installed has less than the indicated minimum floor area, install it in a well ventilated room.
For details refer to the WARNING list in the "Safety Precautions" section of the installation manual for the outdoor unit.
- Comply with national gas regulations.
- Do not pump down when the refrigerant has leaked, otherwise the compressor may be damaged.

SAFETY PRECAUTIONS (2)

⚠ CAUTION

- Do not install the air conditioner at any place where there is a danger of flammable gas leakage. In the event of a gas leakage, build-up of gas near the air conditioner may cause a fire to break out.
- While following the instructions in this installation manual, install drain piping to ensure proper drainage and insulate the piping to prevent condensation. Improper drain piping may result in indoor water leakage and property damage.
- Tighten the flare nut as specified, such as with a torque wrench. If the flare nut is too tight, it may crack after prolonged use, causing refrigerant leakage.
- Only qualified personnel can handle, fill, purge and dispose of the refrigerant.
- Make sure to provide for adequate measure in order to prevent that the outdoor unit be used as a shelter by small animals. Small animals making contact with electrical parts can cause malfunctions, smoke or fire. Please instruct the customer to keep the area around the unit clean.
- The temperature of refrigerant circuit will be high, please keep the inter-unit wiring away from copper pipes that are not thermally insulated.

ACCESSORIES

Ⓐ Mounting plate	1	Ⓑ Mounting plate fixing screw M4 × 25L	7	Ⓒ Wireless remote controller	1
Ⓓ Remote controller holder	1	Ⓔ Remote controller holder fixing screw M3 × 20L	2	Ⓕ Dry battery AAA. LR03 (alkaline)	2
Ⓔ Indoor unit fixing screw M4 × 12L	2	Ⓕ Insulation tape	1	Ⓖ Titanium apatite deodorizing filter	2
Ⓖ Operation manual	1	Ⓖ Installation manual	1		

CHOOSING AN INSTALLATION SITE

- Before choosing the installation site, obtain user approval.

1. Indoor unit

- The indoor unit should be positioned in a place where:
 - the restrictions on the installation requirements specified in "INDOOR UNIT INSTALLATION DIAGRAM" are met,
 - both the air inlet and air outlet are unobstructed,
 - the unit is not exposed to direct sunlight,
 - the unit is away from sources of heat or steam,
 - there is no source of machine oil vapour (this may shorten the indoor unit service life),
 - cool or warm air is circulated throughout the room,
 - the unit is away from electronic ignition type fluorescent lamps (inverter or rapid start type) as they may affect the remote controller range,

- the unit is at least 1m away from any television or radio set (the unit may cause interference with the picture or sound),
- install at the recommended height (not less than 1.8 m),
- no laundry equipment is nearby.

2. Wireless remote controller

- Turn on all the fluorescent lamps in the room, if any, and find a location where the remote controller signals are properly received by the indoor unit (within 7 m).

INSTALLATION TIPS (1)

1. Removing and installing the front panel

• Removal method

- Place your fingers in the indentations on the main unit (one each on the left and right sides), and open the front panel until it stops.
- While pushing the left side front panel shaft outward, push up the front panel and remove it. (Remove the right side front panel shaft in the same manner.)
- After removing both front panel shafts, pull the front panel toward yourself and remove it.



• Installation method

Align the tabs of the front panel with the grooves, and push all the way in, then close slowly. Push the centre of the lower panel surface firmly to engage the tabs.

2. Removing and installing the front grille

• Removal method

- Remove the front panel and air filters.
- Remove the 2 screws from the front grille. (The 50/60/71 class model have 3 screws.)



- In front of the ○○○ mark on the front grille, there are 3 upper hooks. Lightly pull the front grille toward you with one hand, and push down on the hooks with the fingers of your other hand.

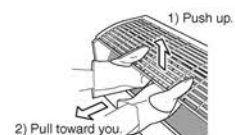


When there is insufficient work space because the unit is close to ceiling

⚠ CAUTION

- Be sure to wear protection gloves.

Place both hands under the centre of the front grille, and while pushing up, pull it toward you.



• Installation method

- Install the front grille and firmly engage the upper hooks (3 locations).
- Install 2 screws of the front grille. (The 50/60/71 class model have 3 screws.)
- Install the air filters and then mount the front panel.

INSTALLATION TIPS (2)

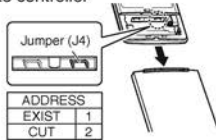
3. How to set the different addresses

When 2 indoor units are installed in one room, the 2 wireless remote controllers can be set for different addresses. Change the address setting of one of the two units.

When cutting the jumper (J4) be careful not to damage any of the surrounding parts.

- 1) Remove the battery cover on the remote controller and cut the address jumper (J4).

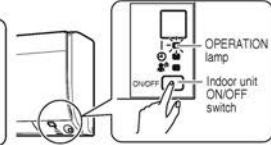
- 2) Press and at the same time.



- 3) Press , select **R**, press .
- (The indoor unit OPERATION lamp will blink for about 1 minute.)

- 4) Press the indoor unit ON/OFF switch while the OPERATION lamp is blinking.

- If setting could not be carried out completely while the OPERATION lamp was blinking, carry out the setting process once again from the beginning.
- After setting is complete, pressing for about 5 seconds will cause the remote controller to return to the previous display.



INDOOR UNIT INSTALLATION DIAGRAM

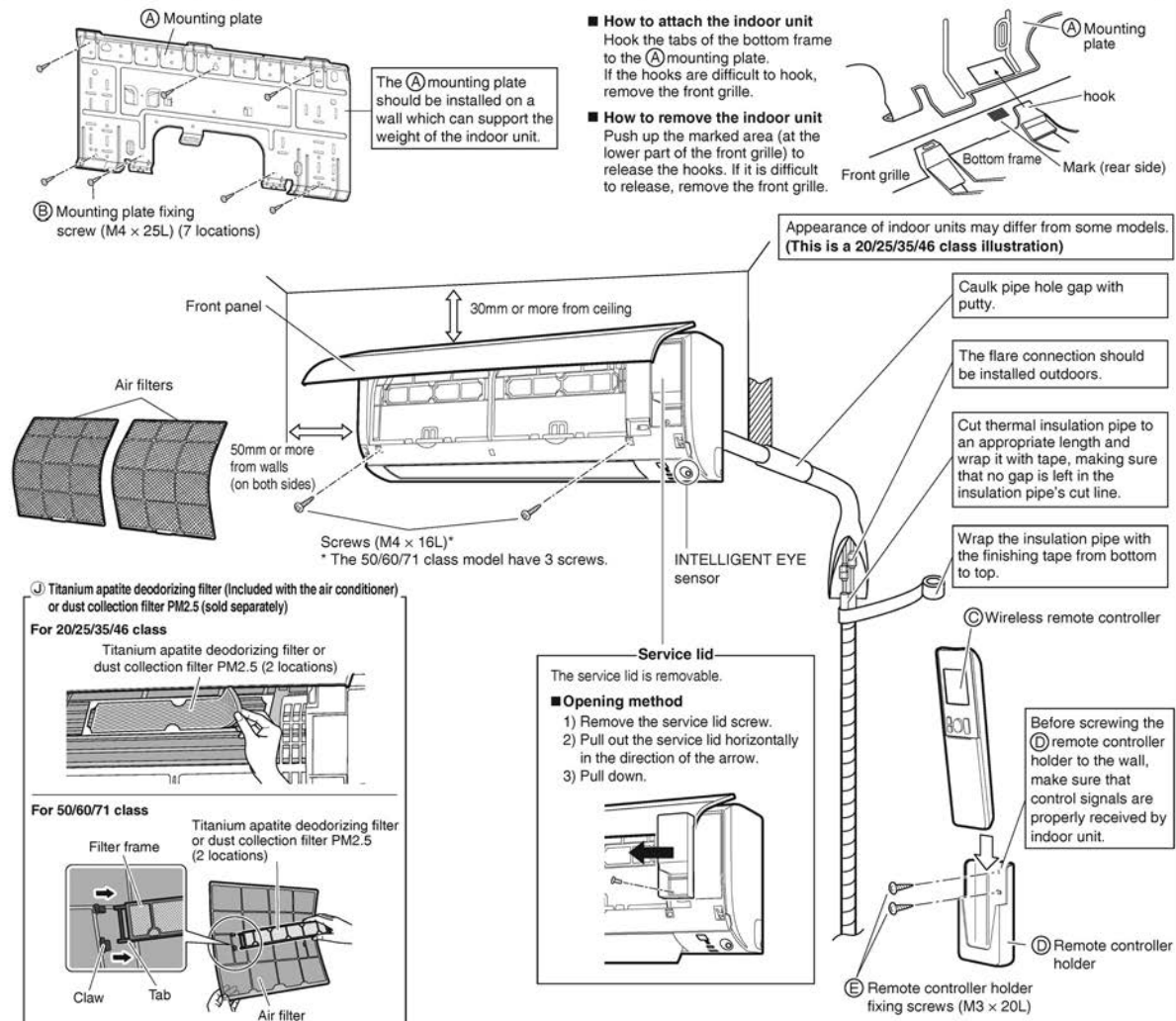
NOTE

When flaring the pipe end and installing the unit, refer to **INDOOR UNIT INSTALLATION (3) "1 FLARING THE PIPE END"**.

INTELLIGENT EYE SENSOR

CAUTION

- Do not hit or violently push the INTELLIGENT EYE sensor. This can lead to damage and malfunction.
- Do not place large objects near the INTELLIGENT EYE sensor. Also keep heating units or humidifiers outside the sensor's detection area.

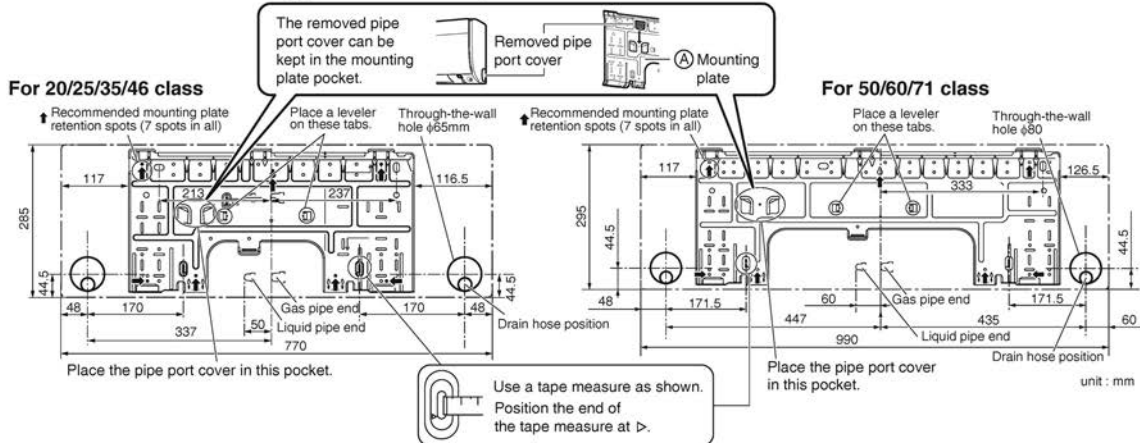


INDOOR UNIT INSTALLATION (1)

1 INSTALLING THE MOUNTING PLATE

- The mounting plate should be installed on a wall which can support the weight of the indoor unit.
 - Temporarily secure the mounting plate to the wall, make sure that the plate is completely level, and mark the drilling points on the wall.
 - Secure the mounting plate to the wall with screws.

Recommended mounting plate retention spots and dimensions

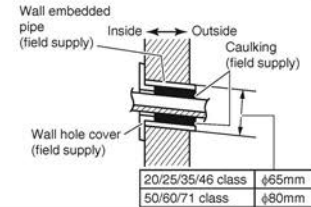


2 DRILLING A WALL HOLE AND INSTALLING WALL EMBEDDED PIPE

WARNING

For metal frame or metal board walls, be sure to use a wall embedded pipe and wall hole cover in the feed-through hole to prevent possible heat, electrical shock, or fire.

- Drill a feed-through hole with a 65mm for 20/25/35/46 class or 80mm for 50/60/71 class diameter through the wall at a downward angle toward the outside.
- Insert a wall embedded pipe into the hole.
- Insert a wall hole cover into wall pipe.
- After completing refrigerant piping, wiring, and drain piping, caulk the pipe hole gap with putty.



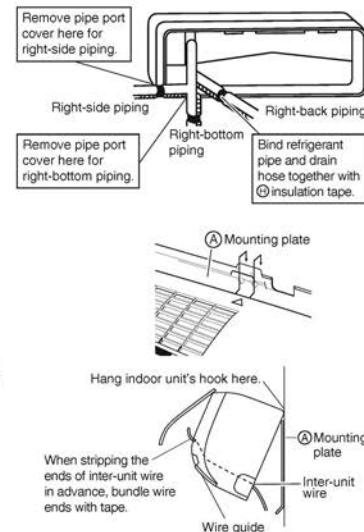
3 INSTALLING THE INDOOR UNIT

In the case of bending or curing refrigerant pipes, keep the following precautions in mind. Abnormal sound may be generated if improper work is conducted.

- Do not strongly press the refrigerant pipes onto the bottom frame.
- Do not strongly press the refrigerant pipes on the front grille, either.

3-1. Right-side, right-back, or right-bottom piping

- Attach the drain hose to the underside of the refrigerant pipes with adhesive vinyl tape.
- Wrap the refrigerant pipes and drain hose together with (H) insulation tape.
- Pass the drain hose and refrigerant pipes through the wall hole, then position the indoor unit on the (A) mounting plate hooks, using the Δ markings at the top of the indoor unit as a guide.
- Open the front panel, then open the service lid. (Refer to "INSTALLATION TIPS (1)".)
- Pass the inter-unit wire from the outdoor unit through the feed-through wall hole and then through the back of the indoor unit. Pull them through the front side. Bend the ends of cable tie wires upward for easier work in advance. (If the inter-unit wire ends are to be stripped first, bundle wire ends with adhesive tape.)
- Press the bottom frame of the indoor unit with both hands until it is firmly caught by the (A) mounting plate hooks. Make sure that the wires do not catch on the edge of the indoor unit.

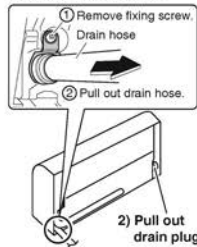


INDOOR UNIT INSTALLATION (2)

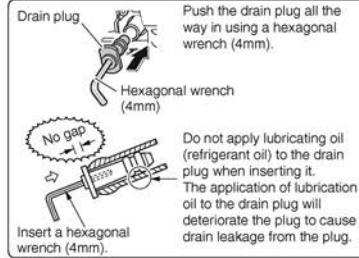
3-2. Left-side, left-back, or left-bottom piping

How to switch around the drain plug and drain hose

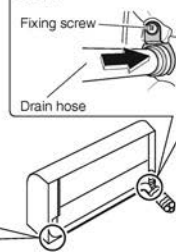
- 1) Remove the fixing screw and pull out the drain hose.



- 3) Switch around the drain hose and drain plug.



- Insert drain hose securely and fix in place with fixing screw.



- 1) Switch around the drain plug and drain hose.
- 2) Attach the drain hose to the underside of the refrigerant pipes with adhesive vinyl tape.

- 3) Shape the refrigerant pipes along the pipe path marking on the (A) mounting plate.

- 4) Pass the drain hose and refrigerant pipes through the wall hole, then position the indoor unit on the (A) mounting plate hooks, using the Δ markings at the top of the indoor unit as a guide.

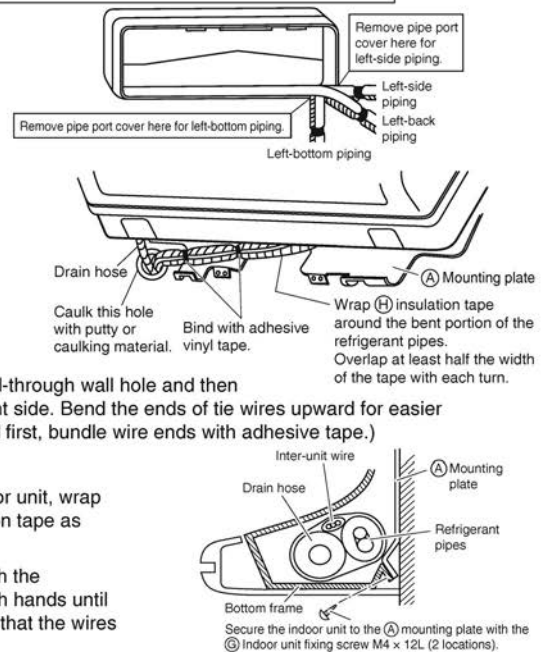
- 5) Open the front panel, then open the service lid. (Refer to "INSTALLATION TIPS (1)")

- 6) Pass the inter-unit wire from the outdoor unit through the feed-through wall hole and then through the back of the indoor unit. Pull them through the front side. Bend the ends of tie wires upward for easier work in advance. (If the inter-unit wire ends are to be stripped first, bundle wire ends with adhesive tape.)

- 7) Connect the refrigerant pipes.

- 8) In case of pulling the drain hose through the back of the indoor unit, wrap the refrigerant pipes and drain hose together with (H) insulation tape as shown in the figure.

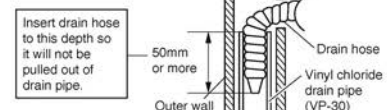
- 9) While exercising care so that the inter-unit wire does not catch the indoor unit, press the bottom frame of the indoor unit with both hands until it is firmly caught by the (A) mounting plate hooks. Make sure that the wires do not catch on the edge of the indoor unit.



3-3. Wall embedded piping

Follow the instructions given under left-side, left-back, or left-bottom piping.

- 1) Insert the drain hose to a depth of 50mm or more so it will not be pulled out of the drain pipe.



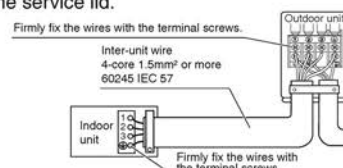
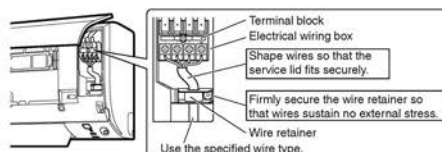
Refer to the installation manual for the outdoor unit also.

4 WIRING

⚠ WARNING

- Do not use tapped wires, extension cords, or starburst connections, as they may cause overheating, electrical shock, or fire.
- Do not use locally purchased electrical parts inside the product. (Do not branch the power for the drain pump, etc., from the terminal block.) Doing so may cause electric shock or fire.
- Do not connect the power wire to the indoor unit. Doing so may cause electric shock or fire.

- 1) Strip wire ends (20mm).
- 2) Match wire colours with terminal numbers on the indoor and outdoor unit's terminal blocks and firmly secure the wires in the corresponding terminals with the screws.
- 3) Connect the earth wire to the corresponding terminals.
- 4) Pull the wires lightly to make sure they are securely connected, then secure them with the wire retainer.
- 5) Shape the wires so that the service lid fits securely, then close the service lid.



INDOOR UNIT INSTALLATION (2)

5 WHEN CONNECTING AN HA SYSTEM

For this procedure, separately sold parts are needed.

- In case there is a work space in the right side of indoor unit, the procedure can be done while fixing electrical wiring box. Skip to remove electrical wiring box if it is possible, in order to make effective work.

(For details, refer fixing manual which attached in HA board)

1) Remove front grille. (For 20/25/35/46 class: 2 screws, for 50/60/71 class: 3 screws)

2) Remove electrical wiring cover. **Figure. 1**

*3) Remove electrical wiring box.

- Remove inter-unit wire.
- Remove fan motor connector (S200), swing motor connector (S6).
- Remove thermistor, earth wire from heat exchanger unit. (1 screw)
- Remove electrical wiring box fixing screw. (1 screw)

4) Fixing HA board (option part). **Figure. 2**

- Fixing HA board (option part) at electrical wiring box.
- Insert HA board (option part) connector to electrical wiring box connector (S403).

5) Fixing HA connection code. **Figure. 1, 2**

- Insert HA connection code to HA connector S21 (white) of HA board (option part). (HA connector S16 that corresponded to JEM-A standard is blue color.)

- Wiring HA connection code as **Figure. 3**

*6) Fix electrical wiring box.

- Fixing electrical wiring box fixing screw. (1 screw)
- Fixing thermistor, earth wire to heat exchanger unit.
- Fixing fan motor connector (S200), swing motor connector (S6).
- Insert connection wiring.

7) Fixing electrical wiring cover.

8) Fixing front grille.

Figure.1 Confirm electrical wiring box position

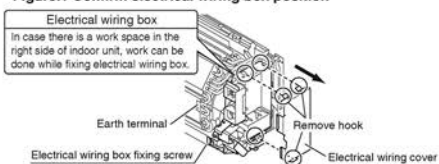
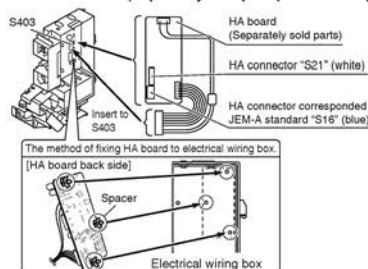
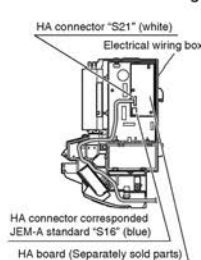


Figure. 2 About HA board (Separately sold parts) connection point



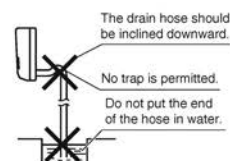
* The picture may differ from some models.

Figure. 3 HA connection code wiring method.

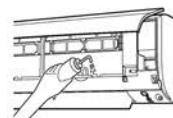


6 DRAIN PIPING

1) Connect the drain hose, as described on the right.



2) Remove the air filters and pour some water into the drain pan to check the water flows smoothly.



3) If drain hose extension or embedded drain piping is required, use appropriate parts that match the hose front end.

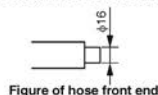
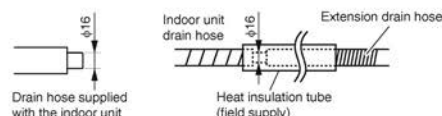


Figure of hose front end

4) When extending the drain hose, use a commercially available extension hose with an inner diameter of 16mm. Be sure to thermally insulate the indoor section of the extension hose.



INDOOR UNIT INSTALLATION (3)

REFRIGERANT PIPING WORK

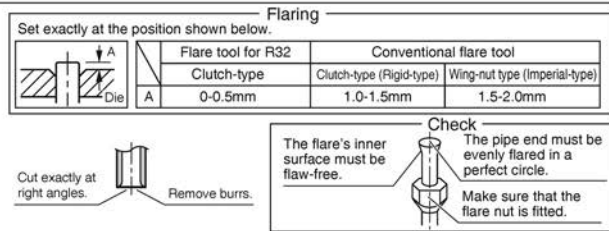
Refer to the installation manual for the outdoor unit also.

1 FLARING THE PIPE END

⚠ WARNING

- Do not apply mineral oil to the flare.
- Prevent mineral oil from getting into the system as this would reduce the service life of the units.
- Never use piping which has been used for previous installations. Only use parts which are delivered with this unit.
- Never install a dryer to this R32 unit in order to guarantee its service life.
- The drying material may dissolve and damage the system.
- Incomplete flaring may result in refrigerant gas leakage.
- Do not reuse joints which have been used once already.

- Cut the pipe end with a pipe cutter.
- Remove burrs with the cut surface facing downward, so that the filings do not enter the pipe.
- Put the flare nut on the pipe.
- Flare the pipe.
- Check that the flaring has been done correctly.



2 REFRIGERANT PIPING

⚠ CAUTION

- Use the flare nut fixed to the main unit. (This is to prevent the flare nut from cracking as a result of deterioration over time.)
- To prevent gas leakage, apply refrigeration oil only to the inner surface of the flare. (Use refrigeration oil for R32.)
- Use a torque wrench when tightening the flare nuts to prevent damage to the flare nuts and gas leakage.
- Align the centres of both flares and tighten the flare nuts 3 or 4 turns by hand, then tighten them fully with a spanner and a torque wrench.

Apply oil

Do not apply refrigeration oil to the outer surface.

Apply refrigeration oil only to the inner surface of the flare.

Do not apply refrigeration oil to the flare nut to avoid tightening with excessive torque.

Tighten

Torque wrench

Spanner

Piping union

Flare nut

	Piping size	Flare nut tightening torque	Valve cap size	
			dimension AA, refer to Fig.1	Valve cap tightening torque
Gas side	O.D. 9.5mm (3/8 inch)	32.7 - 39.9N • m (333 - 407kgf • cm)	17mm	15.7 ± 1.6N • m (160 ± 16kgf • cm)
	O.D. 12.7mm (1/2 inch)	49.5 - 60.3N • m (505 - 615kgf • cm)	19mm	19.0 ± 1.9N • m (193 ± 20kgf • cm)
	O.D. 15.9mm (5/8 inch)	61.8 - 75.4N • m (630 - 770kgf • cm)	22mm	24.5 ± 3.9N • m (250 ± 40kgf • cm)
Liquid side	O.D. 6.4mm (1/4 inch)	14.2 - 17.2N • m (144 - 175kgf • cm)	27mm	53.9 ± 5.9N • m (550 ± 60kgf • cm)
	Service port cap tightening torque	10.7 - 14.7N • m (110 - 150kgf • cm)	32mm	68.6 ± 6.9N • m (700 ± 70kgf • cm)

Fig.1

Be sure to place a cap.

If no flare cap is available, cover the flare mouth with tape to keep dirt and water out.

Rain

Gas pipe

Inter-unit wire

Liquid pipe

Gas pipe insulation

Liquid pipe insulation

Finishing tape

Drain hose

2-1. Caution on piping handling

- Protect the open end of the pipe against dust and moisture.
- All pipe bends should be as gentle as possible. Use a pipe bender for bending.

2-2. Selection of copper and heat insulation materials

When using commercial copper pipes and fittings, observe the following:

- Insulation material: Polyethylene foam
Heat transfer rate: 0.041 to 0.052W/mK (0.035 to 0.045kcal/mh°C)
Refrigerant gas pipe's surface temperature reaches 110°C max.
Choose heat insulation materials that will withstand this temperature.
- Be sure to insulate both the gas and liquid piping and observe the insulation dimensions as below.

	Piping size	Minimum bend radius	Piping thickness	Thermal insulation size	Thermal insulation thickness
Gas side	O.D. 9.5mm	30mm or more	0.8mm (C1220T-O)	I.D. 12-15mm	10mm Min.
	O.D. 12.7mm	40mm or more		I.D. 14-16mm	
	O.D. 15.9mm	50mm or more	1.0mm (C1220T-O)	I.D. 16-20mm	
Liquid side	O.D. 6.4mm	30mm or more	0.8mm (C1220T-O)	I.D. 8-10mm	

- Use separate thermal insulation pipes for gas and liquid refrigerant pipes.

INDOOR UNIT INSTALLATION (3)

TRIAL OPERATION AND TESTING

- Before starting the trial operation, measure the voltage at the primary side of the circuit breaker.
- Check that all liquid and gas stop valves are fully open.
- Check that piping and wiring all match.

■ Trial operation should be carried out in COOL or HEAT operation.

1-1 Measure the supply voltage and make sure that it is within the specified range.

1-2 In COOL operation, select the lowest programmable temperature; in HEAT operation, select the highest programmable temperature.

1-3 Carry out the trial operation following the instructions in the operation manual to ensure that all functions and parts, such as the movement of the flap, are working properly.

- To protect the air conditioner, restart operation is disabled for 3 minutes after the system has been turned off.
- When trial operation is conducted in HEAT operation directly after the circuit breaker is turned on, in some cases no air will be output for about 10 minutes in order to protect the air conditioner.

1-4 After trial operation is complete, set the temperature to a normal level (26°C to 28°C in COOL operation, 20°C to 24°C in HEAT operation).

- When operating the air conditioner in COOL operation in winter, or HEAT operation in summer, set it to the trial operation mode using the following method.

1) Press  to turn on the system.

2) Press  and  at the same time.

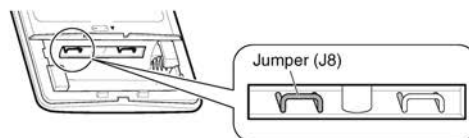
3) Press , select "T", and press  for confirmation.

- Trial operation will stop automatically after about 30 minutes. To stop the operation, press .
- Some of the functions cannot be used in the trial operation mode.
- The air conditioner draws a small amount of power in its standby mode. If the system is not to be used for some time after installation, shut off the circuit breaker to eliminate unnecessary power consumption.
- If the circuit breaker trips to shut off the power to the air conditioner, the system will restore the original operation mode when the circuit breaker is turned on again.

Test items

Test items	Symptom	Check
Indoor and outdoor units are installed securely.	Fall, vibration, noise	
No refrigerant gas leaks.	Incomplete cooling/heating function	
Refrigerant gas and liquid pipes and indoor drain hose extension are thermally insulated.	Water leakage	
Draining line is properly installed.	Water leakage	
System is properly earthed.	Electrical leakage	
Only specified wires are used for all wiring, and all wires are connected correctly.	No operation or burn damage	
Indoor or outdoor unit's air inlet or air outlet are unobstructed.	Incomplete cooling/heating function	
Stop valves are opened.	Incomplete cooling/heating function	
Indoor unit properly receives remote control commands.	No operation	
<For heat pump model>  will be displayed when the MODE button is pressed.*	No heating	

* Check that the jumper (J8) has not been cut. If it has been cut, contact the service shop.



ATTENTION

- Even when the air conditioner is not operating, it consumes some electric power. If the customer is not going to use the unit soon after it is installed, turn off the circuit breaker to avoid wasting electricity.

3P412209-16K

11.2 Outdoor Unit

11.2.1 20/25/35 Class

SAFETY PRECAUTIONS (1)



Read the precautions in this manual carefully before operating the unit.



This appliance is filled with R32.

- The precautions described herein are classified as WARNING and CAUTION. They both contain important information regarding safety. Be sure to observe all precautions without fail.
- Meaning of WARNING and CAUTION notices

	WARNING	Failure to follow these instructions properly may result in personal injury or loss of life.
	CAUTION	Failure to follow these instructions properly may result in property damage or personal injury, which may be serious depending on the circumstances.

- The safety marks shown in this manual have the following meanings:

	Be sure to follow the instructions.		Be sure to establish an earth connection.		Never attempt.
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- After completing installation, conduct a trial operation to check for faults and explain to the user how to operate the air conditioner and take care of it with the aid of the operation manual.



WARNING

- Ask your dealer or qualified personnel to carry out installation work.
Do not attempt to install the air conditioner yourself. Improper installation may result in water leakage, electric shock or fire.
- Install the air conditioner in accordance with the instructions in this installation manual.
Improper installation may result in water leakage, electric shock or fire.
- Be sure to use only the specified accessories and parts for installation work.
Failure to use the specified parts may result in the unit falling, water leakage, electric shock or fire.
- Install the air conditioner on a foundation strong enough to hold the weight of the unit. A foundation of insufficient strength may result in the equipment falling and causing injury.
- Electrical work must be performed in accordance with relevant local and national regulations and with the instructions in this installation manual.
Be sure to use a dedicated power supply circuit only. Insufficient power supply and improper workmanship may result in electric shock or fire.
- Use a cable of suitable length. Do not use tapped wires or an extension lead, as this may cause overheating, electric shock or fire.
- Make sure that all wiring is secured, the specified wires are used, and that there is no strain on the terminal connections or wires.
Improper connections or securing of wires may result in abnormal heat build-up or fire.
- When wiring the power supply and connecting the wiring between the indoor and outdoor units, position the wires so that the electrical wiring box cover can be securely fastened.
Improper positioning of the electrical wiring box cover may result in electric shock, fire or overheating terminals.
- If refrigerant gas leaks during installation, ventilate the area immediately.
Toxic gas may be produced if the refrigerant comes into contact with fire.
- After completing installation, check for refrigerant gas leakage.
Toxic gas may be produced if the refrigerant gas leaks into the room and comes into contact with a source of fire, such as a fan heater, stove or cooker.
- When installing or relocating the air conditioner, do not let any other substances besides R32, such as air, enter the refrigerant circuit.
The presence of air or foreign matter in the refrigerant circuit causes an abnormal pressure rise, which may result in equipment damage and even injury.
- During installation, attach the refrigerant piping securely before operating the compressor. If the refrigerant pipes are not attached and the stop valve is open when the compressor is operated, air will be sucked in, causing abnormal pressure in the refrigeration cycle, which may result in equipment damage and even injury.
- During pump down, stop the compressor before removing the refrigerant piping. If the compressor is still operating and the stop valve is open during pump down, air will be sucked in when the refrigerant piping is removed, causing abnormal pressure in the refrigeration cycle, which may result in equipment damage and even injury.
- Be sure to earth the air conditioner.
Do not earth the unit to a utility pipe, lightning conductor or telephone earth lead. Imperfect earthing may result in electric shock.
- Be sure to install an earth leakage circuit breaker. Failure to install an earth leakage circuit breaker may result in electric shock or fire.
- Do not use means to accelerate the defrosting process (if possible) or to clean, other than those recommended by the manufacturer.
- The appliance must be stored in a room without continuously operating ignition sources (for example: open flames, an operating gas appliance or an operating electric heater).
- Do not pierce or burn.
- Be aware that refrigerants may not contain an odour.
- The appliance must be installed, operated and stored in a room with a floor area larger than $\sim 4\text{m}^2$ for 20/25/35 class.
* Since the maximum refrigerant charge of the models is below 1.22kg, it does not have any limitation for the minimum floor area.
- Comply with national gas regulations.
- Do not pump down when the refrigerant has leaked, otherwise the compressor may be damaged.
- Do not install the air conditioner at any place where there is a danger of flammable gas leakage.
In the event of a gas leakage, build-up of gas near the air conditioner may cause a fire to break out.
- When flared joints are reused indoors, the flare part shall be re-fabricated.

SAFETY PRECAUTIONS (2)



CAUTION

- While following the instructions in this installation manual, install drain piping to ensure proper drainage and insulate the piping to prevent condensation.
Improper drain piping may result in indoor water leakage and property damage.
- Tighten the flare nut as specified, such as with a torque wrench.
If the flare nut is too tight, it may crack after prolonged use, causing refrigerant leakage.
- Take adequate steps to prevent the outdoor unit being used as a shelter by small animals.
If small animals or birds come into contact with electrical parts, this can cause malfunctions, smoke or fire. Please instruct the customer to always keep the area around the unit clean.
- The refrigerant circuit temperature will be high, therefore the inter-unit wires must be kept away from copper pipes that are not thermally insulated.
- Only qualified personnel can handle, fill, purge and dispose of the refrigerant.

ACCESSORIES

Ⓐ Drain socket



1

There is on the bottom packing case.

Ⓑ Installation manual

1

PRECAUTIONS FOR SELECTING THE LOCATION

NOTE

Cannot be installed suspended from a ceiling or stacked.



CAUTION

When operating the air conditioner in a low outdoor ambient temperature, be sure to follow the instructions described below.

- To prevent exposure to wind, install the outdoor unit with its suction side facing the wall.
- Never install the outdoor unit at a site where the suction side may be exposed directly to wind.
- To prevent exposure to wind, it is recommended to install a baffle plate on the air discharge side of the outdoor unit.
- In heavy snowfall areas, select an installation site where the snow will not affect the unit.



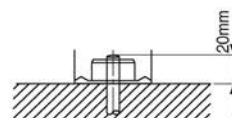
- Construct a large canopy.
- Construct a pedestal.

Install the unit high enough off the ground to prevent burying in snow.

- Choose a place solid enough to bear the weight and vibration of the unit, where the operating sound will not be amplified.
- Choose a location where the hot air discharged from the unit or the operating sound will not cause a nuisance to the neighbours of the user.
- Avoid locations, such as near bedrooms, where the operating sound may cause disturbance.
- There must be sufficient space to carry the unit into and out of the site.
- There must be sufficient space for air passage and no obstructions around the air inlet and the air outlet.
- The site must not be prone to flammable gas leaks in the surrounding area.
- Install units, power cords and inter-unit wire at least 3m away from television and radio sets. (This is to prevent interference to images and sounds. Noise may be produced even if they are more than 3m away depending on radio wave conditions.)
- In coastal areas or other places with a salty atmosphere or one containing sulphate gas, corrosion may shorten the life of the air conditioner.
- Since water will flow from the drain of the outdoor unit, do not place under the unit anything which must be kept away from moisture.

PRECAUTIONS ON INSTALLATION

- Check the strength and level of the installation surface so that the unit does not cause any operating vibrations or noise after installation.
- Fix the unit in place securely using foundation bolts, as in the figure. (Prepare 4 sets of M8 or M10 foundation bolts, nuts and washers; all sold separately.)
- It is best to screw in the foundation bolts until their ends are 20mm from the foundation surface.



IMPORTANT INFORMATION REGARDING THE REFRIGERANT USED

This product contains fluorinated greenhouse gases.

Refrigerant type: **R32**

GWP⁽¹⁾ value: **675**

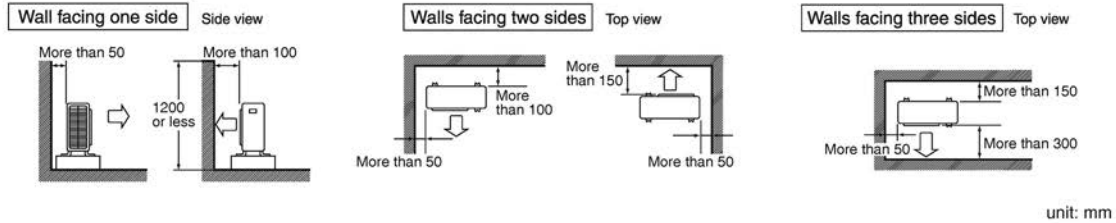
⁽¹⁾GWP = global warming potential

The refrigerant quantity is indicated on the unit name plate.

OUTDOOR UNIT INSTALLATION DIAGRAM

INSTALLATION SPACE REQUIREMENTS

- Where a wall or other obstacle is in the path of the outdoor unit's intake or exhaust airflow, follow the installation space requirements below.
- For any of the below installation patterns, the wall height on the outlet side should be 1200mm or less.



NOTE

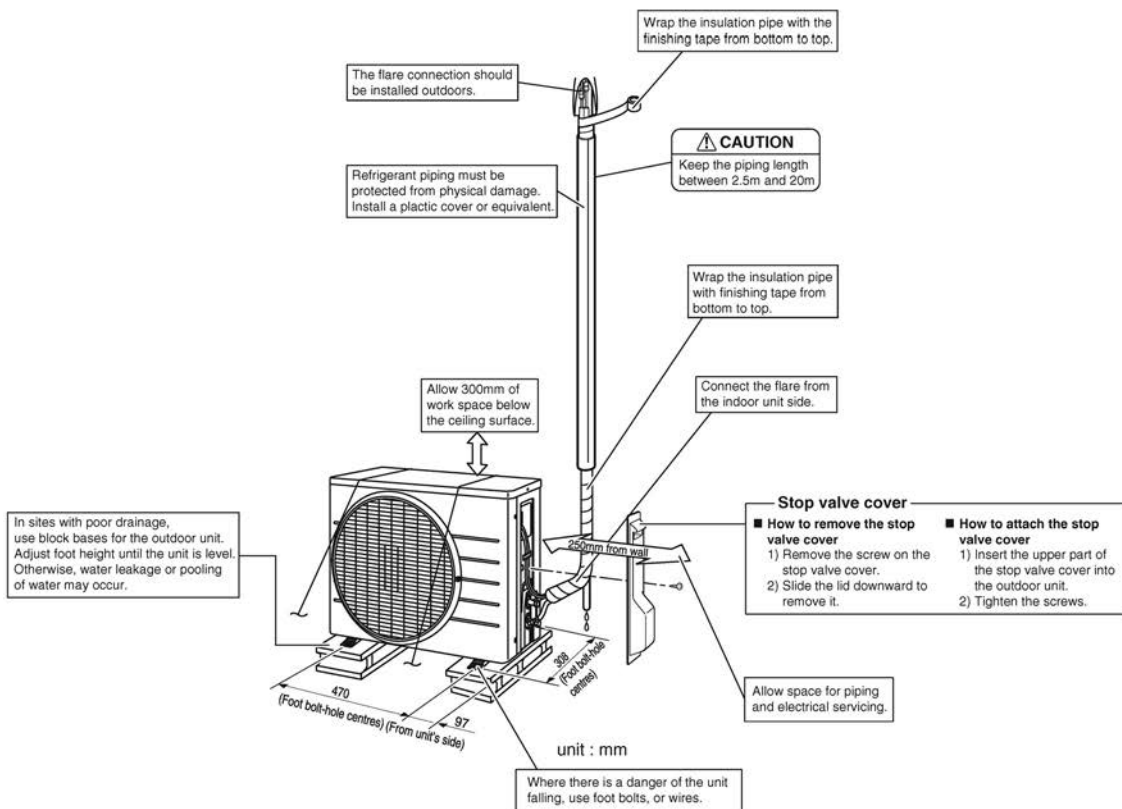
When flaring the pipe end and installing the unit, refer to "❸ FLARING THE PIPE END".

Class	20	25	35
Max. allowable piping length	20m		
Min. allowable piping length	2.5m		
Max. allowable piping height	15m		
Additional refrigerant required for refrigerant pipe exceeding 10m in length.	20g/m		
Gas pipe	O.D. 9.5mm		
Liquid pipe	O.D. 6.4mm		
Max. allowable refrigerant charge amount	0.93 kg	1.05 kg	

Refrigerant piping must be kept to a minimum.

- * It is suggested that the minimum allowable length be used for the pipes connecting the indoor units to the outdoor unit in order to prevent noise and vibration from the outdoor unit.
(Mechanical noise and vibration may occur depending on how the unit is installed and the environment in which it is used.)

- ** Be sure to add the proper amount of additional refrigerant.
Failure to do so may result in reduced performance.



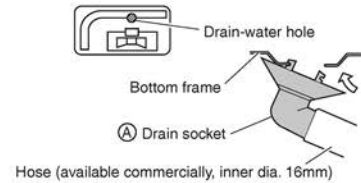
OUTDOOR UNIT INSTALLATION (1)

1 INSTALLING THE OUTDOOR UNIT

- When installing the outdoor unit, refer to "PRECAUTIONS FOR SELECTING THE LOCATION" and the "OUTDOOR UNIT INSTALLATION DIAGRAM".

2 DRAIN WORK

- Use the (A) drain socket for drainage.
- When attaching the drain socket to the bottom frame make sure to first connect the drain hose to the drain socket.
- If the drain port is covered by a mounting base or floor surface, place additional foot bases of at least 30mm in height under the outdoor unit's feet.
- In cold areas, do not use a drain hose with the outdoor unit.
(Otherwise, drain water may freeze, impairing heating performance.)

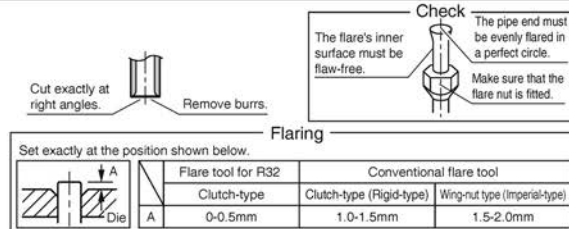


3 FLARING THE PIPE END

WARNING

- Do not apply mineral oil to the flare.
- Prevent mineral oil from getting into the system as this would reduce the service life of the units.
- Never use piping which has been used for previous installations. Only use parts which are delivered with this unit.
- Never install a dryer to this R32 unit in order to guarantee its service life.
- The drying material may dissolve and damage the system.
- Incomplete flaring may result in refrigerant gas leakage.
- When flared joints are reused indoors, the flare part shall be re-fabricated.

- Cut the pipe end with a pipe cutter.
- Remove burrs with the cut surface facing downward, so that the filings do not enter the pipe.
- Put the flare nut on the pipe.
- Flare the pipe.
- Check that the flaring has been done correctly.

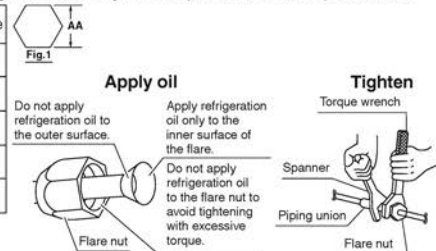


4 REFRIGERANT PIPING

CAUTION

- Use the flare nut fixed to the main unit. (This is to prevent the flare nut from cracking as a result of deterioration over time.)
- To prevent gas leakage, apply refrigeration oil only to the inner surface of the flare. (Use refrigeration oil for R32.)
- Use a torque wrench when tightening the flare nuts to prevent damage to the flare nuts and gas leakage.
- Align the centres of both flares and tighten the flare nuts 3 or 4 turns by hand, then tighten them fully with a spanner and a torque wrench.

	Piping size	Flare nut tightening torque	Valve cap size	Valve cap tightening torque
			dimension AA, refer to Fig.1	
Gas side	O.D. 9.5mm (3/8 inch)	32.7-39.9N • m (333-407kgf • cm)	17mm	15.7 ± 1.6N • m (160 ± 16kgf • cm)
	O.D. 12.7mm (1/2 inch)	49.5-60.3N • m (505-615kgf • cm)	19mm	19.0 ± 1.9N • m (193 ± 20kgf • cm)
	O.D. 15.9mm (5/8 inch)	61.8-75.4N • m (630-770kgf • cm)	22mm	24.5 ± 3.9N • m (250 ± 40kgf • cm)
Liquid side	O.D. 6.4mm (1/4 inch)	14.2-17.2N • m (144-175kgf • cm)	27mm	53.9 ± 5.9N • m (550 ± 60kgf • cm)
			32mm	68.6 ± 6.9N • m (700 ± 70kgf • cm)
Service port cap tightening torque		10.7-14.7N • m (110-150kgf • cm)		



Cautions on pipe handling

- Protect the open end of the pipe from dust and moisture.
- All pipe bends should be as gentle as possible. Use a pipe bender for bending.

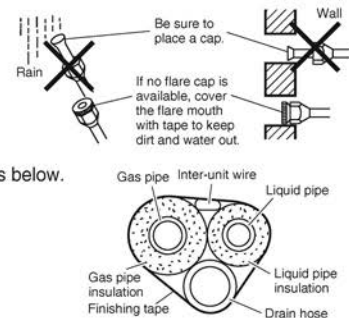
Selection of copper and heat insulation materials

When using commercial copper pipes and fittings, observe the following:

- Insulation material: Polyethylene foam
Heat transfer rate: 0.041 to 0.052W/mK (0.035 to 0.045kcal/mh°C)
Refrigerant gas pipe's surface temperature reaches 110°C max.
Choose heat insulation materials that will withstand this temperature.
- Be sure to insulate both the gas and liquid piping and observe the insulation dimensions as below.

	Piping size	Minimum bend radius	Piping thickness	Thermal insulation size	Thermal insulation thickness
Gas side	O.D. 9.5mm	30mm or more	0.8mm (C1220T-O)	I.D. 12-15mm	10mm Min.
	O.D. 12.7mm	40mm or more		I.D. 14-16mm	
	O.D. 15.9mm	50mm or more	1.0mm (C1220T-O)	I.D. 16-20mm	
Liquid side	O.D. 6.4mm	30mm or more	0.8mm (C1220T-O)	I.D. 8-10mm	

- Use separate thermal insulation pipes for gas and liquid refrigerant pipes.



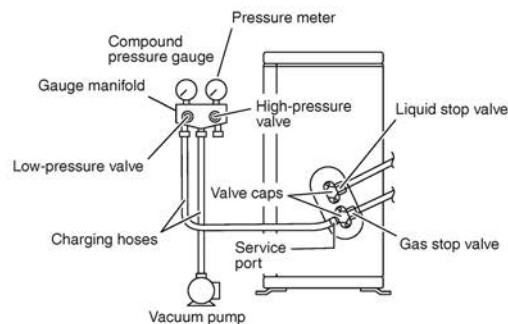
OUTDOOR UNIT INSTALLATION (2)

5 EVACUATING THE AIR WITH A VACUUM PUMP AND CHECKING FOR GAS LEAKAGE

⚠ WARNING

- Do not mix any substance other than the specified refrigerant (R32) into the refrigerant cycle.
- If refrigerant gas leaks should occur, ventilate the room as soon and as much as possible.
- R32, as well as other refrigerants, should always be recovered and never be released directly into the environment.
- Use a vacuum pump for R32 exclusively. Using the same vacuum pump for different refrigerants may damage the vacuum pump or the unit.
- **Use tools for R32 (such as the gauge manifold, charging hose, or vacuum pump adapter).**

- When piping work is complete, it is necessary to evacuate the air with a vacuum pump and check for gas leakage.
- If using additional refrigerant, perform air purging of the refrigerant pipes and indoor unit using a vacuum pump, then charge additional refrigerant.
- Use a hexagonal wrench (4mm) to operate the stop valve rod.
- All refrigerant pipe joints should be tightened with a torque wrench to the specified tightening torque.



- 1) Connect projection side of charging hose (which comes from gauge manifold) to gas stop valve's service port.
- 2) Fully open gauge manifold's low-pressure valve (Lo) and completely close its high-pressure valve (Hi).
(High-pressure valve subsequently requires no operation.)
- 3) Perform vacuum pumping and make sure that the compound pressure gauge reads -0.1MPa (-760mmHg). *1
- 4) Close gauge manifold's low-pressure valve (Lo) and stop vacuum pump.
(Keep this state for a few minutes to make sure that the compound pressure gauge pointer does not swing back.) *2
- 5) Remove valve caps from liquid stop valve and gas stop valve.
- 6) Turn the liquid stop valve's rod 90° counter-clockwise with a hexagonal wrench to open the valve.
Close it after 5 seconds, and check for gas leakage.
Using soapy water, check for gas leakage from indoor unit's flare and outdoor unit's flare and valve rods.
After the check is complete, wipe all soapy water off.
- 7) Disconnect charging hose from gas stop valve's service port, then fully open liquid and gas stop valves.
(Do not attempt to turn valve rod beyond its stop.)
- 8) Tighten valve caps and service port cap for the liquid and gas stop valves with a torque wrench to the specified torques.

*1. Pipe length vs. vacuum pump run time

Pipe length	Up to 15m	More than 15m
Run time	Not less than 10 min.	Not less than 15 min.

*2. If the compound pressure gauge pointer swings back, refrigerant may have water content or a loose pipe joint may exist.
Check all pipe joints and retighten nuts as needed, then repeat steps 2) through 4).

OUTDOOR UNIT INSTALLATION (2)

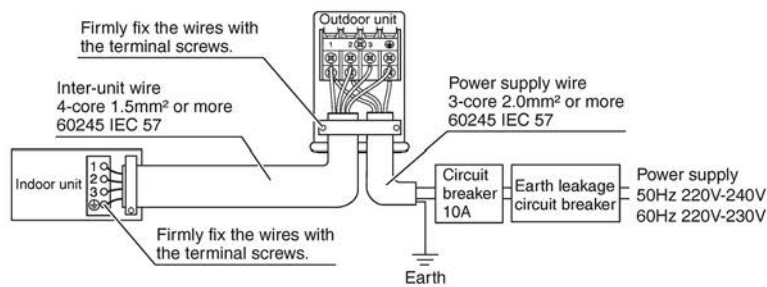
6 WIRING (1)

⚠ WARNING

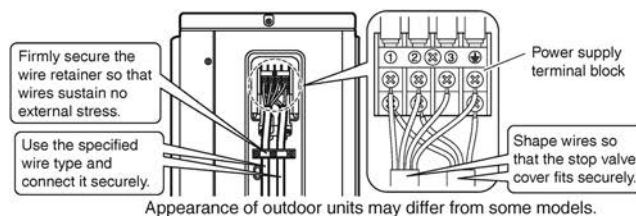
- Do not use tapped wires, extension cords, or starburst connections, as they may cause overheating, electric shock, or fire.
- Do not use locally purchased electrical parts inside the product. (Do not branch the power for the drain pump, etc., from the terminal block.) Doing so may cause electric shock or fire.
- Be sure to install an earth leakage detector. (One that can handle higher harmonics.)
(This unit uses an inverter. Therefore, a earth leakage circuit breaker capable of handling higher harmonics must be used in order to prevent the earth leakage circuit breaker malfunctioning.)
- Use an all-pole disconnection type circuit breaker with at least 3mm between the contact point gaps.
- Do not connect the power wire to the indoor unit. Doing so may cause electric shock or fire.
- There is a risk of electric shock if the earth wires are disconnected, so please make sure the earth wires are longer than the other wires.
(The earth wires should be the last wires to be disconnected if the inter-unit wires are pulled.)

- Do not turn on the circuit breaker until all work is completed.
- Precautions to be taken for power supply wiring, please see **7 WIRING (2)**.

- Strip the insulation from the wire (20mm).
- Connect the inter-unit wires between the indoor and outdoor units so that the terminal numbers match.
Tighten the terminal screws securely. We recommend a flathead screwdriver be used to tighten the screws.
The screws are packed with the terminal block.



- If the stranded wires must be used, make sure to use the round crimp-style terminal for connection to the power supply terminal block. Place the round crimp-style terminals on the wires up to the covered part and secure in place.
- Earth terminal installation. Use the following method **7 WIRING (2)** when installing the round crimp-style terminal.
- Pull the wire and make sure that it does not disconnect. Then fix the wire in place with a wire retainer.



Appearance of outdoor units may differ from some models.

Observe the notes mentioned below when wiring to the power supply terminal block.

OUTDOOR UNIT INSTALLATION (3)

7 WIRING (2)

⚠ CAUTION

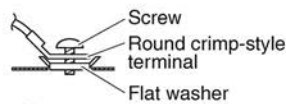
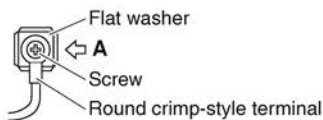
Precautions to be taken for power supply wiring

- When using stranded wires, make sure to use the round crimp-style terminal for connection to the power supply terminal block.

Round crimp-style terminal

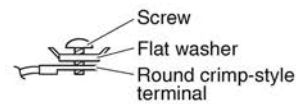


- Use the following method when installing a round crimp-style terminal.



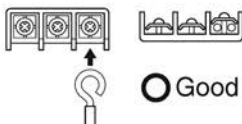
○ Good

Arrow view A



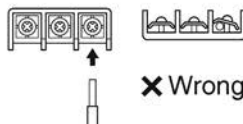
✗ Wrong

- When connecting the inter-unit wire to the terminal block using a single core wire, be sure to curl the end of the lead. Improper work may cause heat and fire.



○ Good

Stripping wire at terminal block



✗ Wrong



○ Good



✗ Wrong

- If the power cord is damaged, it should be replaced by the manufacturer, authorized dealer or qualified technician, in order to avoid hazards.

Earthing

This air conditioner must be earthed. For earthing, follow the applicable local standard for electrical installations.

FACILITY SETTING SWITCH (COOLING AT LOW OUTDOOR TEMPERATURE)

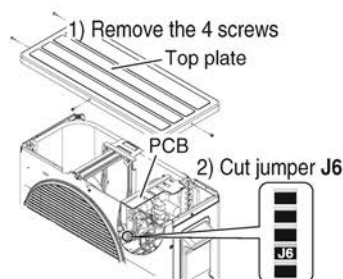
This function is designed for facilities such as equipment or computer rooms. It is never to be used in a residence or office where people occupy the space.

⚠ CAUTION

- If the outdoor unit is installed where the heat exchanger of the unit is exposed to direct wind, provide a windbreak wall.
- Intermittent noises may be produced by the indoor unit due to the outdoor fan turning on and off when using facility settings.
- Do not place humidifiers or other items which might raise the humidity in rooms where facility settings are being used. A humidifier might cause dew condensation from the indoor unit outlet vent.
- Cutting jumper J6 sets the indoor fan tap to the highest position. Notify the user about this.

■ Cutting jumper J6 on the circuit board will expand the operation range down to -15°C . However it will stop if the outdoor temperature drops below -18°C and start back up once the temperature rises again.

- Remove the screws on the side and remove the top plate of the outdoor unit.
- Cut the jumper J6 of the PCB inside.



OUTDOOR UNIT INSTALLATION (3)

PUMP DOWN OPERATION

"Pump Down" means to collect refrigerant in the outdoor unit.

CAUTION

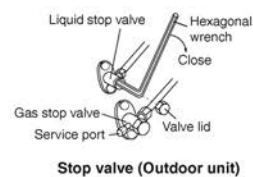
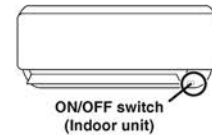
- When pressing the switch, do not touch the terminal block. It has a high voltage, and touching it could cause electric shock.
- After closing the liquid stop valve, close the gas stop valve within 3 minutes, then stop the forced cooling operation.

In order to protect the environment, be sure to pump down when relocating or disposing of the unit.

- Remove the valve cap from liquid stop valve and gas stop valve.
- Carry out forced cooling operation following the method below.
 - Using the indoor unit ON/OFF switch
Press the indoor unit ON/OFF switch for at least 5 seconds. (The operation will start.)^[1]
- After 5 to 10 minutes, close the liquid stop valve with a hexagonal wrench.
- After 2 to 3 minutes, close the gas stop valve and stop forced cooling operation.

Note

- [1] Forced cooling operation will stop automatically after around 15 minutes.
To stop operation before 15 minutes have elapsed, press the indoor unit ON/OFF switch.



TRIAL OPERATION AND TESTING

- Before starting the trial operation, measure the voltage at the primary side of the circuit breaker.
- Check that all liquid and gas stop valves are fully open.
- Check that piping and wiring all match.

■ Trial operation should be carried out in COOL or HEAT operation.

1-1 Measure the supply voltage and make sure that it is within the specified range.

1-2 In COOL operation, select the lowest programmable temperature; in HEAT operation, select the highest programmable temperature.

1-3 Carry out the trial operation following the instructions in the operation manual to ensure that all functions and parts, such as the movement of the flap, are working properly.

- To protect the air conditioner, restart operation is disabled for 3 minutes after the system has been turned off.

1-4 After trial operation is complete, set the temperature to a normal level (26°C to 28°C in COOL operation, 20°C to 24°C in HEAT operation).

- When operating the air conditioner in COOL operation in winter, or HEAT operation in summer, set it to the trial operation mode using the following method.

- Press  to turn on the system.
- Press  and  at the same time.
- Press , select "T", and press  for confirmation.

- Trial operation will stop automatically after about 30 minutes. To stop the operation, press .
- Some of the functions cannot be used in the trial operation mode.

- The air conditioner draws a small amount of power in its standby mode. If the system is not to be used for some time after installation, shut off the circuit breaker to eliminate unnecessary power consumption.
- If the circuit breaker trips to shut off the power to the air conditioner, the system will restore the original operation mode when the circuit breaker is turned on again.

Test items

Test items	Symptom	Check
Indoor and outdoor units are installed securely.	Fall, vibration, noise	
No refrigerant gas leaks.	Incomplete cooling/heating function	
Refrigerant gas and liquid pipes and indoor drain hose extension are thermally insulated.	Water leakage	
Draining line is properly installed.	Water leakage	
System is properly earthed.	Electrical leakage	
Only specified wires are used for all wiring, and all wires are connected correctly.	No operation or burn damage	
Indoor or outdoor unit's air inlet or air outlet are unobstructed.	Incomplete cooling/heating function	
Stop valves are opened.	Incomplete cooling/heating function	
Indoor unit properly receives remote control commands.	No operation	

ATTENTION

- Even when the air conditioner is not operating, it consumes some electric power. If the customer is not going to use the unit soon after it is installed, turn off the circuit breaker to avoid wasting electricity.

3P412209-20M

11.2.2 46/50/60/71 Class

SAFETY PRECAUTIONS (1)



Read the precautions in this manual carefully before operating the unit.



This appliance is filled with R32.

- The precautions described herein are classified as WARNING and CAUTION. They both contain important information regarding safety. Be sure to observe all precautions without fail.
- Meaning of WARNING and CAUTION notices

	WARNING	Failure to follow these instructions properly may result in personal injury or loss of life.
	CAUTION	Failure to follow these instructions properly may result in property damage or personal injury, which may be serious depending on the circumstances.

- The safety marks shown in this manual have the following meanings:

	Be sure to follow the instructions.		Be sure to establish an earth connection.		Never attempt.
--	-------------------------------------	--	---	--	----------------

- After completing installation, conduct a trial operation to check for faults and explain to the user how to operate the air conditioner and take care of it with the aid of the operation manual.

**WARNING**

- Ask your dealer or qualified personnel to carry out installation work.
Do not attempt to install the air conditioner yourself. Improper installation may result in water leakage, electric shock or fire.
- Install the air conditioner in accordance with the instructions in this installation manual. Improper installation may result in water leakage, electric shock or fire.
- Be sure to use only the specified accessories and parts for installation work.
Failure to use the specified parts may result in the unit falling, water leakage, electric shock or fire.
- Install the air conditioner on a foundation strong enough to hold the weight of the unit. A foundation of insufficient strength may result in the equipment falling and causing injury.
- Electrical work must be performed in accordance with relevant local and national regulations and with the instructions in this installation manual.
Be sure to use a dedicated power supply circuit only. Insufficient power supply and improper workmanship may result in electric shock or fire.
- Use a cable of suitable length. Do not use tapped wires or an extension lead, as this may cause overheating, electric shock or fire.
- Make sure that all wiring is secured, the specified wires are used, and that there is no strain on the terminal connections or wires.
Improper connections or securing of wires may result in abnormal heat build-up or fire.
- When wiring the power supply and connecting the wiring between the indoor and outdoor units, position the wires so that the electrical wiring box cover can be securely fastened.
Improper positioning of the electrical wiring box cover may result in electric shock, fire or overheating terminals.
- If refrigerant gas leaks during installation, ventilate the area immediately.
Toxic gas may be produced if the refrigerant comes into contact with fire.
- After completing installation, check for refrigerant gas leakage.
Toxic gas may be produced if the refrigerant gas leaks into the room and comes into contact with a source of fire, such as a fan heater, stove or cooker.
- When installing or relocating the air conditioner, do not let any other substances besides R32, such as air, enter the refrigerant circuit.
The presence of air or foreign matter in the refrigerant circuit causes an abnormal pressure rise, which may result in equipment damage and even injury.
- During installation, attach the refrigerant piping securely before operating the compressor. If the refrigerant pipes are not attached and the stop valve is open when the compressor is operated, air will be sucked in, causing abnormal pressure in the refrigeration cycle, which may result in equipment damage and even injury.
- During pump down, stop the compressor before removing the refrigerant piping. If the compressor is still operating and the stop valve is open during pump down, air will be sucked in when the refrigerant piping is removed, causing abnormal pressure in the refrigeration cycle, which may result in equipment damage and even injury.
- Be sure to earth the air conditioner.
Do not earth the unit to a utility pipe, lightning conductor or telephone earth lead. Imperfect earthing may result in electric shock.
- Be sure to install an earth leakage circuit breaker. Failure to install an earth leakage circuit breaker may result in electric shock or fire.
- Do not use means to accelerate the defrosting process (if possible) or to clean, other than those recommended by the manufacturer.
- The appliance must be stored in a room without continuously operating ignition sources (for example: open flames, an operating gas appliance or an operating electric heater).
- Do not pierce or burn.
- Be aware that refrigerants may not contain an odour.
- The appliance must be installed, operated and stored in a room with a floor area larger than 2.5 m² for 46 class, 3.5 m² for 50/60 class, 3.9 m² for 71 class.
- Comply with national gas regulations.
- Do not pump down when the refrigerant has leaked, otherwise the compressor may be damaged.
- Do not install the air conditioner at any place where there is a danger of flammable gas leakage.
In the event of a gas leakage, build-up of gas near the air conditioner may cause a fire to break out.
- When flared joints are reused indoors, the flare part shall be re-fabricated.

SAFETY PRECAUTIONS (2)



CAUTION

- While following the instructions in this installation manual, install drain piping to ensure proper drainage and insulate the piping to prevent condensation.
Improper drain piping may result in indoor water leakage and property damage.
- Tighten the flare nut as specified, such as with a torque wrench.
If the flare nut is too tight, it may crack after prolonged use, causing refrigerant leakage.
- Take adequate steps to prevent the outdoor unit being used as a shelter by small animals.
If small animals or birds come into contact with electrical parts, this can cause malfunctions, smoke or fire. Please instruct the customer to always keep the area around the unit clean.
- The refrigerant circuit temperature will be high, therefore the inter-unit wires must be kept away from copper pipes that are not thermally insulated.
- Only qualified personnel can handle, fill, purge and dispose of the refrigerant.

ACCESSORIES

Ⓐ Drain socket



1

There is on the bottom packing case.

Ⓑ Installation manual

1

PRECAUTIONS FOR SELECTING THE LOCATION

NOTE

Cannot be installed suspended from a ceiling or stacked.



CAUTION

When operating the air conditioner in a low outdoor ambient temperature, be sure to follow the instructions described below.

- To prevent exposure to wind, install the outdoor unit with its suction side facing the wall.
- Never install the outdoor unit at a site where the suction side may be exposed directly to wind.
- To prevent exposure to wind, it is recommended to install a baffle plate on the air discharge side of the outdoor unit.
- In heavy snowfall areas, select an installation site where the snow will not affect the unit.



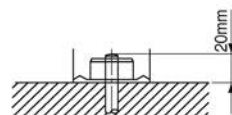
- Construct a large canopy.
- Construct a pedestal.

Install the unit high enough off the ground to prevent burying in snow.

- Choose a place solid enough to bear the weight and vibration of the unit, where the operating sound will not be amplified.
- Choose a location where the hot air discharged from the unit or the operating sound will not cause a nuisance to the neighbours of the user.
- Avoid locations, such as near bedrooms, where the operating sound may cause disturbance.
- There must be sufficient space to carry the unit into and out of the site.
- There must be sufficient space for air passage and no obstructions around the air inlet and the air outlet.
- The site must not be prone to flammable gas leaks in the surrounding area.
- Install units, power cords and inter-unit wire at least 3m away from television and radio sets. (This is to prevent interference to images and sounds. Noise may be produced even if they are more than 3m away depending on radio wave conditions.)
- In coastal areas or other places with a salty atmosphere or one containing sulphate gas, corrosion may shorten the life of the air conditioner.
- Since water will flow from the drain of the outdoor unit, do not place under the unit anything which must be kept away from moisture.

PRECAUTIONS ON INSTALLATION

- Check the strength and level of the installation surface so that the unit does not cause any operating vibrations or noise after installation.
- Fix the unit in place securely using foundation bolts, as in the figure. (Prepare 4 sets of M8 or M10 foundation bolts, nuts and washers; all sold separately.)
- It is best to screw in the foundation bolts until their ends are 20mm from the foundation surface.



IMPORTANT INFORMATION REGARDING THE REFRIGERANT USED

This product contains fluorinated greenhouse gases.

Refrigerant type: **R32**

GWP⁽¹⁾ value: **675**

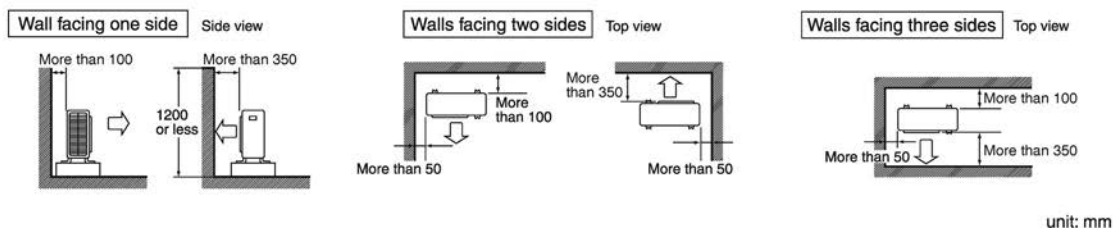
⁽¹⁾GWP = global warming potential

The refrigerant quantity is indicated on the unit name plate.

OUTDOOR UNIT INSTALLATION DIAGRAM

INSTALLATION SPACE REQUIREMENTS

- Where a wall or other obstacle is in the path of the outdoor unit's intake or exhaust airflow, follow the installation space requirements below.
- For any of the below installation patterns, the wall height on the outlet side should be 1200mm or less.



NOTE

When flaring the pipe end and installing the unit, refer to "❸ FLARING THE PIPE END".

Class	46/50/60	71
Max. allowable piping length	30m	
* Min. allowable piping length	2.5m	
Max. allowable piping height	20m	
** Additional refrigerant required for refrigerant pipe exceeding 10m in length.	20g/m	
Gas pipe	O.D. 12.7mm	O.D. 15.9mm
Liquid pipe	O.D. 6.4mm	
Max. allowable refrigerant charge amount	***	

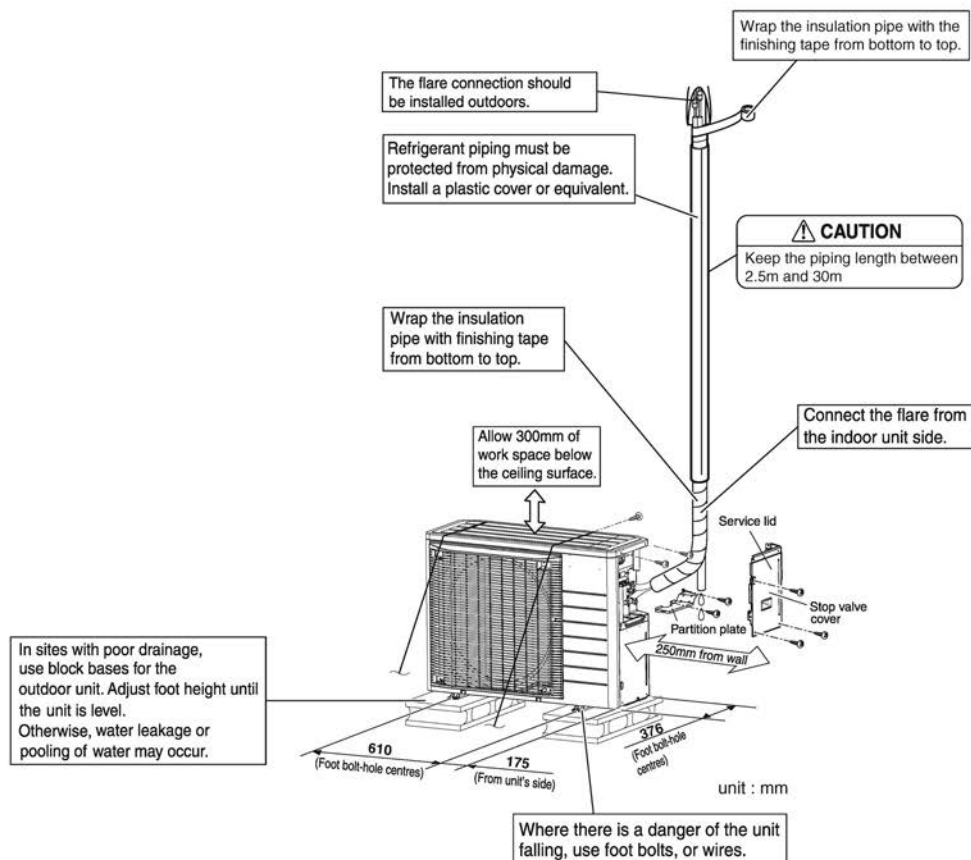
Refrigerant piping must be kept to a minimum.

* It is suggested that the minimum allowable length be used for the pipes connecting the indoor units to the outdoor unit in order to prevent noise and vibration from the outdoor unit.

(Mechanical noise and vibration may occur depending on how the unit is installed and the environment in which it is used.)

** Be sure to add the proper amount of additional refrigerant. Failure to do so may result in reduced performance.

*** 46 class	1.60kg
50 class	1.90kg
60 class	1.90kg
71 class	2.00kg



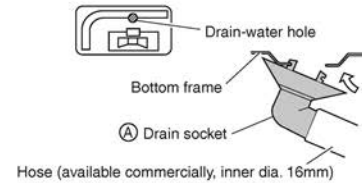
OUTDOOR UNIT INSTALLATION (1)

1 INSTALLING THE OUTDOOR UNIT

- When installing the outdoor unit, refer to "PRECAUTIONS FOR SELECTING THE LOCATION" and the "OUTDOOR UNIT INSTALLATION DIAGRAM".

2 DRAIN WORK

- Use the (A) drain socket for drainage.
- When attaching the drain socket to the bottom frame make sure to first connect the drain hose to the drain socket.
- If the drain port is covered by a mounting base or floor surface, place additional foot bases of at least 30mm in height under the outdoor unit's feet.
- In cold areas, do not use a drain hose with the outdoor unit.
(Otherwise, drain water may freeze, impairing heating performance.)

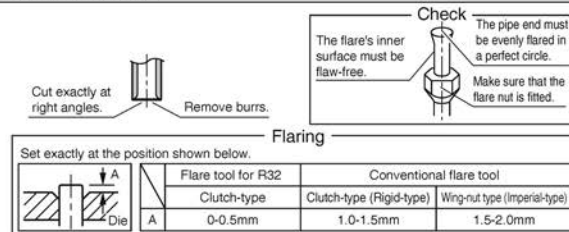


3 FLARING THE PIPE END

WARNING

- Do not apply mineral oil to the flare.
- Prevent mineral oil from getting into the system as this would reduce the service life of the units.
- Never use piping which has been used for previous installations. Only use parts which are delivered with this unit.
- Never install a dryer to this R32 unit in order to guarantee its service life.
- The drying material may dissolve and damage the system.
- Incomplete flaring may result in refrigerant gas leakage.
- When flared joints are reused indoors, the flare part shall be re-fabricated.

- Cut the pipe end with a pipe cutter.
- Remove burrs with the cut surface facing downward, so that the filings do not enter the pipe.
- Put the flare nut on the pipe.
- Flare the pipe.
- Check that the flaring has been done correctly.

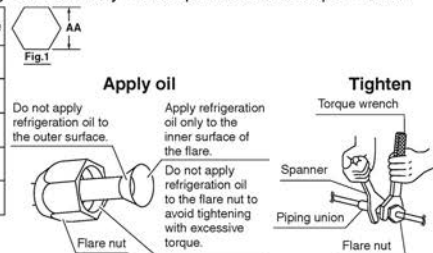


4 REFRIGERANT PIPING

CAUTION

- Use the flare nut fixed to the main unit. (This is to prevent the flare nut from cracking as a result of deterioration over time.)
- To prevent gas leakage, apply refrigeration oil only to the inner surface of the flare. (Use refrigeration oil for R32.)
- Use a torque wrench when tightening the flare nuts to prevent damage to the flare nuts and gas leakage.
- Align the centres of both flares and tighten the flare nuts 3 or 4 turns by hand, then tighten them fully with a spanner and a torque wrench.

	Piping size	Flare nut tightening torque	Valve cap size		Valve cap tightening torque
			dimension AA, refer to Fig.1		
Gas side	O.D. 9.5mm (3/8 inch)	32.7-39.9N • m (333-407kgf • cm)	17mm		15.7 ± 1.6N • m (160 ± 16kgf • cm)
	O.D. 12.7mm (1/2 inch)	49.5-60.3N • m (505-615kgf • cm)	19mm		19.0 ± 1.9N • m (193 ± 20kgf • cm)
	O.D. 15.9mm (5/8 inch)	61.8-75.4N • m (630-770kgf • cm)	22mm		24.5 ± 3.9N • m (250 ± 40kgf • cm)
Liquid side	O.D. 6.4mm (1/4 inch)	14.2-17.2N • m (144-175kgf • cm)	27mm		53.9 ± 5.9N • m (550 ± 60kgf • cm)
			32mm		68.6 ± 6.9N • m (700 ± 70kgf • cm)
Service port cap tightening torque		10.7-14.7N • m (110-150kgf • cm)			



Cautions on pipe handling

- Protect the open end of the pipe from dust and moisture.
- All pipe bends should be as gentle as possible. Use a pipe bender for bending.

Selection of copper and heat insulation materials

When using commercial copper pipes and fittings, observe the following:

- Insulation material: Polyethylene foam

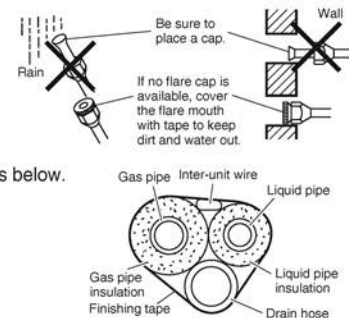
Heat transfer rate: 0.041 to 0.052W/mK (0.035 to 0.045kcal/mh°C)

Refrigerant gas pipe's surface temperature reaches 110°C max.

Choose heat insulation materials that will withstand this temperature.

- Be sure to insulate both the gas and liquid piping and observe the insulation dimensions as below.

	Piping size	Minimum bend radius	Piping thickness	Thermal insulation size	Thermal insulation thickness
Gas side	O.D. 9.5mm	30mm or more	0.8mm (C1220T-O)	I.D. 12-15mm	10mm Min.
	O.D. 12.7mm	40mm or more		I.D. 14-16mm	
	O.D. 15.9mm	50mm or more	1.0mm (C1220T-O)	I.D. 16-20mm	
Liquid side	O.D. 6.4mm	30mm or more	0.8mm (C1220T-O)	I.D. 8-10mm	



- Use separate thermal insulation pipes for gas and liquid refrigerant pipes.

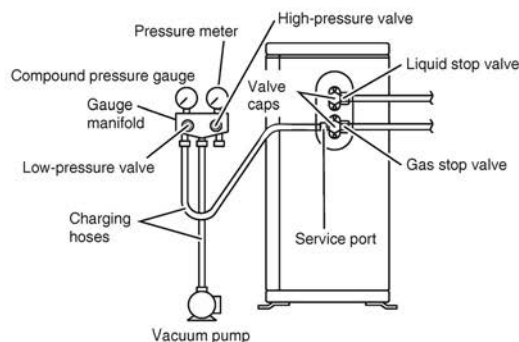
OUTDOOR UNIT INSTALLATION (2)

5 EVACUATING THE AIR WITH A VACUUM PUMP AND CHECKING FOR GAS LEAKAGE

⚠ WARNING

- Do not mix any substance other than the specified refrigerant (R32) into the refrigerant cycle.
- If refrigerant gas leaks should occur, ventilate the room as soon and as much as possible.
- R32, as well as other refrigerants, should always be recovered and never be released directly into the environment.
- Use a vacuum pump for R32 exclusively. Using the same vacuum pump for different refrigerants may damage the vacuum pump or the unit.
- Use tools for R32 (such as the gauge manifold, charging hose, or vacuum pump adapter).

- When piping work is complete, it is necessary to evacuate the air with a vacuum pump and check for gas leakage.
- If using additional refrigerant, perform air purging of the refrigerant pipes and indoor unit using a vacuum pump, then charge additional refrigerant.
- Use a hexagonal wrench (4mm) to operate the stop valve rod.
- All refrigerant pipe joints should be tightened with a torque wrench to the specified tightening torque.



- 1) Connect projection side of charging hose (which comes from gauge manifold) to gas stop valve's service port.
- 2) Fully open gauge manifold's low-pressure valve (Lo) and completely close its high-pressure valve (Hi). (High-pressure valve subsequently requires no operation.)
- 3) Perform vacuum pumping and make sure that the compound pressure gauge reads -0.1MPa (-760mmHg). *1
- 4) Close gauge manifold's low-pressure valve (Lo) and stop vacuum pump. (Keep this state for a few minutes to make sure that the compound pressure gauge pointer does not swing back.) *2
- 5) Remove valve caps from liquid stop valve and gas stop valve.
- 6) Turn the liquid stop valve's rod 90° counter-clockwise with a hexagonal wrench to open the valve. Close it after 5 seconds, and check for gas leakage. Using soapy water, check for gas leakage from indoor unit's flare and outdoor unit's flare and valve rods. After the check is complete, wipe all soapy water off.
- 7) Disconnect charging hose from gas stop valve's service port, then fully open liquid and gas stop valves. (Do not attempt to turn valve rod beyond its stop.)
- 8) Tighten valve caps and service port cap for the liquid and gas stop valves with a torque wrench to the specified torques.

*1. Pipe length vs. vacuum pump run time

Pipe length	Up to 15m	More than 15m
Run time	Not less than 10 min.	Not less than 15 min.

*2. If the compound pressure gauge pointer swings back, refrigerant may have water content or a loose pipe joint may exist. Check all pipe joints and retighten nuts as needed, then repeat steps 2) through 4).

OUTDOOR UNIT INSTALLATION (2)

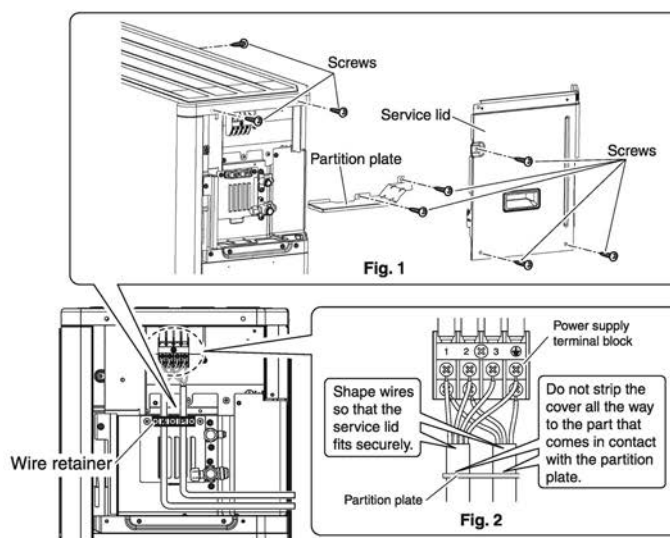
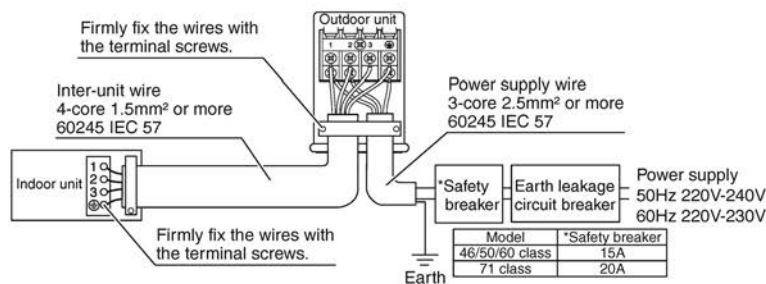
6 WIRING (1)

⚠ WARNING

- Do not use tapped wires, extension cords, or starburst connections, as they may cause overheating, electric shock, or fire.
- Do not use locally purchased electrical parts inside the product. (Do not branch the power for the drain pump, etc., from the terminal block.) Doing so may cause electric shock or fire.
- Be sure to install an earth leakage detector. (One that can handle higher harmonics.)
(This unit uses an inverter. Therefore, a earth leakage circuit breaker capable of handling higher harmonics must be used in order to prevent the earth leakage circuit breaker malfunctioning.)
- Use an all-pole disconnection type circuit breaker with at least 3mm between the contact point gaps.
- Do not connect the power wire to the indoor unit. Doing so may cause electric shock or fire.
- There is a risk of electric shock if the earth wires are disconnected, so please make sure the earth wires are longer than the other wires. (The earth wires should be the last wires to be disconnected if the inter-unit wires are pulled.)

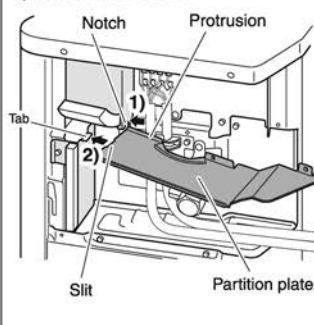
- Do not turn on the circuit breaker until all work is completed.
- Precautions to be taken for power supply wiring, please see **7 WIRING (2)**

- Dismount the service lid and the partition plate. (Refer to Fig.1)
- Strip the insulation from the wire (20mm.)
- Connect the inter-unit wire between the indoor and outdoor units **so that the terminal numbers match**.
Tighten the terminal screws securely. We recommend a flathead screwdriver be used to tighten the screws. (Refer to Fig. 2)
- Pull the wire lightly to make sure that it does not disconnect. Then fix the wire in place with the wire retainer.
- After completing the work, reattach the partition plate and the service lid to their original positions.
(Refer to "How to attach the partition plate.")



■ How to attach the partition plate.

- Insert the protrusion of the partition plate into the notch in the metal sheet.
- Slide the partition plate so that the tab on the metal sheet fits into the slit on the partition plate.
- Secure with 2 screws.



OUTDOOR UNIT INSTALLATION (3)

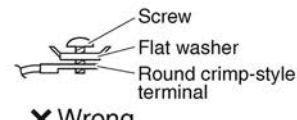
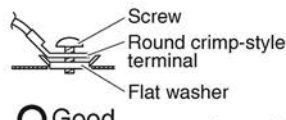
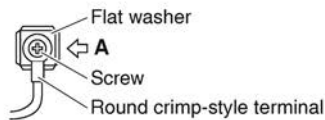
7 WIRING (2)

⚠ CAUTION

Precautions to be taken for power supply wiring

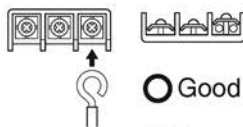
- When using stranded wires, make sure to use the round crimp-style terminal for connection to the power supply terminal block.

- Use the following method when installing a round crimp-style terminal.

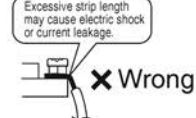
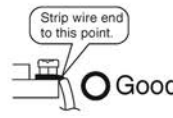
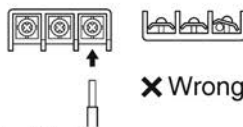


Arrow view A

- When connecting the inter-unit wire to the terminal block using a single core wire, be sure to curl the end of the lead. Improper work may cause heat and fire.



Stripping wire at terminal block



- If the power cord is damaged, it should be replaced by the manufacturer, authorized dealer or qualified technician, in order to avoid hazards.

Earthing

This air conditioner must be earthed. For earthing, follow the applicable local standard for electrical installations.

FACILITY SETTING SWITCH (COOLING AT LOW OUTDOOR TEMPERATURE)

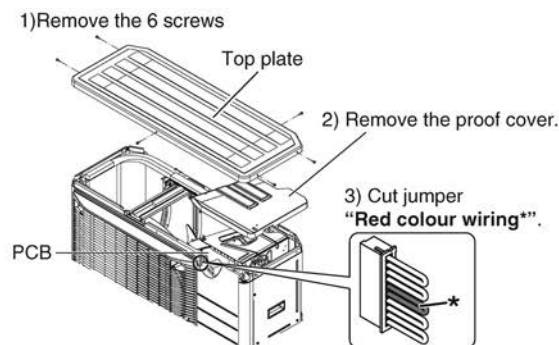
This function is designed for facilities such as equipment or computer rooms. It is never to be used in a residence or office where people occupy the space.

⚠ CAUTION

- If the outdoor unit is installed where the heat exchanger of the unit is exposed to direct wind, provide a windbreak wall.
- Intermittent noises may be produced by the indoor unit due to the outdoor fan turning on and off when using facility settings.
- Do not place humidifiers or other items which might raise the humidity in rooms where facility settings are being used. A humidifier might cause dew condensation from the indoor unit outlet vent.
- Cutting jumper "Red colour wiring*" sets the indoor fan tap to the highest position. Notify the user about this.

- Cutting jumper "Red colour wiring*" on the circuit board will expand the operation range down to -15°C . However it will stop if the outdoor temperature drops below -18°C and start back up once the temperature rises again.

- Remove the screws on the side and remove the top plate of the outdoor unit.
- Remove the drip proof cover.
- Cut the jumper "Red colour wiring*" of the PCB inside.



OUTDOOR UNIT INSTALLATION (3)

PUMP DOWN OPERATION

"Pump Down" means to collect refrigerant in the outdoor unit.

CAUTION

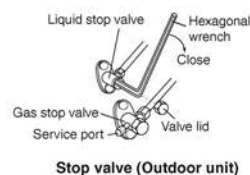
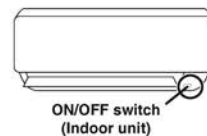
- When pressing the switch, do not touch the terminal block. It has a high voltage, and touching it could cause electric shock.
- After closing the liquid stop valve, close the gas stop valve within 3 minutes, then stop the forced cooling operation.

In order to protect the environment, be sure to pump down when relocating or disposing of the unit.

- 1) Remove the valve cap from liquid stop valve and gas stop valve.
- 2) Carry out forced cooling operation following the method below.
 - Using the indoor unit ON/OFF switch
Press the indoor unit ON/OFF switch for at least 5 seconds. (The operation will start.)^[1]
- 3) After 3 to 4 minutes, close the liquid stop valve with a hexagonal wrench.
- 4) After 5 to 6 minutes, close the gas stop valve and stop forced cooling operation.

Note

- [1] Forced cooling operation will stop automatically after around 15 minutes.
To stop operation before 15 minutes have elapsed, press the indoor unit ON/OFF switch.



TRIAL OPERATION AND TESTING

- Before starting the trial operation, measure the voltage at the primary side of the circuit breaker.
- Check that all liquid and gas stop valves are fully open.
- Check that piping and wiring all match.

■ Trial operation should be carried out in COOL or HEAT operation.

1-1 Measure the supply voltage and make sure that it is within the specified range.

1-2 In COOL operation, select the lowest programmable temperature; in HEAT operation, select the highest programmable temperature.

1-3 Carry out the trial operation following the instructions in the operation manual to ensure that all functions and parts, such as the movement of the flap, are working properly.

- To protect the air conditioner, restart operation is disabled for 3 minutes after the system has been turned off.
- When trial operation is conducted in HEAT operation directly after the circuit breaker is turned on, in some cases no air will be output for about 10 minutes in order to protect the air conditioner.

1-4 After trial operation is complete, set the temperature to a normal level (26°C to 28°C in COOL operation, 20°C to 24°C in HEAT operation).

- When operating the air conditioner in COOL operation in winter, or HEAT operation in summer, set it to the trial operation mode using the following method.

1) Press  to turn on the system.

2) Press  and  at the same time.

3) Press , select "T", and press  for confirmation.

• Trial operation will stop automatically after about 30 minutes. To stop the operation, press .

• Some of the functions cannot be used in the trial operation mode.

- The air conditioner draws a small amount of power in its standby mode. If the system is not to be used for some time after installation, shut off the circuit breaker to eliminate unnecessary power consumption.
- If the circuit breaker trips to shut off the power to the air conditioner, the system will restore the original operation mode when the circuit breaker is turned on again.

Test items

Test items	Symptom	Check
Indoor and outdoor units are installed securely.	Fall, vibration, noise	
No refrigerant gas leaks.	Incomplete cooling/heating function	
Refrigerant gas and liquid pipes and indoor drain hose extension are thermally insulated.	Water leakage	
Draining line is properly installed.	Water leakage	
System is properly earthed.	Electrical leakage	
Only specified wires are used for all wiring, and all wires are connected correctly.	No operation or burn damage	
Indoor or outdoor unit's air inlet or air outlet are unobstructed.	Incomplete cooling/heating function	
Stop valves are opened.	Incomplete cooling/heating function	
Indoor unit properly receives remote control commands.	No operation	

ATTENTION

- Even when the air conditioner is not operating, it consumes some electric power. If the customer is not going to use the unit soon after it is installed, turn off the circuit breaker to avoid wasting electricity.

3P412209-21M

12. Operation Manual

12.1 20/25/35/46 Class

Read Before Operation

Safety Precautions

	Read the precautions in this manual carefully before operating the unit.		This appliance is filled with R32.
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- Keep this manual where the user can easily find it.
- The precautions described herein are classified as WARNING and CAUTION. They both contain important information regarding safety. Be sure to observe all precautions without fail.

	WARNING	Failure to follow these instructions properly may result in personal injury or loss of life.
	CAUTION	Failure to observe these instructions properly may result in property damage or personal injury, which may be serious depending on the circumstances.

	Never attempt.		Be sure to follow the instructions.		Be sure to establish an earth connection.
---	----------------	---	-------------------------------------	--	---

- After reading, keep this manual in a convenient place so that you can refer to it whenever necessary. If the equipment is transferred to a new user, be sure also to hand over the manual.

WARNING

- Do not use means to accelerate the defrosting process or to clean, other than those recommended by the manufacturer.
- The appliance must be stored in a room without continuously operating ignition sources (for example: open flames, an operating gas appliance or an operating electric heater).
- Do not pierce or burn.
- Be aware that refrigerants may not contain an odour.
- The appliance must be installed, operated and stored in a room with floor area larger than $\sim 2\text{m}^2$ for 20/25/35 class, 2.5m^2 for 46 class.
- * Since the maximum refrigerant charge of the models is below 1.22kg, it does not have any limitation for the minimum floor area. Maintain an installation height of 1.8m or more from the floor surface to the bottom of the appliance. If the size of the room where the appliance is to be installed has less than the indicated minimum floor area, install it in a well ventilated room. For details refer to the WARNING list in the "Safety Precautions" section of the installation manual for the outdoor unit.
- To avoid fire, explosion or injury, do not operate the unit when harmful gases (e.g. flammable or corrosive) are detected near the unit.
- Be aware that prolonged, direct exposure to cool or warm air from the air conditioner, or to air that is too cool or too warm, can be harmful to your physical condition and health.
- Do not place objects, including rods, your fingers, etc., in the air inlet or outlet. Product damage or personal injury may result due to contact with the air conditioner's high-speed fan blades.
- Do not attempt to repair, dismantle, reinstall or modify the air conditioner yourself as this may result in water leakage, electric shock or fire hazards.
- Do not use flammable spray near the air conditioner, or otherwise fire may result.
- Do not use a refrigerant other than the one indicated on the outdoor unit (R32) when installing, moving or repairing. Using other refrigerants may cause trouble or damage to the unit, and personal injury.
- To avoid electric shock, do not operate with wet hands.

- Beware of fire in case of refrigerant leakage. If the air conditioner is not operating correctly, i.e. not generating cool or warm air, refrigerant leakage could be the cause. Consult your dealer for assistance. The refrigerant within the air conditioner is safe and normally does not leak. However, in the event of a leakage, contact with a naked burner, heater or cooker may result in generation of noxious gas. Do not use the air conditioner until a qualified service person confirms that the leakage has been repaired.
- Do not attempt to install or repair the air conditioner yourself. Improper workmanship may result in water leakage, electric shock or fire hazards. Please contact your local dealer or qualified personnel for installation and maintenance work.
- If the air conditioner is malfunctioning (giving off a burning odour, etc.), turn off power to the unit and contact your local dealer. Continued operation under such circumstances may result in a failure, electric shock or fire hazards.
- Be sure to install an earth leakage circuit breaker. Failure to install an earth leakage circuit breaker may result in electric shock or fire.
- Be sure to earth the unit. Do not earth the unit to a utility pipe, lightning conductor or telephone earth lead. Imperfect earthing may result in electric shock.

CAUTION

- Do not use the air conditioner for purposes other than those for which it is intended. Do not use the air conditioner for cooling precision instruments, food, plants, animals or works of art as this may adversely affect the performance, quality and/or longevity of the object concerned.
- Do not expose plants or animals directly to the airflow from the unit as this may cause adverse effects.
- Do not place appliances that produce naked flames in places exposed to the airflow from the unit as this may impair combustion of the burner.
- Do not block the air inlets nor outlets. Impaired airflow may result in insufficient performance or trouble.

Read Before Operation



- Do not sit on the outdoor unit, put things on the unit, or pull the unit. Doing so may cause accidents, such as falling or toppling down, thus resulting in injury, product malfunctioning, or product damage.
- Do not place objects that are susceptible to moisture directly beneath the indoor or outdoor units. Under certain conditions, condensation on the main unit or refrigerant pipes, air filter dirt or drain blockage may cause dripping, resulting in fouling or failure of the object concerned.
- After prolonged use, check the unit stand and its mounts for damage. If they are left in a damaged condition, the unit may fall and cause injury.
- To avoid injury, do not touch the air inlet or aluminium fins of the indoor or outdoor units.
- The appliance is not intended for use by unattended young children or infirm persons. Impairment of bodily functions and harm to health may result.
- Children should be supervised to ensure that they do not play with the unit or its remote controller. Accidental operation by a child may result in impairment of bodily functions and harm health.
- Avoid impacts to the indoor and outdoor units, or otherwise product damage may result.
- Do not place flammable items, such as spray cans, within 1m of the air outlet.
- The spray cans may explode as a result of hot air from the indoor or outdoor units.
- Be careful not to let pets urinate on the air conditioner. Urination on the air conditioner may result in electric shock or fire.
- Do not wash the air conditioner with water, as this may result in electric shock or fire.
- Do not place water containers (vases, etc.) above the unit, as this may result in electric shock or fire hazards if they should topple over.
- Do not insert the batteries in the wrong polarity (+/-) orientation as this may result in short circuiting, fire, or battery leakage.
- Do not sit or hang on the panel. The panel may fall, and injury or product malfunctioning may result.
- Do not sway the panel. The panel may hit people or objects, and injury or property damage may result.
- Do not let children play around the panel. Injury or property damage may result.
- Do not pull the wires. The wires may be broken and the panel may fall, and injury or property damage may result.
- Do not locate obstacles in the route. The panel may fall, and injury or property damage may result.
- Do not bend or damage the wires. The wires may be broken and the panel may fall, and injury or property damage may result.
- Do not put objects on the panel, or otherwise production malfunctioning may result.
- Do not use an unstable stand at the time of operating or maintaining the air conditioner, or otherwise you may topple over or injury yourself.
- Locate the remote controller in places out of reach of children. The wrong operation of the remote controller may result in injury.



- To avoid oxygen depletion, ensure that the room is adequately ventilated if equipment such as a burner is used together with the air conditioner.
- Before cleaning, be sure to stop unit operation and turn off the circuit breaker. Otherwise, an electric shock and injury may result.
- Only connect the air conditioner to the specified power supply circuit. Power supplies other than the one specified may result in electric shock, overheating and fires.
- Arrange the drain hose to ensure smooth drainage. Imperfect drainage may cause wetting of the building, furniture, etc.
- Do not place objects in direct proximity of the outdoor unit and do not let leaves and other debris accumulate around the unit. Leaves are a hotbed for small animals which can enter the unit. Once inside the unit, such animals can cause malfunctions, smoke or fire if they come into contact with electrical parts.
- Do not place objects around the indoor unit.
Doing so may have an adverse influence on the performance, product quality, and life of the air conditioner.
- This appliance is not intended to be used by persons with reduced physical, sensory or mental capabilities, or with lack of operation knowledge, unless they have been given supervision or instruction concerning the appliance use by person responsible for their safety. Keep out of children's reach to ensure that they do not play with the appliance.

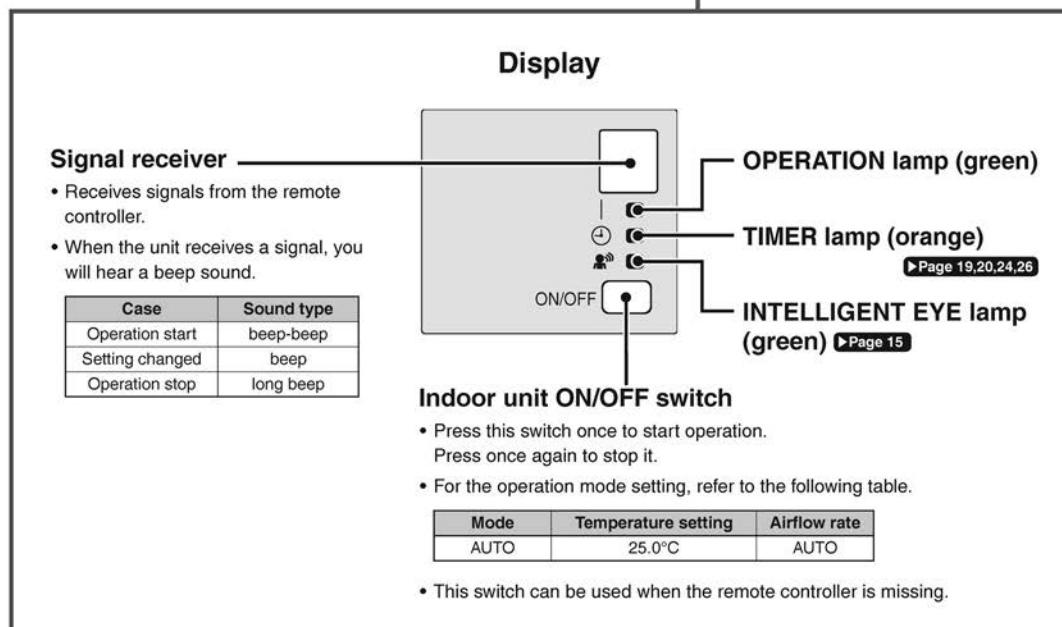
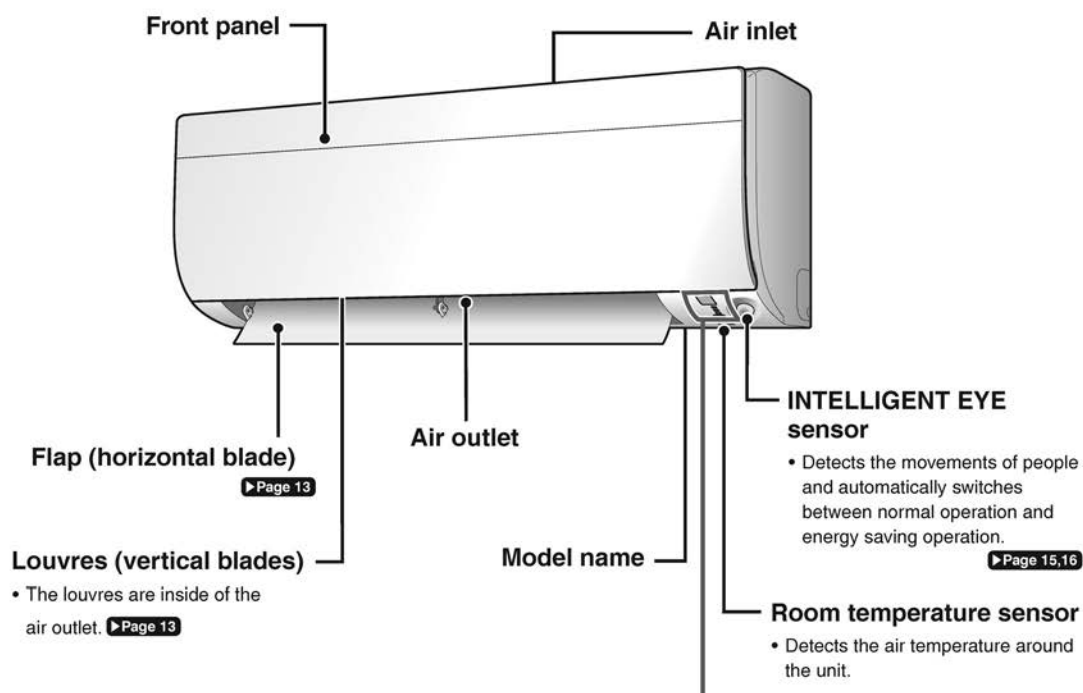
Precautions relating to area surrounding the indoor and outdoor units

- Be sure to follow the instructions below.
 - The indoor unit is at least 1m away from any television or radio set (unit may cause interference with the picture or sound).
 - Refrain from using the units in areas prone to high levels of oily smoke, such as a kitchen. Water leakage may result.

Read Before Operation

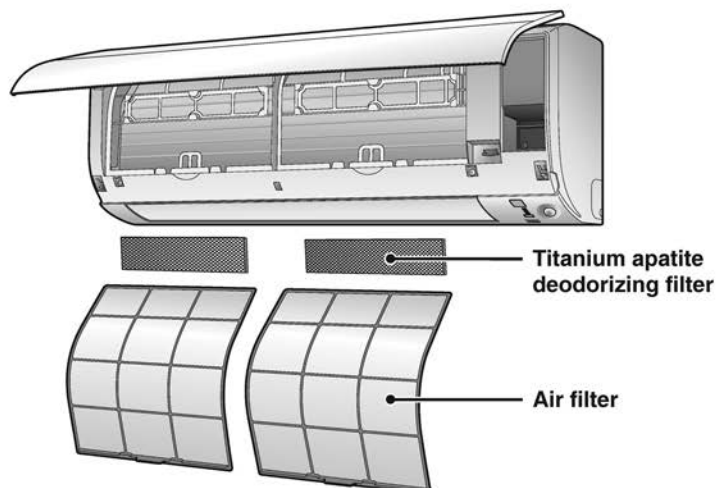
Names of Parts

Indoor Unit

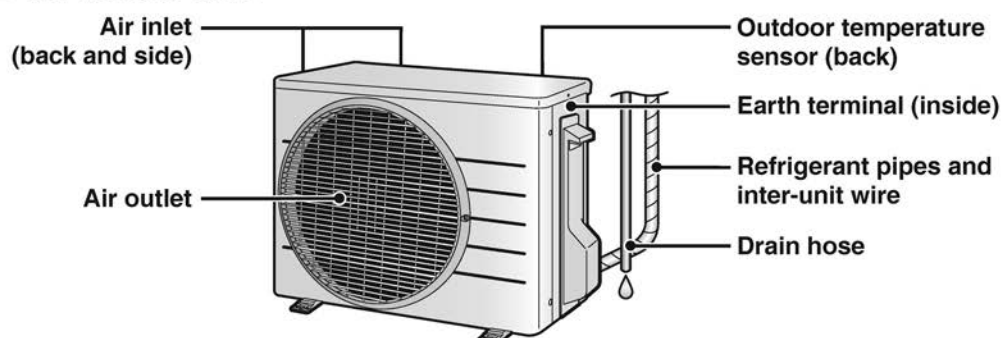


Read Before Operation

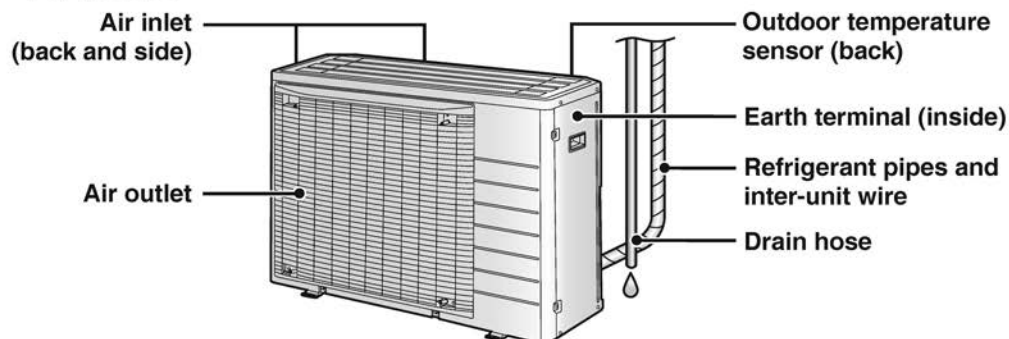
■ Open the front panel

**Outdoor Unit**

■ For 20/25/35 class



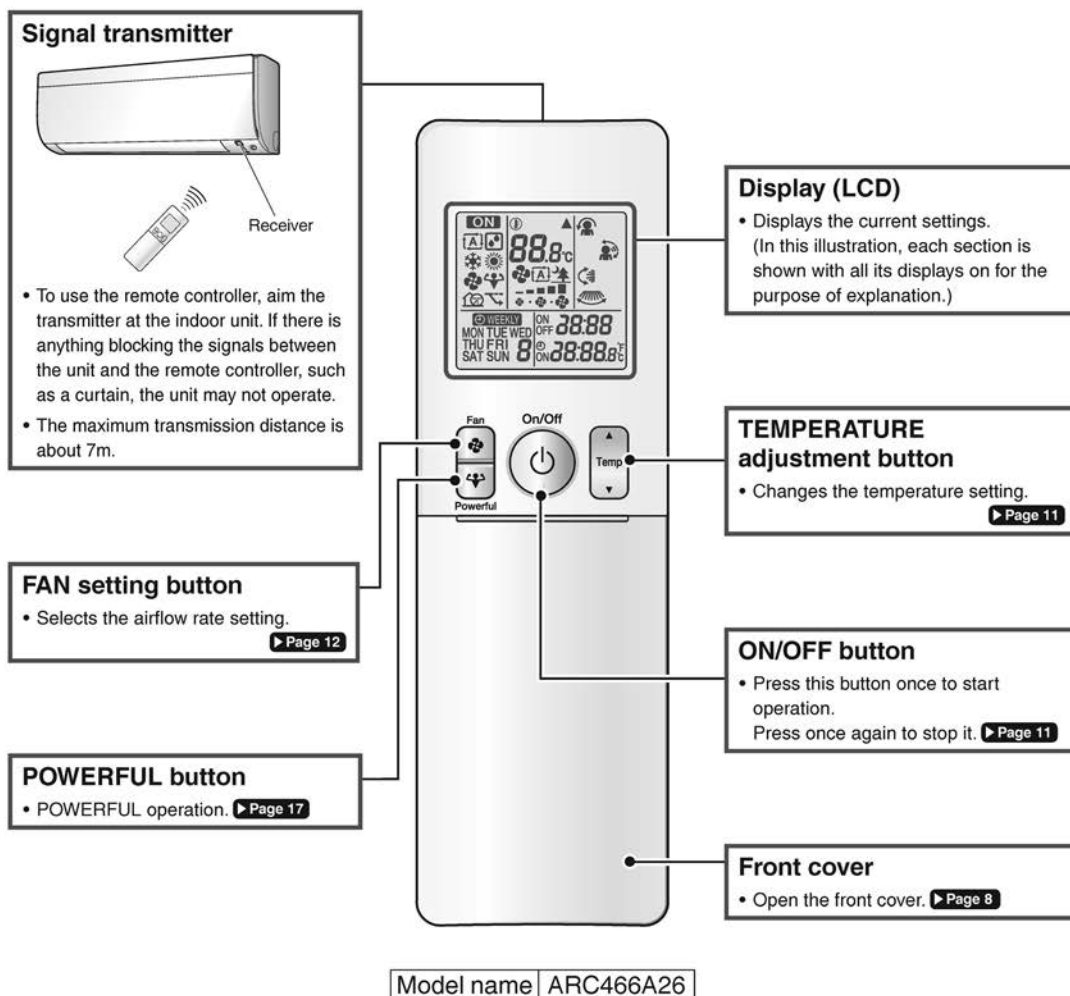
■ For 46 class



Read Before Operation

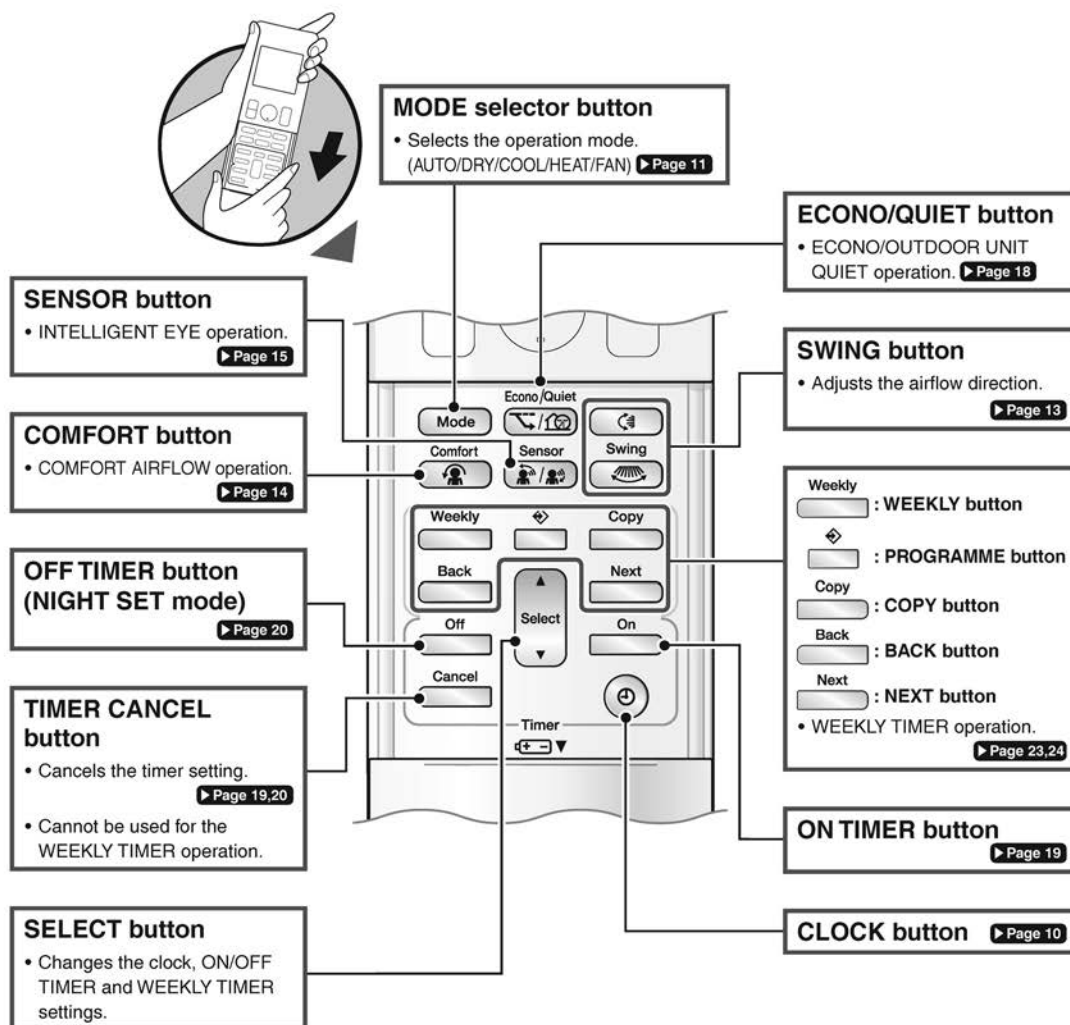
Names of Parts

Remote Controller



Read Before Operation

■ Open the front cover



Read Before Operation

Preparation Before Operation

⚠ CAUTION

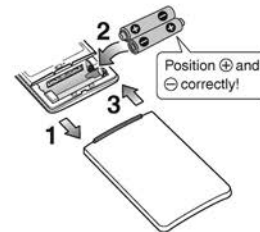
Incorrect handling of batteries can result in injury from battery leakage, rupturing or heating, or lead to equipment failure.

Please observe the following precautions and use safely.

- If the alkaline solution from the batteries should get in the eyes, do not rub the eyes. Instead, immediately flush the eyes with tap water and seek the attention of a medical professional.
- Keep batteries out of reach of children. In the event that batteries are swallowed, seek the immediate attention of a medical professional.
- Do not expose batteries to heat or fire. Do not disassemble or modify batteries. The insulation or gas release vent inside the battery may be damaged, resulting in battery leakage, rupturing, or heating.
- Do not damage or peel off labels on the batteries.

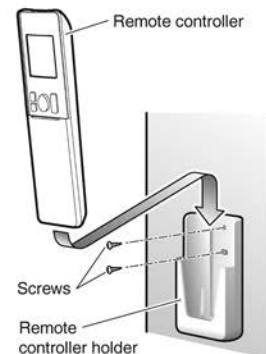
■ To insert the batteries

1. Slide the front cover to take it off.
2. Insert 2 dry batteries AAA.LR03 (alkaline).
3. Replace the front cover.



■ To fix the remote controller holder to a wall

1. Choose a place from where the signals reach the unit.
2. Fix the holder to a wall, a pillar, or similar location with the screws supplied with the holder.
3. Place the remote controller in the remote controller holder.



■ Turn on the circuit breaker

- After the power is turned on, the flap of the indoor unit opens and closes once to set the reference position.

NOTE

Notes on batteries

- To avoid possible injury or damage from battery leakage or rupturing, remove the batteries when not using the product for long periods of time.
- The standard replacement time is 1 year. Both batteries should be replaced at the same time. Be sure to replace them with new size AAA.LR03 (alkaline) batteries.
- When battery power has run out, the LCD will begin blinking as an alert that the batteries need replacing. In some cases, when battery consumption is accelerated owing usage conditions, signal reception may decline before the LCD begins blinking.
- The batteries supplied with the remote controller are for initial operation. The batteries may run out in less than 1 year.

Note on remote controller

- Do not drop the remote controller. Do not get it wet.

Read Before Operation



■ To set the clock

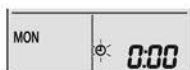
1. Press .



"0:00" appears on the LCD.
"MON" and "⌚" blink.

2. Press to set the current day of the week.

3. Press .



"⌚" blinks.

4. Press to set the clock to the present time.

- Holding down ▲ or ▼ rapidly increases or decreases the displayed time.

5. Press .

- Point the remote controller at the indoor unit when pressing the buttons.



"." blinks.

NOTE

Note on setting the clock

- If the indoor unit's internal clock is not set to the correct time, the ON/OFF TIMER and WEEKLY TIMER will not operate punctually.

Basic Operation



AUTO · DRY · COOL · HEAT · FAN Operation

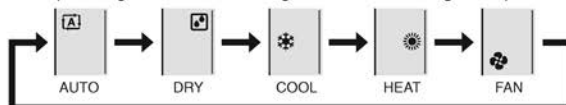


The air conditioner operates with the operation mode of your choice.
From the next time on, the air conditioner will operate with the same operation mode.

■ To start

1. Press and select an operation mode.

- Each pressing of the button changes the mode setting in sequence.



2. Press .

- "ON" appears on the LCD.
- The OPERATION lamp lights green.



Display

■ To stop

Press again.

- "ON" disappears from the LCD.
- The OPERATION lamp goes off.

■ To change the temperature setting

Press .

- Press ▲ to raise the temperature and press ▼ to lower the temperature.
Each pressing of the button changes the temperature by 0.5°C.

COOL operation	HEAT operation	AUTO operation	DRY or FAN operation
18.0-32.0°C	10.0-30.0°C	18.0-30.0°C	The temperature setting cannot be changed.

NOTE

Notes on AUTO operation

- In AUTO operation, the system selects an appropriate operation mode (COOL or HEAT) based on the indoor temperature and starts the operation.
- The system automatically reselects setting at a regular interval to bring the indoor temperature to the user-setting level.

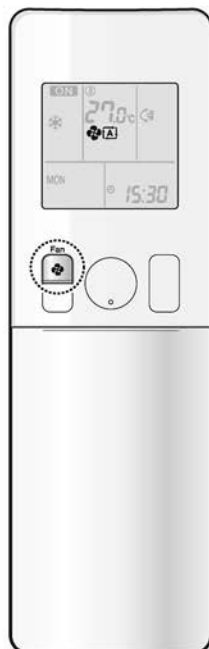
Note on DRY operation

- Eliminates humidity while maintaining the indoor temperature as much as possible. It automatically controls temperature and airflow rate, so manual adjustment of these functions is unavailable.

Basic Operation



Adjusting the Airflow Rate

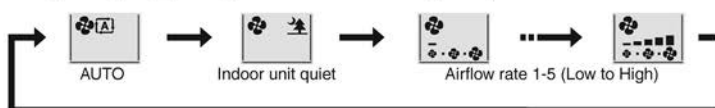


You can adjust the airflow rate to increase your comfort.

■ To adjust the airflow rate setting

▶ Press .

- Each pressing of  changes the airflow rate setting in sequence.



- When the airflow is set to "  ", quiet operation starts and noise from the indoor unit will become quieter.
- In the quiet operation mode, the airflow rate is set to a weak level.
- In DRY operation, the airflow rate setting cannot be changed.
- If COOL/DRY operation is started with airflow settings on "AUTO", the indoor unit will initially deodorise, which functions to suppress any internal odours. As a result, the indoor unit will not emit air immediately. (Wait for about 1 minute.)
- At smaller airflow rates, the cooling (heating) effect is also smaller.

Tips for saving energy

Keeping the temperature setting at a moderate level helps save energy.

- Recommended temperature setting for cooling: 26.0-28.0°C, for heating: 20.0-24.0°C

Cover windows with a blind or a curtain.

- Blocking sunlight and air from outdoors increases the cooling (heating) effect.

Keep the air filters clean.

- Clogged air filters cause inefficient operation and waste energy. Clean them once in about every 2 weeks. [▶ Page 28](#)

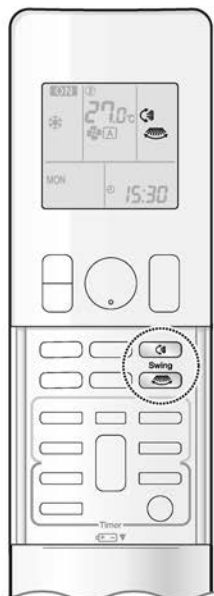
If you are not going to use the air conditioner for a long period, for example in spring or autumn, turn off the circuit breaker.

- The air conditioner always consumes a small amount of electricity even while it is not operating.

Basic Operation



Adjusting the Airflow Direction



You can adjust the airflow direction to increase your comfort.

⚠ CAUTION

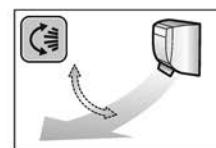
- Always use a remote controller to adjust the angles of the flap and louvres.
 - If you attempt to move the flap and louvres forcibly by hand when they are swinging, the mechanism may be damaged.
 - Inside the air outlet, a fan is rotating at a high speed.

■ To start auto swing

Up and down airflow direction

Press .

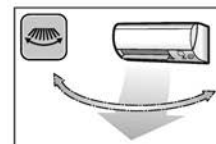
- “” appears on the LCD.
- The flap (horizontal blade) will begin to swing.



Right and left airflow direction

Press .

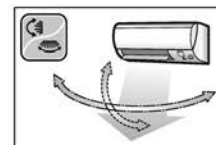
- “” appears on the LCD.
- The louvres (vertical blades) will begin to swing.



The 3-D airflow direction

Press  and .

- “” and “” appear on the LCD.
- The flap and louvres move in turn.
- To cancel 3-D airflow, press either  or  again. The flap or louvres will stop moving.



■ To set the flap or louvres at the desired position

- This function is effective while the flap or louvres are in auto swing mode.

Press  and  when the flap or louvres reach the desired position.

- In the 3-D airflow, the flap and louvres move in turn.
- “” or “” disappears from the LCD.

NOTE

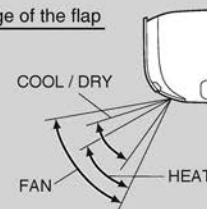
Notes on airflow direction setting

- The movable range of the flap varies according to the operation mode.

Note on 3-D airflow

- Using 3-D airflow circulates cold air, which tends to collect at the bottom of the room, and hot air, which tends to collect near the ceiling, throughout the room, preventing areas of cold and hot developing.

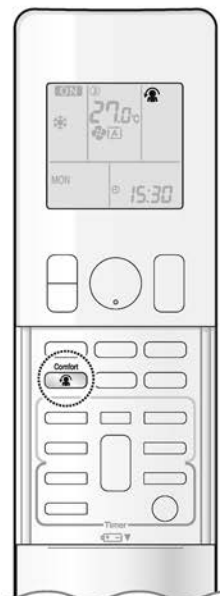
Movable range of the flap



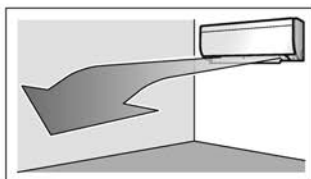
Useful Functions



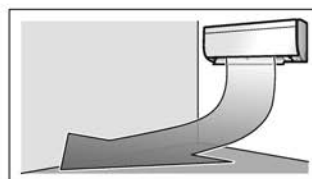
COMFORT AIRFLOW Operation



The flow of air will be in the upward direction while in COOL operation and in the downward direction while in HEAT operation, which will provide a comfortable wind that will not come in direct contact with people.



COOL operation



HEAT operation

■ To start

Press  .

- “” appears on the LCD.

	COOL and DRY operation	HEAT operation	FAN ONLY operation
Flap direction	Goes up	Goes down	Not available
Airflow rate	AUTO		Not available

■ To cancel

Press  again.

- “” disappears from the LCD.
- The flap will return to the memory position from before COMFORT AIRFLOW operation.

NOTE

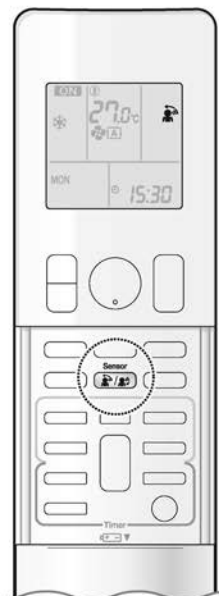
Notes on COMFORT AIRFLOW Operation

- The flap position will change, preventing air from blowing directly on the occupants of the room.
- POWERFUL operation and COMFORT AIRFLOW operation cannot be used at the same time.
- The airflow rate will be set to AUTO.
- If the upward and downward airflow direction is selected, the COMFORT AIRFLOW function will be canceled. Priority is given to the function of whichever button is pressed last.

Useful Functions



INTELLIGENT EYE Operation



The INTELLIGENT EYE sensor detects human movement and adjusts airflow direction. It can be set to avoid people or to direct airflow specifically at people. If no one is in the room for more than 20 minutes, the operation automatically changes to energy saving operation.

CAUTION

- Do not place large objects near the INTELLIGENT EYE sensor. Also keep heating units and humidifiers outside the sensor's detection area. This sensor can detect undesirable objects.
- Do not hit or violently push the INTELLIGENT EYE sensor. This can lead to damage and malfunction.

To start

Press  and select the desired mode.

- Each time  is pressed, a different setting option appears on the LCD.
- The INTELLIGENT EYE lamp lights green.



Display



- When the flap (horizontal blade) is swinging, selecting any of the modes above will cause the flap (horizontal blade) to stop.
- The lamp lights when human movement is detected.

To cancel

Press  until no icon is displayed.

- The INTELLIGENT EYE lamp goes off.

NOTE

Note on INTELLIGENT EYE operation

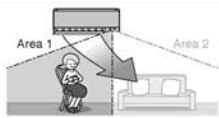
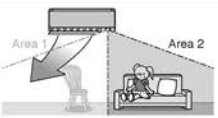
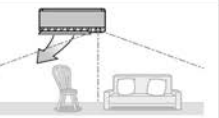
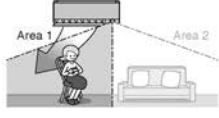
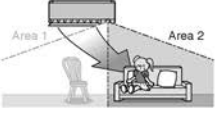
- INTELLIGENT EYE operation is set to blow directly on people, the air conditioner tracks you horizontally, without the air blowing directly on you.
- When the INTELLIGENT EYE sensor detects the movement of people, it adjusts the airflow direction upward (while in COOL operation) and downward (while in HEAT operation), by adjusting the flap. Depending on the setting, the louvres (vertical blades) will either direct airflow directly toward or away from people.

Useful Functions

■ How the INTELLIGENT EYE operation works

The INTELLIGENT EYE sensor detects human movement and adjusts the right and left airflow direction. If no one is in the room for more than 20 minutes, the operation automatically changes to energy saving operation. The INTELLIGENT EYE sensor works differently depending on the situation.

[Example]

	A person is detected in area 1.	A person is detected in area 2.	People are detected in both areas.	No people are detected in the areas.
				
			When the INTELLIGENT EYE operation and the COMFORT AIRFLOW operation are used together, the unit will not blow air directly at people (only in their general direction).	The air conditioner will switch to energy-saving mode after 20 minutes.

* The airflow direction may differ from the illustrated direction depending on the actions and movements of the people in the areas.

INTELLIGENT EYE operation is useful for energy saving

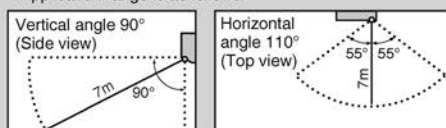
Energy saving operation

- If no presence detected in the room for 20 minutes, the energy saving operation will start, and the INTELLIGENT EYE lamp goes off. If human movement is detected again, the INTELLIGENT EYE lamp lights up and energy saving operation terminates.
- This operation changes the temperature by -2°C in HEAT / $+2^{\circ}\text{C}$ in COOL / $+2^{\circ}\text{C}$ in DRY operation from the set temperature. When the room temperature exceeds 30°C , the operation changes the temperature by $+1^{\circ}\text{C}$ in COOL / $+1^{\circ}\text{C}$ in DRY operation from the set temperature.
- This operation decreases the airflow rate slightly in FAN operation only.

NOTE

Notes on INTELLIGENT EYE operation

- Application range is as follows.



- While the air conditioner is in INTELLIGENT EYE operation, the louvers adjust the airflow direction if there are people in the sensing areas of the INTELLIGENT EYE so that the leftward or rightward airflow will not be directed to the people.
- If no people are detected in either area 1 or 2 for 20 minutes, the air conditioner switches to the energy-saving mode with the set temperature shifted by 2°C .
- The air conditioner may switch to the energy-saving operation even if there are people in the areas.
- This may occur depending on the clothes the people are wearing, if there is no movement of the people in the areas.
- The airflow direction from the louvers will be leftward if there are people in both areas 1 and 2. The air will also flow left if there is a person right in front of the sensor as the sensor judges that there are people in both areas.
- Due to the position of the sensor, people might be exposed to the airflow of the indoor unit if they are close to the front side of the indoor unit. If there are people close to the front side of the indoor unit or in both areas, it is recommended to use the COMFORT AIRFLOW and INTELLIGENT EYE operations simultaneously. Using both modes together, the air conditioner will not direct the airflow towards the people.
- The sensor could also mistakenly detect pets, sunlight, fluttering curtains and light reflected off of mirrors as passers-by.
- The sensor may not detect moving objects further than 7m away. (Please see the application range)
- Sensor detection sensitivity changes according to the indoor unit location, the speed of passers-by, temperature range, etc.
- The sensor could also mistakenly detect pets, sunlight, fluttering curtains and light reflected off of mirrors as passers-by.
- NIGHT SET mode [▶Page 20](#) will not switch on during use of INTELLIGENT EYE operation.

Useful Functions



POWERFUL Operation



POWERFUL operation quickly maximises the cooling (heating) effect in any operation mode. In this mode, the air conditioner operates at maximum capacity.

■ To start

▶ Press  during operation.

- “” appears on the LCD.
- POWERFUL operation ends in 20 minutes. Then the system automatically operates again with the previous settings which were used before POWERFUL operation.

■ To cancel

▶ Press  again.

- “” disappears from the LCD.

NOTE

Notes on POWERFUL operation

- Pressing  causes the settings to be cancelled, and “” disappears from the LCD.
- POWERFUL operation will not increase the capacity of the air conditioner if the air conditioner is already in operation with its maximum capacity demonstrated.
- In COOL, HEAT and AUTO operation
To maximise the cooling (heating) effect, the capacity of outdoor unit increases and the airflow rate becomes fixed at the maximum setting. The temperature and airflow settings cannot be changed.
- In DRY operation
The temperature setting is lowered by 2.5°C and the airflow rate is slightly increased.
- In FAN operation
The airflow rate is fixed at the maximum setting.

Regarding the combination of POWERFUL and other operations

POWERFUL + COMFORT AIRFLOW	Not available*
POWERFUL + ECONO	
POWERFUL + OUTDOOR UNIT QUIET	

*Priority is given to the function of whichever button is pressed last.

Useful Functions



ECONO / OUTDOOR UNIT QUIET Operation



ECONO operation enables efficient operation by limiting the maximum power consumption. This function is useful to prevent the circuit breaker from tripping when the unit operates alongside other appliances on the same circuit.

OUTDOOR UNIT QUIET operation lowers the noise level of the outdoor unit by changing the frequency and fan speed of the outdoor unit. This function is convenient during the night-time operation.

■ To start

➤ Press  and select the desired mode.

- Each time  is pressed, a different setting option appears on the LCD.



■ To cancel

➤ Press  until no icon is displayed.

NOTE

Notes on ECONO operation

- This operation is performed with lower power and therefore may not provide a sufficient cooling (heating) effect.
- Pressing  causes the settings to be cancelled, and "ECONO" disappears from the LCD.
- If the power consumption level is already low, switching to ECONO operation will not reduce the power consumption.

Notes on OUTDOOR UNIT QUIET operation

- Even if the operation is stopped by using the remote controller or the indoor unit ON/OFF switch when using OUTDOOR UNIT QUIET operation, "OUTDOOR UNIT QUIET" will remain displayed on the remote controller.
- OUTDOOR UNIT QUIET operation will not reduce the frequency nor fan speed if they already are operating at reduced levels.
- This operation is performed with lower power and therefore may not provide a sufficient cooling (heating) effect.

Possible combinations of ECONO / OUTDOOR UNIT QUIET operation and basic operations

	Operation mode			
	DRY	COOL	HEAT	FAN
ECONO	✓	✓	✓	—
OUTDOOR UNIT QUIET	—	✓	✓	—

TIMER Operation



ON/OFF TIMER Operation



Timer functions are useful for automatically switching the air conditioner on or off at night or in the morning. You can also use the ON TIMER and OFF TIMER together.

[ON TIMER with a 24 hour clock display] (24 HOUR ON/OFF TIMER)

■ To set

- Check that the clock is correct.
If not, set the clock to the present time. [▶Page 10](#)

1. Press .



"6:00" appears on the LCD.
"ON" blinks.

- "⌚" and day of the week disappear from the LCD.

2. Press until the time setting reaches the point you like.

- Each pressing of either button increases or decreases the time setting by 10 minutes.
Holding down either button changes the setting rapidly.

3. Press again.

- "ON" and setting time appear on the LCD.
- The TIMER lamp lights orange.



Display

■ To cancel

▶ Press .

- "ON" and setting time disappear from the LCD.
- "⌚" and day of the week appear on the LCD.
- The TIMER lamp goes off.

NOTE

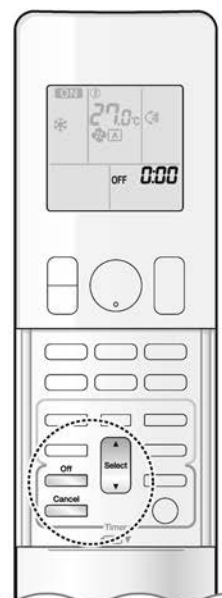
Notes on TIMER operation

- When TIMER is set, the present time is not displayed.
- When using the ON/OFF TIMER to start/stop operation, the actual operation start/stop time may differ from the time set.
(Maximum of about 10 minutes)

In the following cases, set the timer again.

- After the circuit breaker has turned off.
- After a power failure.
- After replacing the batteries in the remote controller.

TIMER Operation



[OFF TIMER with a 24 hour clock display] (24 HOUR ON/OFF TIMER)

■ To set

- Check that the clock is correct.
If not, set the clock to the present time. ▶ [Page 10](#)

1. Press .



"0:00" appears on the LCD.
"OFF" blinks.

- "⊕" and day of the week disappear from the LCD.

2. Press until the time setting reaches the point you like.

- Each pressing of either button increases or decreases the time setting by 10 minutes.
Holding down either button changes the time setting rapidly.

3. Press again.

- "OFF" and setting time appear on the LCD.
- The TIMER lamp lights orange.



Display

■ To cancel

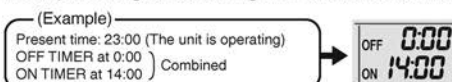
Press .

- "OFF" and setting time disappear from the LCD.
- "⊕" and day of the week appear on the LCD.
- The TIMER lamp goes off.

[ON/OFF TIMER combination]

■ To combine

- A sample setting for combining the 2 timers is shown below.



NOTE

NIGHT SET mode

- When the OFF TIMER is set, the air conditioner automatically adjusts the temperature setting (0.5°C up in COOL, 2.0°C down in HEAT) to prevent excessive cooling (heating) during sleeping hours.

TIMER Operation



ON/OFF TIMER Operation

[ON/OFF TIMER with an hours remaining display]
(COUNT UP-DOWN ON/OFF TIMER)

- Check that the clock is correct. If not, set the clock to the present time. [▶Page 10](#)

1. Press and hold or for about 5 seconds, when signal indicator (▲) blink, release your finger from the button.



For ON TIMER
"1hr" appears on the LCD.
"ON" blinks.



For OFF TIMER
"1hr" appears on the LCD.
"OFF" blinks.

- "⊙" and day of the week disappear from the LCD.

2. Press until the time setting reaches the point you like.

- Each pressing of the button increases or decreases the time setting by 1 hour.
- The time can be set between 1 and 12 hours.

3. Press or again.

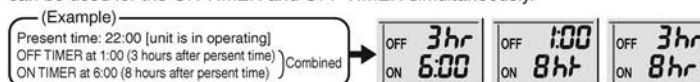
- "ON" and "OFF", and setting time appear on the LCD.
- The TIMER lamp lights orange.
- To change the display mode from COUNT UP-DOWN ON/OFF TIMER to 24 HOUR ON/OFF TIMER, repeat **STEP 1**, and continue from **STEP 2** of "24 HOUR ON/OFF TIMER". [▶Page 19, 20](#)
- Timer display mode cannot be changed while "ON" or "OFF" are blinking.
Press or to stop the blinking.



Display

■ Combining display modes

- Different display modes (24 HOUR ON/OFF TIMER and COUNT UP-DOWN ON/OFF TIMER) can be used for the ON TIMER and OFF TIMER simultaneously.



NOTE

Notes on COUNT UP-DOWN ON/OFF TIMER

- After setting, the time displayed on the LCD will automatically count down by 1 hour for every 1 hour that passes.
For example, if the OFF TIMER is set to 3 hours, the display will change as follows: 3hr → 2hr → 1hr → blank (OFF)

Switching between ON/OFF TIMER display modes (24 hour clock and hours remaining display)

- When changing timer display modes (from 24 HOUR ON/OFF TIMER to COUNT UP-DOWN ON/OFF TIMER or vice versa), the timer cancels. The timer must be set again to activate it.

TIMER Operation



WEEKLY TIMER Operation

Up to 4 timer settings can be saved for each day of the week. This is convenient to adapt the WEEKLY TIMER to your family's life style.

■ Setting example of the WEEKLY TIMER

The same timer settings are used from Monday through Friday, while different timer settings are used for the weekend.

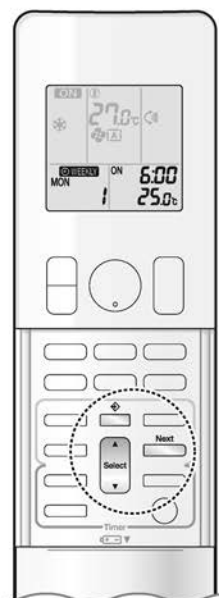
[Monday]	Make timer settings for programmes 1-4. ▶ Page 23 Programme 1 ON 25.0°C 6:00 Programme 2 OFF 8:30 Programme 3 ON 17:30 27.0°C Programme 4 OFF 22:00
[Tuesday] to [Friday]	Use the copy mode to make settings for Tuesday to Friday, because these settings are the same as those for Monday. ▶ Page 25 Programme 1 ON 25.0°C 6:00 Programme 2 OFF 8:30 Programme 3 ON 17:30 27.0°C Programme 4 OFF 22:00
[Saturday]	No timer settings
[Sunday]	Make timer settings for programmes 1-4. ▶ Page 23 Programme 1 ON 8:00 25.0°C Programme 2 OFF 10:00 Programme 3 OFF 19:00 27.0°C Programme 4 ON 21:00 27.0°C

- Up to 4 reservations per day and 28 reservations per week can be set using the WEEKLY TIMER. The effective use of the copy mode simplifies timer programming.
- The use of ON-ON-ON-ON settings, for example, makes it possible to schedule operating mode and set temperature changes. Furthermore, by using OFF-OFF-OFF-OFF settings, only the turn off time of each day can be set. This will turn off the air conditioner automatically if you forget to turn it off.

TIMER Operation



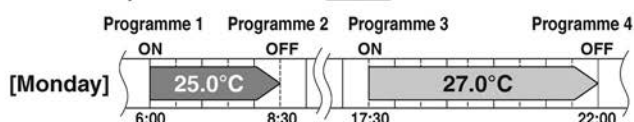
WEEKLY TIMER Operation



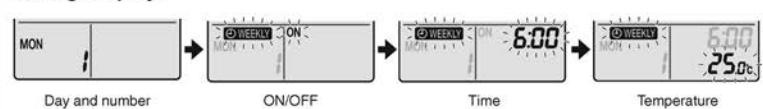
■ To use WEEKLY TIMER operation

Setting mode

- Make sure the day of the week and time are set.
If not, set the day of the week and time. ▶ **Page 10**



Setting Displays



1. Press .

- The day of the week and the reservation number of the current day will be displayed.
- 1 to 4 settings can be made per day.

2. Press to select the desired day of the week and reservation number.

- Pressing changes the reservation number and the day of the week.

3. Press .

- The day of the week and reservation number will be set.
- "⌚ WEEKLY" and "ON" blink.

4. Press to select the desired mode.

- Pressing changes the "ON" or "OFF" setting in sequence.

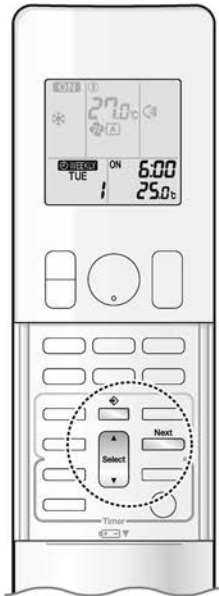


- In case the reservation has already been set, selecting "blank" deletes the reservation.
- Proceed to **STEP 9** if "blank" is selected.
- To return to the day of the week and reservation number setting, press .

5. Press .

- The ON/OFF TIMER mode will be set.
- "⌚ WEEKLY" and the time blink.

TIMER Operation



6. Press to select the desired time.

- The time can be set between 0:00 and 23:50 in 10-minute intervals.
- To return to the ON/OFF TIMER mode setting, press .
- Proceed to **STEP 9** when setting the OFF TIMER.

7. Press .

- The time will be set.
- "WEEKLY" and the temperature blink.

8. Press to select the desired temperature.

- The temperature can be set between 10.0°C and 32.0°C.
COOL or AUTO: The unit operates at 18.0°C even if it is set to 10.0°C to 17.0°C. [▶Page 11](#)
HEAT or AUTO: The unit operates at 30.0°C even if it is set to 31.0°C to 32.0°C. [▶Page 11](#)
- To return to the time setting, press .
- The set temperature is only displayed when the mode setting is on.

9. Press .

- The temperature will be set and go to the next reservation setting.
- The temperature is set while in ON TIMER operation, and the time is set while in OFF TIMER operation.
- The next reservation screen will appear.
- To continue further settings, repeat the procedure from **STEP 4**.

10. Press to complete the setting.

- Be sure to direct the remote controller toward the indoor unit and check for a receiving tone and blinking of the OPERATION lamp.
- "WEEKLY" appears on the LCD and WEEKLY TIMER operation is activated.
- The TIMER lamp lights orange.



Display

- A reservation made once can be easily copied and the same settings used for another day of the week. Refer to **Copy mode**. [▶Page 25](#)

NOTE

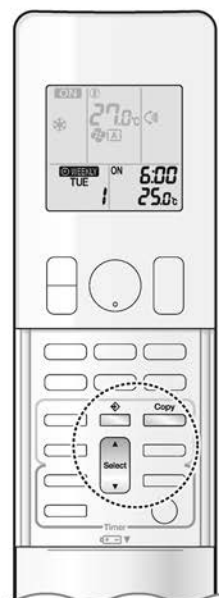
Notes on WEEKLY TIMER operation

- Do not forget to set the clock on the remote controller first. [▶Page 10](#)
- The day of the week, ON/OFF TIMER mode, time and set temperature (only for ON TIMER mode) can be set with the WEEKLY TIMER. Other settings for the ON TIMER are based on the settings just before the operation.
- WEEKLY TIMER and ON/OFF TIMER operation cannot be used at the same time. The ON/OFF TIMER operation has priority if it is set while WEEKLY TIMER is still active. The WEEKLY TIMER will enter the standby state, and "WEEKLY" will disappear from the LCD. When the ON/OFF TIMER is up, the WEEKLY TIMER will automatically become active.
- Only the time and set temperature with the WEEKLY TIMER are sent with the . Set the WEEKLY TIMER only after setting the operation mode, the airflow rate and the airflow direction ahead of time.
- Turning off the circuit breaker, power failure, and other similar events will render operation of the indoor unit's internal clock inaccurate. Reset the clock. [▶Page 10](#)
-  can be used only for the time and temperature settings. It cannot be used to go back to the reservation number.

TIMER Operation

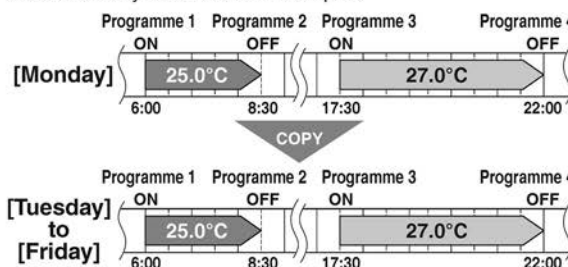


WEEKLY TIMER Operation

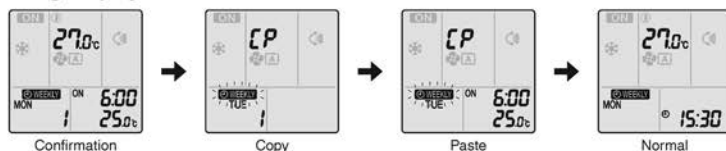


Copy mode

- A reservation made once can be copied to another day of the week. The whole reservation of the selected day of the week will be copied.



Setting Displays



1. Press .

2. Press to confirm the day of the week to be copied.

3. Press .

- The whole reservation of the selected day of the week will be copied.

4. Press to select the destination day of the week.

5. Press .

- The reservation will be copied to the selected day of the week. The whole reservation of the selected day of the week will be copied.
- To continue copying the settings to other days of the week, repeat **STEP 4** and **STEP 5**.

6. Press to complete the setting.

- " WEEKLY" appears on the LCD and WEEKLY TIMER operation is activated.

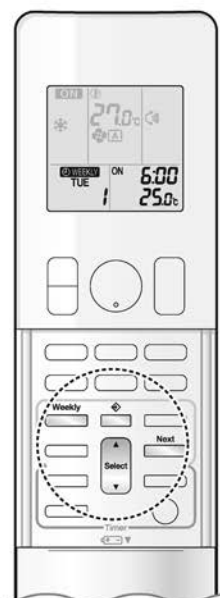
NOTE

Note on COPY MODE

- The entire reservation of the source day of the week is copied in the copy mode.

In the case of making a reservation change for any day of the week individually after copying the content of weekly reservations, press and change the settings in the steps of **Setting mode**. ▶ **Page 23**

TIMER Operation



■ Confirming a reservation

- The reservation can be confirmed.



1. Press .

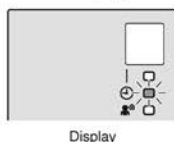
- The day of the week and the reservation number of the current day will be displayed.

2. Press to select the day of the week and the reservation number to be confirmed.

- Pressing  displays the reservation details.
- To change the confirmed reserved settings, select the reservation number and press . The mode is switched to setting mode. Proceed to **Setting mode STEP 4.** [▶Page 23](#)

3. Press to exit the confirmation mode.

- “ WEEKLY” appears on the LCD and WEEKLY TIMER operation is activated.
- The TIMER lamp lights orange.



■ To deactivate WEEKLY TIMER operation

▶ Press while “ WEEKLY” is displayed on the LCD.

- “ WEEKLY” disappears from the LCD.
- The TIMER lamp goes off.
- To reactivate the WEEKLY TIMER operation, press  again.
- If a reservation deactivated with  is activated once again, the last reservation mode will be used.

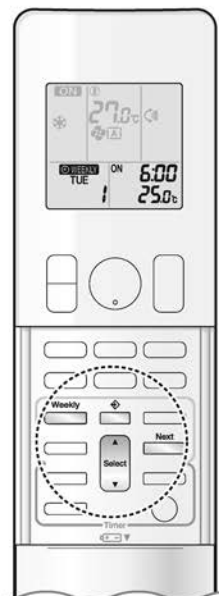
NOTE

- If not all the reservation settings are reflected, deactivate the WEEKLY TIMER operation once. Then press  again to reactivate the WEEKLY TIMER operation.

TIMER Operation



WEEKLY TIMER Operation



■ To delete a reservation

An individual reservation

1. Press .

- The day of the week and the reservation number will be displayed.

2. Press to select the day of the week and the reservation number to be deleted.

3. Press .

- "WEEKLY" and "ON" or "OFF" blink.

4. Press until no icon is displayed.

- Pressing changes the ON/OFF TIMER mode in sequence.
- Selecting "blank" will cancel any reservation you may have.



5. Press .

- The selected reservation will be deleted.

6. Press .

- If there are still other reservations, WEEKLY TIMER operation will be activated.

Reservations for each day of the week

- This function can be used for deleting reservations for each day of the week.
- It can be used while confirming or setting reservations.

1. Press .

- The day of the week and the reservation number will be displayed.

2. Press to select the day of the week to be deleted.

3. Hold for about 5 seconds.

- The reservation of the selected day of the week will be deleted.

4. Press .

- If there are still other reservations, WEEKLY TIMER operation will be activated.

All reservations

■ Hold for about 5 seconds with the normal display.

- Be sure to direct the remote controller toward the indoor unit and check for a receiving tone.
- This operation cannot be used for the WEEKLY TIMER setting display.
- All reservations will be deleted.

Care

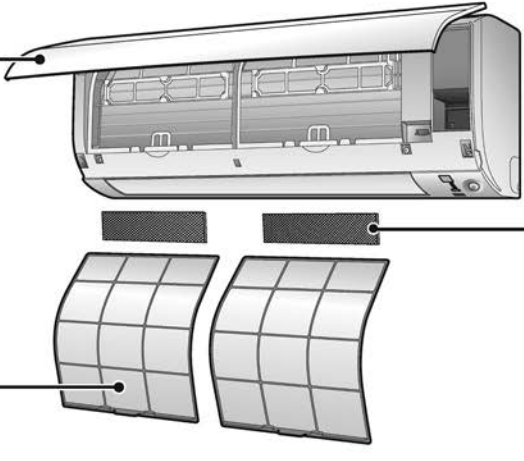
Care and Cleaning

⚠ CAUTION

- Before cleaning, be sure to stop the operation and turn off the circuit breaker.
- Do not touch the aluminum fins of the indoor unit. If you touch those parts, this may cause an injury.

■ Quick reference

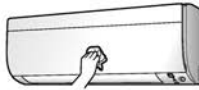
Cleaning parts



Front panel

- Wipe it with a soft damp cloth.
- Only neutral detergent may be used.

If dirty



Air filter

- Vacuum dust or wash the filter.

Once every 2 weeks

▶ Page 29

Indoor unit, outdoor unit and remote controller

- Wipe them with a soft cloth.

If dirty

Titanium apatite deodorizing filter (included with the air conditioner) and dust collection filter (PM2.5) (sold separately)

To purchase, contact the dealer where you bought the air conditioner.

Item	Part No.
Titanium apatite deodorizing filter - without frame (set of 2)	KAF970A46
Dust collection filter (PM2.5) - without frame (set of 2)	BAFP046A41

Titanium apatite deodorizing filter

- Vacuum dust or replace the filter.

[Cleaning]

Once every 6 months

▶ Page 30

[Replacement]

Once every 3 years

▶ Page 30

Dust collection filter (PM2.5)

- No cleaning is required. Please replace the filter every 6 months. ▶ Page 30

Notes on cleaning

For cleaning, do not use any of the following:

- Water hotter than 40°C
- Volatile liquid such as benzene, petrol and thinner
- Polishing compounds
- Rough materials such as a scrubbing brush



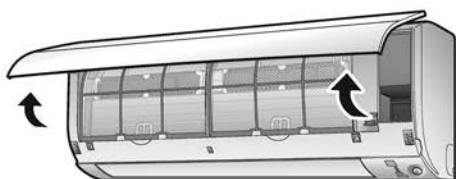
Care

Care and Cleaning

■ Air filter

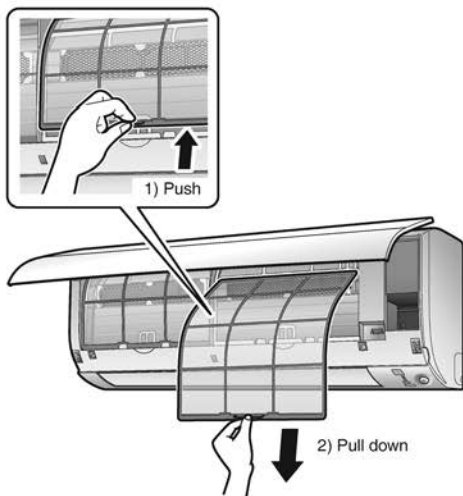
1. Open the front panel.

- Hold the front panel by the sides and open it.



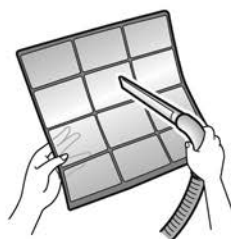
2. Pull out the air filters.

- Push the filter tab at the centre of each air filter a little upwards, then pull it down.



3. Wash the air filters with water or clean them with vacuum cleaner.

- It is recommended to clean the air filters every 2 weeks.



If the dust does not come off easily

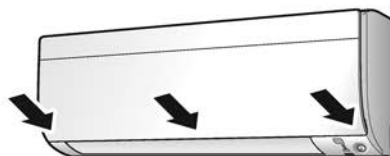
- Wash the air filters with neutral detergent thinned with lukewarm water, then dry them up in the shade.



4. Reattach the filters.

5. Close the front panel slowly.

- Press the panel at both sides and the centre.



- Make sure that the front panel is securely fixed.

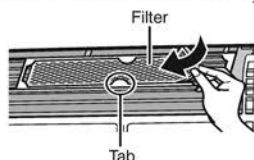
Care

■ Titanium apatite deodorizing filter (included with the air conditioner) / dust collection filter (PM2.5) (sold separately)

Attaching

1. Open the front panel and pull out the air filters. **▶ Page 29**
2. Insert the titanium apatite deodorizing filters or dust collection filters (PM2.5).

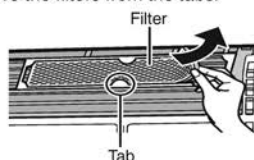
- Check that the filters are inserted securely under the tabs.



Cleaning and Replacement

1. Open the front panel and pull out the air filters. **▶ Page 29**
2. Take off the titanium apatite deodorizing filters or dust collection filters (PM2.5).

- Remove the filters from the tabs.



3. Clean or replace the titanium apatite deodorizing filters or dust collection filters (PM2.5)

[Cleaning for titanium apatite deodorizing filter (Blue-Black)]

- 3-1 Vacuum dust, and soak in lukewarm water or water for about 10 to 15 minutes if very dirty.

- 3-2 After washing, shake off remaining water and let them dry in the shade.

- This filter can be renewed by washing it with water once **every 6 months** and we recommend replacing it once **every 3 years**.
- Do not wring out the filter when removing water from it.

[Cleaning for dust collection filter (PM2.5) (White)]

- Do not wash the filters as this will reduce their efficiency.
- It is recommended to replace it once **every 6 months**.

[Replacement]

Remove each filter from the front grille by detaching them from the 6 tabs and then install the new filters.

- When attaching the filter, check that the filter is properly set in the tabs.

4. Set the filters as they were and close the front panel.

- Press the front panel at both sides and the center.

NOTE

- Operation with dirty filters:
 - cannot deodorise the air,
 - cannot clean the air,
 - results in poor cooling,
 - may cause odour.
- Dispose of old filters as non-flammable waste.

Care

Care and Cleaning

■ Prior to a long period of non-use

1. Operate the FAN mode for several hours on a fine day to dry out the inside.

- 1) Press  and select "  ".
- 2) Press  and start the operation.

2. After operation stops, turn off the circuit breaker for the room air conditioner.

3. Clean the air filters and reattach them.

4. To prevent battery leakage, take out the batteries from the remote controller.

■ We recommend periodical maintenance

- In certain operating conditions, the inside of the air conditioner may get foul after several seasons of use, resulting in poor performance. It is recommended to have periodical maintenance by a specialist in addition to regular cleaning by the user.
- For specialist maintenance, please contact the dealer where you bought the air conditioner.
- The maintenance cost must be borne by the user.

When the Need Arises

FAQ

Indoor unit

The flap does not start swinging immediately.

- The air conditioner is adjusting the position of the flap. The flap will start moving soon.

The air conditioner stops generating airflow during HEAT operation.

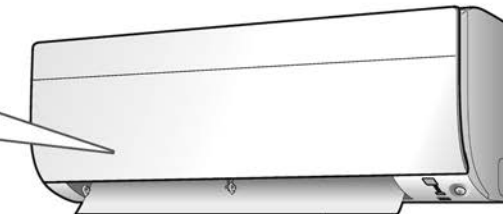
- Once the set temperature is reached, the airflow rate is reduced and operation stopped in order to avoid generating a cool airflow. Operation will resume automatically when the indoor temperature falls.

HEAT operation stops suddenly and a flowing sound is heard.

- The outdoor unit is defrosting. HEAT operation starts after the frost on the outdoor unit has been removed. This can take about 4 to 12 minutes.

Operation does not start soon.

- When  was pressed soon after operation was stopped.
- When the mode was reselected.
 - This is to protect the air conditioner. You should wait for about 3 minutes.

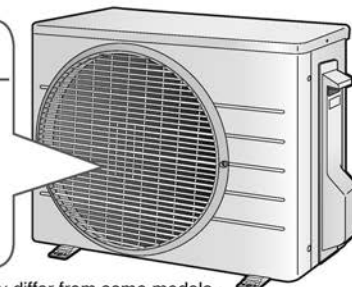
**Different sounds are heard.**

- **A sound like flowing water**
 - This sound is generated because the refrigerant in the air conditioner is flowing.
 - This is a pumping sound of the water in the air conditioner and can be heard when the water is pumped out from the air conditioner during COOL or DRY operation.
- **Blowing sound**
 - This sound is generated when the flow of the refrigerant in the air conditioner is switched over.
- **Ticking sound**
 - This sound is generated when the cabinet and frame of the air conditioner slightly expand or shrink as a result of temperature changes.
- **Whistling sound**
 - This sound is generated when refrigerant flows during defrosting operation.
- **Clicking sound during operation or idle time**
 - This sound is generated when the refrigerant control valves or the electrical parts operate.
- **Clopping sound**
 - This sound is heard from the inside of the air conditioner when the exhaust fan is activated while the room doors are closed. Open the window or turn off the exhaust fan.

Outdoor unit

The outdoor unit emits water or steam.

- **In HEAT operation**
 - The frost on the outdoor unit melts into water or steam when the air conditioner is in defrosting operation.
- **In COOL or DRY operation**
 - Moisture in the air condenses into water on the cool surface of the outdoor unit piping and drips.



* The appearance of the outdoor unit may differ from some models.

When the Need Arises

Troubleshooting

Before making an inquiry or a request for repair, please check the following.
If the problem persists, consult your dealer.



Not a problem

This case is not a problem.



Check

Please check again before requesting repairs.

The air conditioner does not operate

Case	Description / what to check
OPERATION lamp is off.	- Has the circuit breaker been tripped or the fuse blown? - Is there a power failure? - Are batteries set in the remote controller?
OPERATION lamp is blinking.	Turn off the power with the circuit breaker and restart operation with the remote controller. If the OPERATION lamp is still blinking, check the error code and consult your dealer. ▶ Page 36

The air conditioner suddenly stops operating

Case	Description / what to check
OPERATION lamp is on.	To protect the system, the air conditioner may stop operating after sudden large voltage fluctuations. It automatically resumes operation in about 3 minutes.
OPERATION lamp is blinking.	Is there anything blocking the air inlet or air outlet of the indoor unit or outdoor unit? Stop operation and after turning off the circuit breaker, remove the obstruction. Then restart operation with the remote controller. If the OPERATION lamp is still blinking, check the error code and consult your dealer. ▶ Page 36

The air conditioner does not stop operating

Case	Description / what to check
The air conditioner continues operating even after operation is stopped.	Immediately after the air conditioner is stopped - The outdoor unit fan continues rotating for about another 1 minute to protect the system. While the air conditioner is not in operation - When the outdoor temperature is high, the outdoor unit fan may start rotating to protect the system.

The room does not cool down/ warm up

Case	Description / what to check
Air does not come out.	In HEAT operation - The air conditioner is warming up. Wait for about 1 to 4 minutes. - During defrosting operation, hot air does not flow out of the indoor unit. In COOL or DRY operation - If COOL/DRY operation is started with airflow settings on "AUTO", the indoor unit will initially deodorise, which functions to suppress any internal odours. As a result, the indoor unit will not emit air immediately. (Wait for about 1 minute.) When the air conditioner operates immediately after the circuit breaker is turned on - The air conditioner is preparing to operate. Wait for about 3 to 15 minutes.
Air does not come out / Air comes out.	Is the airflow rate setting appropriate? - Is the airflow rate setting low, such as "Indoor unit quiet" or "Airflow rate 1"? Increase the airflow rate setting. Is the set temperature appropriate? Is the adjustment of the airflow direction appropriate?
Air comes out.	- Is there any furniture directly under or beside the indoor unit? - Is the air conditioner in ECONO operation or OUTDOOR UNIT QUIET operation? ▶ Page 18 - Are the air filters dirty? - Is there anything blocking the air inlet or air outlet of the indoor unit or outdoor unit? - Is a window or door open? - Is an exhaust fan turning?

When the Need Arises

Mist comes out

Case	Description / what to check
Mist comes out of the indoor unit.	☒ • This happens when the air in the room is cooled into mist by the cold airflow during COOL or other operation.

Remote controller

Case	Description / what to check
The unit does not receive signals from the remote controller or has a limited operating range.	☒ • The batteries may be exhausted. Replace both batteries with new dry batteries AAA.LR03 (alkaline). For details, refer to "Preparation Before Operation". ▶Page 9 ☒ • Signal communication may be disabled if an electronic-starter-type fluorescent lamp (such as inverter-type lamps) is in the room. Consult your dealer if that is the case. ☒ • The remote controller may not function correctly if the transmitter is exposed to direct sunlight.
LCD is faint, is not working, or the display is erratic.	☒ • The batteries may be exhausted. Replace both batteries with new dry batteries AAA.LR03 (alkaline). For details, refer to "Preparation Before Operation". ▶Page 9
The LCD is blinking and the remote controller cannot be operated.	☒ • Battery power has run out. Replace both batteries at the same time with new size AAA.LR03 (alkaline) batteries. Leaving exhausted batteries in the remote controller can result in injury due to battery leakage, rupturing or heating, or lead to equipment failure. (Even when the LCD is blinking, the OFF button remains functional.)
Other electric devices start operating.	☒ • If the remote controller activates other electric devices, move them away or consult your dealer.

Air has an odour

Case	Description / what to check
The air conditioner gives off an odour.	☒ • The room odour absorbed in the unit is discharged with the airflow. We recommend you to have the indoor unit cleaned. Please consult your dealer.

Others

Case	Description / what to check
The air conditioner suddenly starts behaving strangely during operation.	☒ • The air conditioner may malfunction due to lightning or radio. If the air conditioner malfunctions, turn off the power with the circuit breaker and restart the operation with the remote controller.
The ON/OFF TIMER does not operate according to the settings.	☒ • Check if the ON/OFF TIMER and the WEEKLY TIMER are set to the same time. Change or deactivate the settings in the WEEKLY TIMER. ▶Page 22

NOTE

Notes on the operating conditions

- If operation continues under any conditions other than those listed in the table,
 - A safety device may activate to stop the operation.
 - Dew may form on the indoor unit and drip from it when COOL or DRY operation is selected.

Mode	Operating conditions
COOL / DRY	Outdoor temperature: -10.0-46.0°C Indoor temperature: 18.0-32.0°C Indoor humidity: 80% max.
HEAT	Outdoor temperature: -15.0-24.0°C Indoor temperature: 10.0-30.0°C

When the Need Arises

Troubleshooting

■ Call your dealer immediately



WARNING

When an abnormality (such as a burning smell) occurs, stop operation and turn off the circuit breaker.

- Continued operation in an abnormal condition may result in problems, electric shock or fire.
- Consult the dealer where you bought the air conditioner.

Do not attempt to repair or modify the air conditioner by yourself.

- Incorrect work may result in electric shock or fire.
- Consult the dealer where you bought the air conditioner.

If one of the following symptoms takes place, call your dealer immediately.

- The power cord is abnormally hot or damaged.
- An abnormal sound is heard during operation.
- The circuit breaker, a fuse, or the earth leakage circuit breaker cuts off the operation frequently.
- A switch or a button often fails to work properly.
- There is a burning smell.
- Water leaks from the indoor unit.

Turn off the circuit breaker and call your dealer.



■ After a power failure

- The air conditioner automatically resumes operation in about 3 minutes. Please wait for a while.

■ Lightning

- If there is a risk lightning could strike in the neighbourhood, stop operation and turn off the circuit breaker to protect the system.

■ Disposal requirements

- Dismantling of the unit, handling of the refrigerant, oil and other parts, should be done in accordance with the relevant local and national regulations.

When the Need Arises



■ Fault diagnosis by remote controller

- In case the air conditioner is malfunctioning, you can check the error code via the remote controller before consulting with service personnel and informing them of the malfunction code.

1. When  is held down for about 5 seconds, “00” blinks in the temperature display section.

2. Press  repeatedly until a long beep is produced.

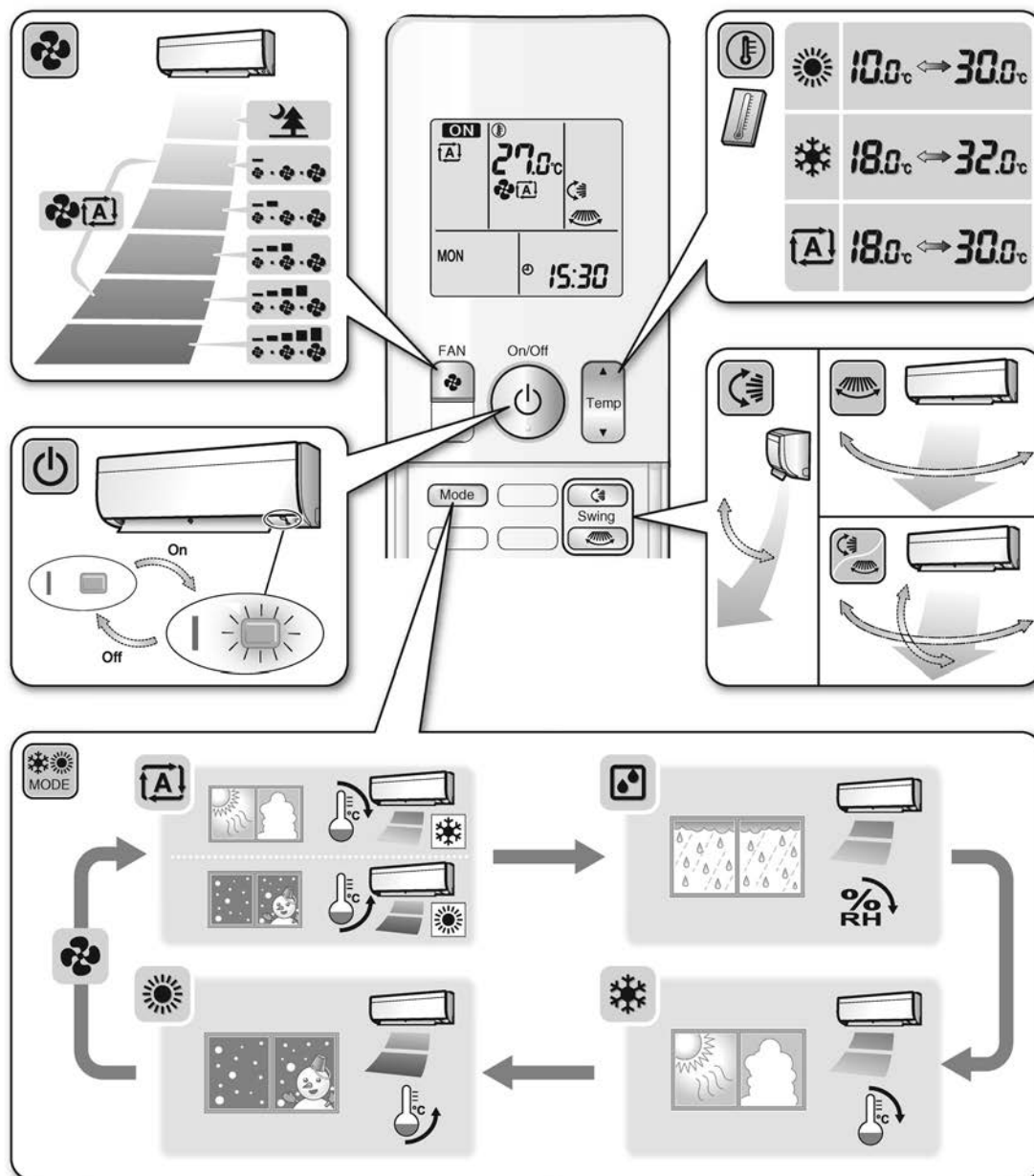
- The code indication changes as shown below, and notifies you with a long beep.

	CODE	MEANING
SYSTEM	00	NORMAL
	UA	INDOOR-OUTDOOR UNIT COMBINATION FAULT
	U0	REFRIGERANT SHORTAGE
	U2	DROP VOLTAGE OR MAIN CIRCUIT OVERVOLTAGE
	U4	FAILURE OF TRANSMISSION (BETWEEN INDOOR UNIT AND OUTDOOR UNIT)
INDOOR UNIT	A1	INDOOR PCB DEFECTIVENESS
	A5	HIGH PRESSURE CONTROL OR FREEZE-UP PROTECTOR
	A6	FAN MOTOR FAULT
	C4	FAULTY HEAT EXCHANGER TEMPERATURE SENSOR
	C9	FAULTY SUCTION AIR TEMPERATURE SENSOR
OUTDOOR UNIT	EA	COOLING-HEATING SWITCHING ERROR
	E1	CIRCUIT BOARD FAULT
	E5	OL STARTED
	E6	FAULTY COMPRESSOR START UP
	E7	DC FAN MOTOR FAULT
	E8	OVERCURRENT INPUT
	F3	HIGH TEMPERATURE DISCHARGE PIPE CONTROL
	F6	HIGH PRESSURE CONTROL (IN COOLING)
	H0	SENSOR FAULT
	H6	OPERATION HALT DUE TO FAULTY POSITION DETECTION SENSOR
	H8	DC CURRENT SENSOR FAULT
	H9	FAULTY SUCTION AIR TEMPERATURE SENSOR
	J3	FAULTY DISCHARGE PIPE TEMPERATURE SENSOR
	J6	FAULTY HEAT EXCHANGER TEMPERATURE SENSOR
	L3	ELECTRICAL PARTS HEAT FAULT
	L4	HIGH TEMPERATURE AT INVERTER CIRCUIT HEATSINK
	L5	OUTPUT OVERCURRENT
	P4	FAULTY INVERTER CIRCUIT HEATSINK TEMPERATURE SENSOR

NOTE

- A short beep indicate non-corresponding codes.
- To cancel the code display, hold  down for about 5 seconds. The code display also clears if no button is pressed for 1 minute.

Quick Reference



12.2 50/60/71 Class

Read Before Operation

Safety Precautions

	Read the precautions in this manual carefully before operating the unit.		This appliance is filled with R32.
--	--	--	------------------------------------

- Keep this manual where the user can easily find it.
- The precautions described herein are classified as WARNING and CAUTION. They both contain important information regarding safety. Be sure to observe all precautions without fail.

	WARNING	Failure to follow these instructions properly may result in personal injury or loss of life.
	CAUTION	Failure to observe these instructions properly may result in property damage or personal injury, which may be serious depending on the circumstances.

	Never attempt.		Be sure to follow the instructions.		Be sure to establish an earth connection.
--	----------------	--	-------------------------------------	--	---

- After reading, keep this manual in a convenient place so that you can refer to it whenever necessary. If the equipment is transferred to a new user, be sure also to hand over the manual.

WARNING

- Do not use means to accelerate the defrosting process or to clean, other than those recommended by the manufacturer.
- The appliance must be stored in a room without continuously operating ignition sources (for example: open flames, an operating gas appliance or an operating electric heater).
- Do not pierce or burn.
- Be aware that refrigerants may not contain an odour.
- The appliance must be installed, operated and stored in a room with floor area larger than 3.5m² for 50/60 class, 3.9m² for 71 class. Maintain an installation height of 1.8m or more from the floor surface to the bottom of the appliance. If the size of the room where the appliance is to be installed has less than the indicated minimum floor area, install it in a well ventilated room. For details refer to the WARNING list in the "Safety Precautions" section of the installation manual for the outdoor unit.
- To avoid fire, explosion or injury, do not operate the unit when harmful gases (e.g. flammable or corrosive) are detected near the unit.
- Be aware that prolonged, direct exposure to cool or warm air from the air conditioner, or to air that is too cool or too warm, can be harmful to your physical condition and health.
- Do not place objects, including rods, your fingers, etc., in the air inlet or outlet. Product damage or personal injury may result due to contact with the air conditioner's high-speed fan blades.
- Do not attempt to repair, dismantle, reinstall or modify the air conditioner yourself as this may result in water leakage, electric shock or fire hazards.
- Do not use flammable spray near the air conditioner, or otherwise fire may result.
- Do not use a refrigerant other than the one indicated on the outdoor unit (R32) when installing, moving or repairing. Using other refrigerants may cause trouble or damage to the unit, and personal injury.
- To avoid electric shock, do not operate with wet hands.

- Beware of fire in case of refrigerant leakage. If the air conditioner is not operating correctly, i.e. not generating cool or warm air, refrigerant leakage could be the cause. Consult your dealer for assistance. The refrigerant within the air conditioner is safe and normally does not leak. However, in the event of a leakage, contact with a naked burner, heater or cooker may result in generation of noxious gas. Do not use the air conditioner until a qualified service person confirms that the leakage has been repaired.
- Do not attempt to install or repair the air conditioner yourself. Improper workmanship may result in water leakage, electric shock or fire hazards. Please contact your local dealer or qualified personnel for installation and maintenance work.
- If the air conditioner is malfunctioning (giving off a burning odour, etc.), turn off power to the unit and contact your local dealer. Continued operation under such circumstances may result in a failure, electric shock or fire hazards.
- Be sure to install an earth leakage circuit breaker. Failure to install an earth leakage circuit breaker may result in electric shock or fire.

- Be sure to earth the unit. Do not earth the unit to a utility pipe, lightning conductor or telephone earth lead. Imperfect earthing may result in electric shock.

CAUTION

- Do not use the air conditioner for purposes other than those for which it is intended. Do not use the air conditioner for cooling precision instruments, food, plants, animals or works of art as this may adversely affect the performance, quality and/or longevity of the object concerned.
- Do not expose plants or animals directly to the airflow from the unit as this may cause adverse effects.
- Do not place appliances that produce naked flames in places exposed to the airflow from the unit as this may impair combustion of the burner.
- Do not block the air inlets nor outlets. Impaired airflow may result in insufficient performance or trouble.

Read Before Operation



- Do not sit on the outdoor unit, put things on the unit, or pull the unit. Doing so may cause accidents, such as falling or toppling down, thus resulting in injury, product malfunctioning, or product damage.
- Do not place objects that are susceptible to moisture directly beneath the indoor or outdoor units. Under certain conditions, condensation on the main unit or refrigerant pipes, air filter dirt or drain blockage may cause dripping, resulting in fouling or failure of the object concerned.
- After prolonged use, check the unit stand and its mounts for damage. If they are left in a damaged condition, the unit may fall and cause injury.
- To avoid injury, do not touch the air inlet or aluminium fins of the indoor or outdoor units.
- The appliance is not intended for use by unattended young children or infirm persons. Impairment of bodily functions and harm to health may result.
- Children should be supervised to ensure that they do not play with the unit or its remote controller. Accidental operation by a child may result in impairment of bodily functions and harm health.
- Avoid impacts to the indoor and outdoor units, or otherwise product damage may result.
- Do not place flammable items, such as spray cans, within 1m of the air outlet.
The spray cans may explode as a result of hot air from the indoor or outdoor units.
- Be careful not to let pets urinate on the air conditioner. Urination on the air conditioner may result in electric shock or fire.
- Do not wash the air conditioner with water, as this may result in electric shock or fire.
- Do not place water containers (vases, etc.) above the unit, as this may result in electric shock or fire hazards if they should topple over.
- Do not insert the batteries in the wrong polarity (+/-) orientation as this may result in short circuiting, fire, or battery leakage.
- Do not sit or hang on the panel. The panel may fall, and injury or product malfunctioning may result.
- Do not sway the panel. The panel may hit people or objects, and injury or property damage may result.
- Do not let children play around the panel. Injury or property damage may result.
- Do not pull the wires. The wires may be broken and the panel may fall, and injury or property damage may result.
- Do not locate obstacles in the route. The panel may fall, and injury or property damage may result.
- Do not bend or damage the wires. The wires may be broken and the panel may fall, and injury or property damage may result.
- Do not put objects on the panel, or otherwise production malfunctioning may result.
- Do not use an unstable stand at the time of operating or maintaining the air conditioner, or otherwise you may topple over or injury yourself.
- Locate the remote controller in places out of reach of children. The wrong operation of the remote controller may result in injury.



- To avoid oxygen depletion, ensure that the room is adequately ventilated if equipment such as a burner is used together with the air conditioner.
- Before cleaning, be sure to stop unit operation and turn off the circuit breaker. Otherwise, an electric shock and injury may result.
- Only connect the air conditioner to the specified power supply circuit. Power supplies other than the one specified may result in electric shock, overheating and fires.
- Arrange the drain hose to ensure smooth drainage. Imperfect drainage may cause wetting of the building, furniture, etc.
- Do not place objects in direct proximity of the outdoor unit and do not let leaves and other debris accumulate around the unit. Leaves are a hotbed for small animals which can enter the unit. Once inside the unit, such animals can cause malfunctions, smoke or fire if they come into contact with electrical parts.
- Do not place objects around the indoor unit.
Doing so may have an adverse influence on the performance, product quality, and life of the air conditioner.
- This appliance is not intended to be used by persons with reduced physical, sensory or mental capabilities, or with lack of operation knowledge, unless they have been given supervision or instruction concerning the appliance use by person responsible for their safety. Keep out of children's reach to ensure that they do not play with the appliance.

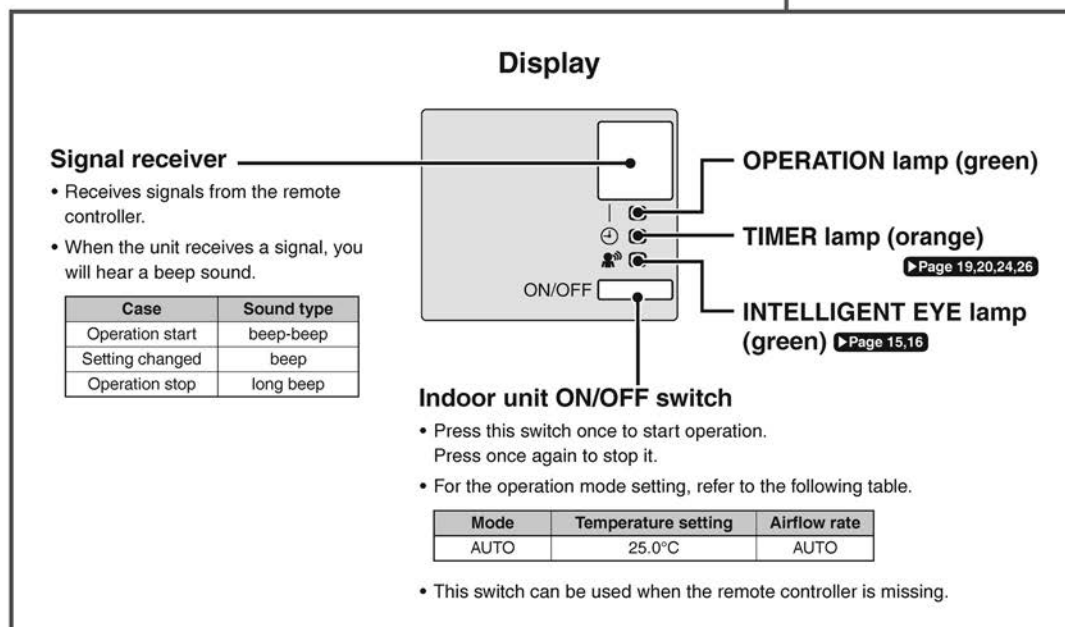
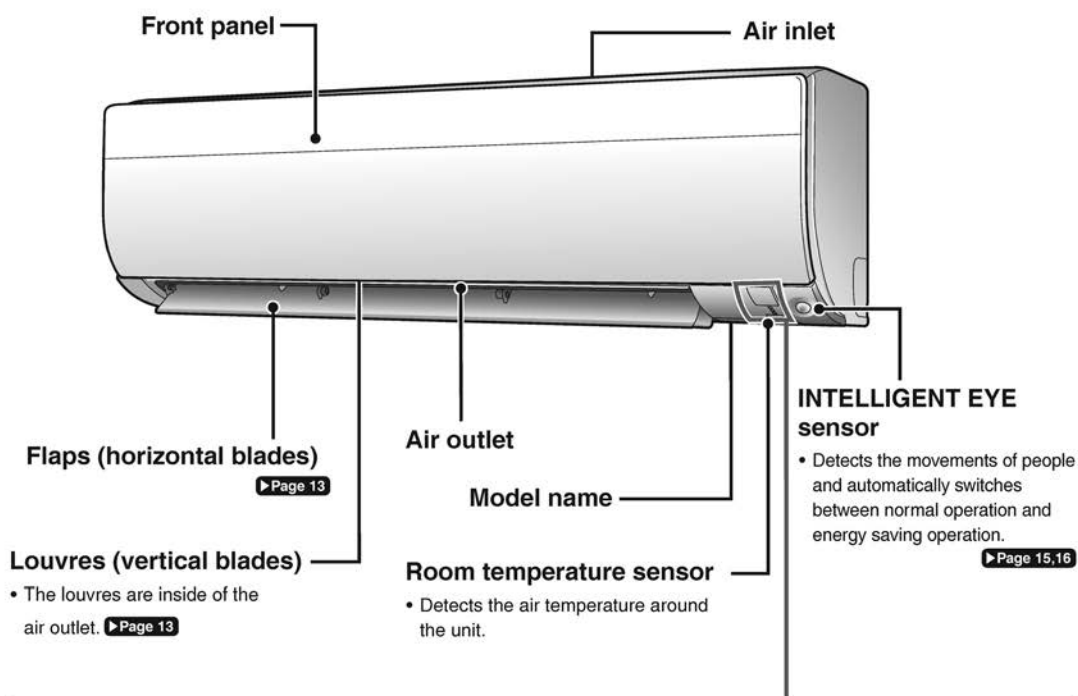
Precautions relating to area surrounding the indoor and outdoor units

- Be sure to follow the instructions below.
 - The indoor unit is at least 1m away from any television or radio set (unit may cause interference with the picture or sound).
 - Refrain from using the units in areas prone to high levels of oily smoke, such as a kitchen. Water leakage may result.

Read Before Operation

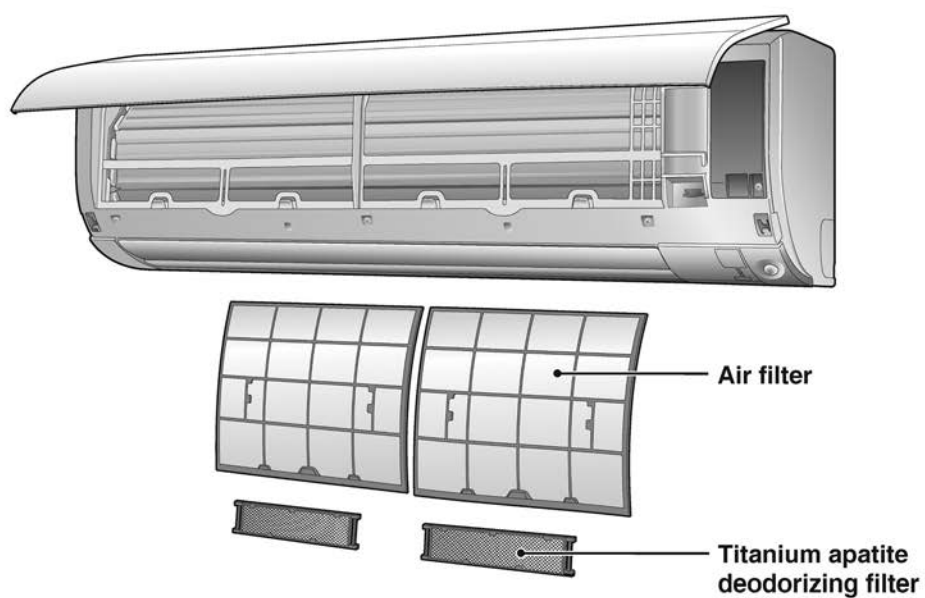
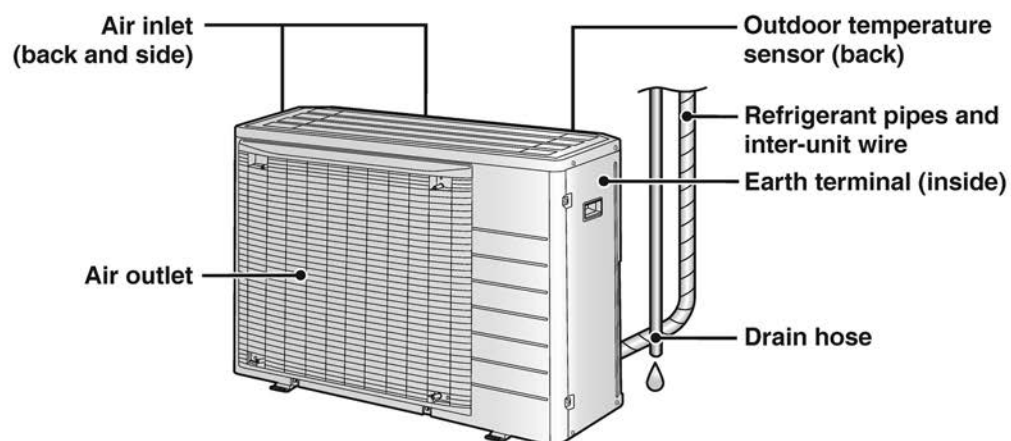
Names of Parts

Indoor Unit



Read Before Operation

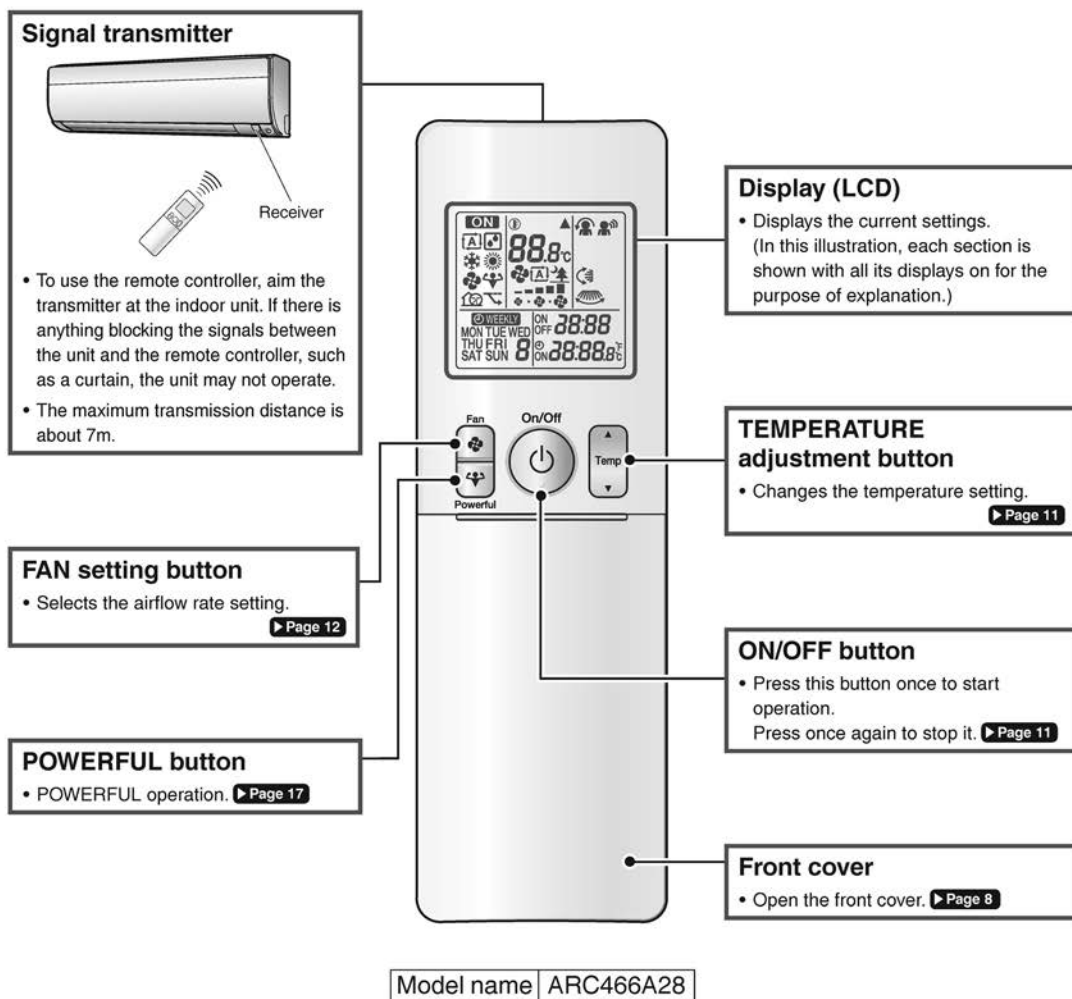
■ Open the front panel

**Outdoor Unit**

Read Before Operation

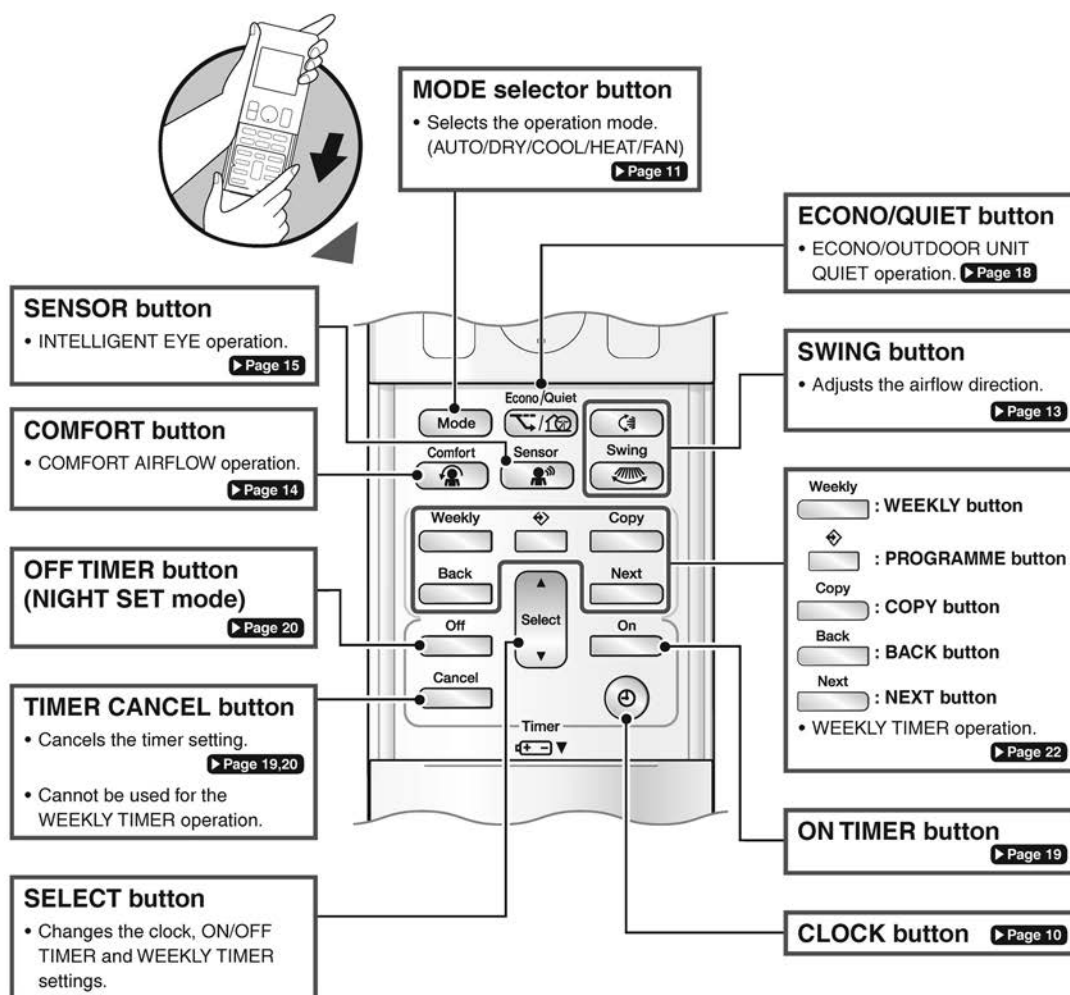
Names of Parts

Remote Controller



Read Before Operation

■ Open the front cover



Read Before Operation

Preparation Before Operation

⚠ CAUTION

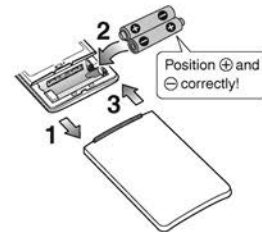
Incorrect handling of batteries can result in injury from battery leakage, rupturing or heating, or lead to equipment failure.

Please observe the following precautions and use safely.

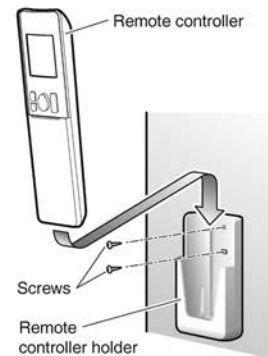
- If the alkaline solution from the batteries should get in the eyes, do not rub the eyes. Instead, immediately flush the eyes with tap water and seek the attention of a medical professional.
- Keep batteries out of reach of children. In the event that batteries are swallowed, seek the immediate attention of a medical professional.
- Do not expose batteries to heat or fire. Do not disassemble or modify batteries. The insulation or gas release vent inside the battery may be damaged, resulting in battery leakage, rupturing, or heating.
- Do not damage or peel off labels on the batteries.

■ To insert the batteries

1. Slide the front cover to take it off.
2. Insert 2 dry batteries AAA.LR03 (alkaline).
3. Replace the front cover.

**■ To fix the remote controller holder to a wall**

1. Choose a place from where the signals reach the unit.
2. Fix the holder to a wall, a pillar, or similar location with the screws supplied with the holder.
3. Place the remote controller in the remote controller holder.

**■ Turn on the circuit breaker**

- After the power is turned on, the flaps of the indoor unit open and close once to set the reference position.

NOTE**Notes on batteries**

- To avoid possible injury or damage from battery leakage or rupturing, remove the batteries when not using the product for long periods of time.
- The standard replacement time is 1 year. Both batteries should be replaced at the same time. Be sure to replace them with new size AAA.LR03 (alkaline) batteries.
- When battery power has run out, the LCD will begin blinking as an alert that the batteries need replacing. In some cases, when battery consumption is accelerated owing usage conditions, signal reception may decline before the LCD begins blinking.
- The batteries supplied with the remote controller are for initial operation. The batteries may run out in less than 1 year.

Note on remote controller

- Do not drop the remote controller. Do not get it wet.

Read Before Operation



■ To set the clock

1. Press .



"0:00" appears on the LCD.
"MON" and "⌚" blink.

2. Press to set the current day of the week.

3. Press .



"⌚" blinks.

4. Press to set the clock to the present time.

- Holding down ▲ or ▼ rapidly increases or decreases the displayed time.

5. Press .

- Point the remote controller at the indoor unit when pressing the buttons.



"." blinks.

NOTE

Note on setting the clock

- If the indoor unit's internal clock is not set to the correct time, the ON/OFF TIMER and WEEKLY TIMER will not operate punctually.

Basic Operation



AUTO · DRY · COOL · HEAT · FAN Operation



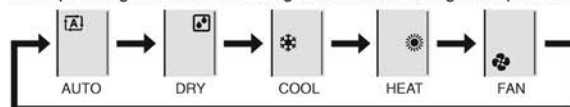
The air conditioner operates with the operation mode of your choice.

From the next time on, the air conditioner will operate with the same operation mode.

■ To start

1. Press and select an operation mode.

- Each pressing of the button changes the mode setting in sequence.



2. Press .

- "ON" appears on the LCD.
- The OPERATION lamp lights green.



Display

■ To stop

Press again.

- "ON" disappears from the LCD.
- The OPERATION lamp goes off.

■ To change the temperature setting

Press .

- Press ▲ to raise the temperature and press ▼ to lower the temperature. Each pressing of the button changes the temperature by 0.5°C.

COOL operation	HEAT operation	AUTO operation	DRY or FAN operation
18.0 - 32.0°C	10.0 - 30.0°C	18.0 - 30.0°C	The temperature setting cannot be changed.

NOTE

Notes on AUTO operation

- In AUTO operation, the system selects an appropriate operation mode (COOL or HEAT) based on the indoor temperature and starts the operation.
- The system automatically reselects setting at a regular interval to bring the indoor temperature to the user-setting level.

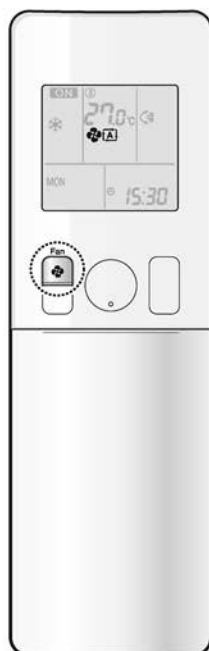
Note on DRY operation

- Eliminates humidity while maintaining the indoor temperature as much as possible. It automatically controls temperature and airflow rate, so manual adjustment of these functions is unavailable.

Basic Operation



Adjusting the Airflow Rate

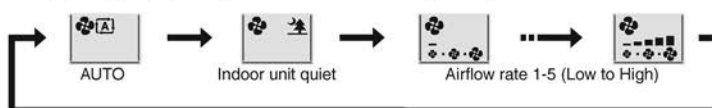


You can adjust the airflow rate to increase your comfort.

■ To adjust the airflow rate setting

Press .

- Each pressing of  changes the airflow rate setting in sequence.



- When the airflow is set to "🌳", quiet operation starts and noise from the indoor unit will become quieter.
- In the quiet operation mode, the airflow rate is set to a weak level.
- In DRY operation, the airflow rate setting cannot be changed.
- If COOL/DRY operation is started with airflow settings on "AUTO", the indoor unit will initially deodorise, which functions to suppress any internal odours. As a result, the indoor unit will not emit air immediately. (Wait for about 1 minute.)
- At smaller airflow rates, the cooling (heating) effect is also smaller.

Tips for saving energy

Keeping the temperature setting at a moderate level helps save energy.

- Recommended temperature setting for cooling: 26.0 - 28.0°C, heating: 20.0 - 24.0°C

Cover windows with a blind or a curtain.

- Blocking sunlight and air from outdoors increases the cooling (heating) effect.

Keep the air filters clean.

- Clogged air filters cause inefficient operation and waste energy. Clean them once in about every 2 weeks. [▶ Page 28](#)

If you are not going to use the air conditioner for a long period, for example in spring or autumn, turn off the circuit breaker.

- The air conditioner always consumes a small amount of electricity even while it is not operating.

Basic Operation



Adjusting the Airflow Direction



You can adjust the airflow direction to increase your comfort.

⚠ CAUTION

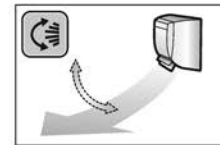
- Always use a remote controller to adjust the angles of the flaps and louvres.
 - If you attempt to move the flaps and louvres forcibly by hand when they are swinging, the mechanism may be damaged.
 - Inside the air outlet, a fan is rotating at a high speed.

■ To start auto swing

Up and down airflow direction

Press .

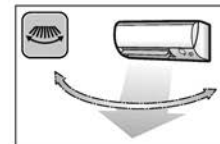
- “” appears on the LCD.
- The flaps (horizontal blades) will begin to swing.



Right and left airflow direction

Press .

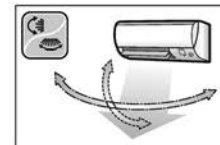
- “” appears on the LCD.
- The louvres (vertical blades) will begin to swing.



The 3-D airflow direction

Press  and .

- “” and “” appear on the LCD.
- The flaps and louvres move in turn.
- To cancel 3-D airflow, press either  or  again. The flaps or louvres will stop moving.



■ To set the flaps or louvres at the desired position

- This function is effective while the flaps or louvres are in auto swing mode.

Press  and  when the flaps or louvres reach the desired position.

- In the 3-D airflow, the flaps and louvres move in turn.
- “” or “” disappears from the LCD.

NOTE

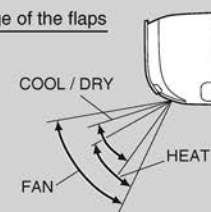
Notes on airflow direction setting

- The movable range of the flaps varies according to the operation mode.

Note on 3-D airflow

- Using 3-D airflow circulates cold air, which tends to collect at the bottom of the room, and hot air, which tends to collect near the ceiling, throughout the room, preventing areas of cold and hot developing.

Movable range of the flaps



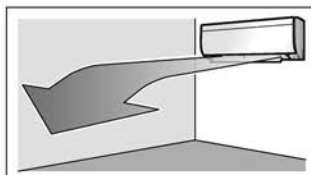
Useful Functions



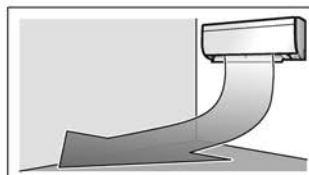
COMFORT AIRFLOW Operation



The flow of air will be in the upward direction while in COOL operation and in the downward direction while in HEAT operation, which will provide a comfortable wind that will not come in direct contact with people.



COOL operation



HEAT operation

■ To start

Press  .

- “” appears on the LCD.

	COOL and DRY operation	HEAT operation	FAN ONLY operation
Flaps direction	Goes up	Goes down	Not available
Airflow rate	AUTO		

■ To cancel

Press  again.

- “” disappears from the LCD.
- The flaps will return to the memory position form before COMFORT AIRFLOW operation.

NOTE

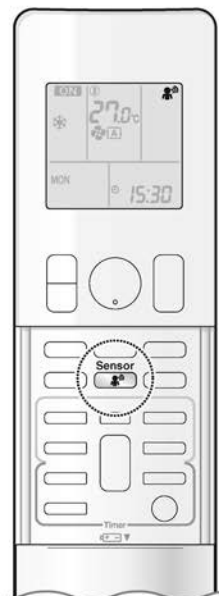
Notes on COMFORT AIRFLOW Operation

- The flaps position will change, preventing air from blowing directly on the occupants of the room.
- POWERFUL operation and COMFORT AIRFLOW operation cannot be used at the same time.
- The airflow rate will be set to AUTO.
- If the upward and downward airflow direction is selected, the COMFORT AIRFLOW function will be canceled. Priority is given to the function of whichever button is pressed last.

Useful Functions



INTELLIGENT EYE Operation



The INTELLIGENT EYE operation detects human movement. If no one is detected in the room for more than 20 minutes, the unit automatically enters energy saving operation.

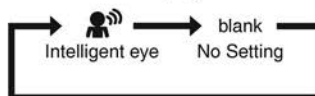
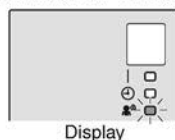
CAUTION

- Do not place large objects near the INTELLIGENT EYE sensor. Also keep heating units and humidifiers outside the sensor's detection area. This sensor can detect undesirable objects.
- Do not hit or violently push the INTELLIGENT EYE sensor. This can lead to damage and malfunction.

To start

Press  and select the desired model.

- Each time "" is pressed, a different setting option is displayed on the LCD.
- The INTELLIGENT EYE lamp lights green.



- The lamp lights when human movement is detected.

To cancel

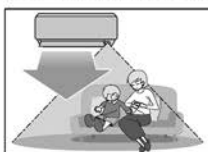
Press  button.

- The INTELLIGENT EYE lamp goes off.

Useful Functions

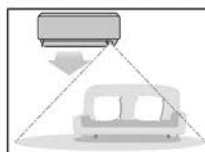
INTELLIGENT EYE operation is useful for energy saving

■ People are detected in the sensing area.



The air conditioner is in normal operation while the sensor is detecting human movement.

■ No people are detected in the sensing area.



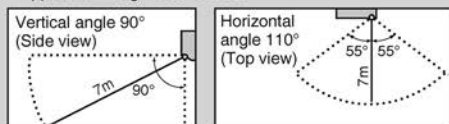
The air conditioner will switch to energy-saving mode after 20 minutes.

Energy saving operation

- If no presence detected in the room for 20 minutes, the energy saving operation will start, and the INTELLIGENT EYE lamp goes off. If human movement is detected again, the INTELLIGENT EYE lamp lights up and energy saving operation terminates.
- This operation changes the temperature by -2°C in HEAT / $+2^{\circ}\text{C}$ in COOL / $+2^{\circ}\text{C}$ in DRY operation from the set temperature. When the room temperature exceeds 30°C , the operation changes the temperature by $+1^{\circ}\text{C}$ in COOL / $+1^{\circ}\text{C}$ in DRY operation from the set temperature.
- This operation decreases the airflow rate slightly in FAN operation only.

NOTE**Notes on INTELLIGENT EYE operation**

- Application range is as follows.



- The sensor may not detect moving objects further than 7m away. (Please see the application range)
- Sensor detection sensitivity changes according to the indoor unit location, the speed of passers-by, temperature range, etc.
- The sensor could also mistakenly detect pets, sunlight, fluttering curtains and light reflected off of mirrors as passers-by.
- NIGHT SET mode [▶Page 20](#) will not switch on during use of INTELLIGENT EYE operation.

Note on combination of COMFORT AIRFLOW operation and INTELLIGENT EYE operation

- The airflow rate will be set to AUTO. If the up and down airflow direction is selected, COMFORT AIRFLOW operation will be cancelled. Priority is given to the function of whichever button is pressed last.

Useful Functions



POWERFUL Operation



POWERFUL operation quickly maximises the cooling (heating) effect in any operation mode. In this mode, the air conditioner operates at maximum capacity.

■ To start

▶ Press  during operation.

- “” appears on the LCD.
- POWERFUL operation ends in 20 minutes. Then the system automatically operates again with the previous settings which were used before POWERFUL operation.

■ To cancel

▶ Press  again.

- “” disappears from the LCD.

NOTE

Notes on POWERFUL operation

- Pressing  causes the settings to be cancelled, and “” disappears from the LCD.
- POWERFUL operation will not increase the capacity of the air conditioner if the air conditioner is already in operation with its maximum capacity demonstrated.
 - In COOL, HEAT and AUTO operation
To maximise the cooling (heating) effect, the capacity of outdoor unit increases and the airflow rate becomes fixed at the maximum setting. The temperature and airflow settings cannot be changed.
 - In DRY operation
The temperature setting is lowered by 2.5°C and the airflow rate is slightly increased.
 - In FAN operation
The airflow rate is fixed at the maximum setting.

Regarding the combination of POWERFUL and other operations

POWERFUL + COMFORT AIRFLOW	Not available*
POWERFUL + ECONO	
POWERFUL + OUTDOOR UNIT QUIET	

*Priority is given to the function of whichever button is pressed last.

Useful Functions



ECONO / OUTDOOR UNIT QUIET Operation



ECONO operation enables efficient operation by limiting the maximum power consumption. This function is useful to prevent the circuit breaker from tripping when the unit operates alongside other appliances on the same circuit.

OUTDOOR UNIT QUIET operation lowers the noise level of the outdoor unit by changing the frequency and fan speed of the outdoor unit. This function is convenient during the night-time operation.

■ To start

➤ Press  and select the desired mode.

- Each time  is pressed, a different setting option appears on the LCD.



■ To cancel

➤ Press  until no icon is displayed.

NOTE

Notes on ECONO operation

- This operation is performed with lower power and therefore may not provide a sufficient cooling (heating) effect.
- Pressing  causes the settings to be cancelled, and "ECONO" disappears from the LCD.
- If the power consumption level is already low, switching to ECONO operation will not reduce the power consumption.

Notes on OUTDOOR UNIT QUIET operation

- Even if the operation is stopped by using the remote controller or the indoor unit ON/OFF switch when using OUTDOOR UNIT QUIET operation, "OUTDOOR UNIT QUIET" will remain displayed on the remote controller.
- OUTDOOR UNIT QUIET operation will not reduce the frequency nor fan speed if they already are operating at reduced levels.
- This operation is performed with lower power and therefore may not provide a sufficient cooling (heating) effect.

Possible combinations of ECONO / OUTDOOR UNIT QUIET operation and basic operations

	Operation mode			
	DRY	COOL	HEAT	FAN
ECONO	✓	✓	✓	—
OUTDOOR UNIT QUIET	—	✓	✓	—

TIMER Operation



ON/OFF TIMER Operation



Timer functions are useful for automatically switching the air conditioner on or off at night or in the morning. You can also use the ON TIMER and OFF TIMER together.

[ON TIMER with a 24 hour clock display] (24 HOUR ON/OFF TIMER)

■ To set

- Check that the clock is correct.
If not, set the clock to the present time. ▶ [Page 10](#)

1. Press .



"6:00" appears on the LCD.
"ON" blinks.

- "⌚" and day of the week disappear from the LCD.

2. Press until the time setting reaches the point you like.

- Each pressing of either button increases or decreases the time setting by 10 minutes.
Holding down either button changes the setting rapidly.

3. Press again.

- "ON" and setting time appear on the LCD.
- The TIMER lamp lights orange.



Display

■ To cancel

▶ Press .

- "ON" and setting time disappear from the LCD.
- "⌚" and day of the week appear on the LCD.
- The TIMER lamp goes off.

NOTE

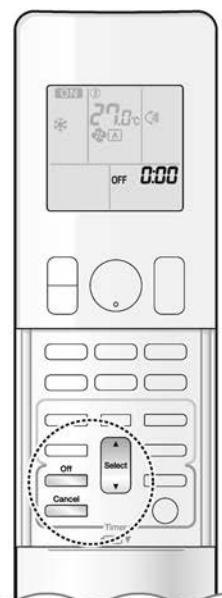
Notes on TIMER operation

- When TIMER is set, the present time is not displayed.
- When using the ON/OFF TIMER to start/stop operation, the actual operation start/stop time may differ from the time set.
(Maximum of about 10 minutes)

In the following cases, set the timer again.

- After the circuit breaker has turned off.
- After a power failure.
- After replacing the batteries in the remote controller.

TIMER Operation



[OFF TIMER with a 24 hour clock display] (24 HOUR ON/OFF TIMER)

■ To set

- Check that the clock is correct.
If not, set the clock to the present time. ▶Page 10

1. Press .



"0:00" appears on the LCD.
"OFF" blinks.

- "⊕" and day of the week disappear from the LCD.

2. Press until the time setting reaches the point you like.

- Each pressing of either button increases or decreases the time setting by 10 minutes.
Holding down either button changes the time setting rapidly.

3. Press again.

- "OFF" and setting time appear on the LCD.
- The TIMER lamp lights orange.



Display

■ To cancel

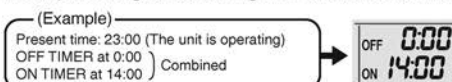
Press .

- "OFF" and setting time disappear from the LCD.
- "⊕" and day of the week appear on the LCD.
- The TIMER lamp goes off.

[ON/OFF TIMER combination]

■ To combine

- A sample setting for combining the 2 timers is shown below.



NOTE

NIGHT SET mode

- When the OFF TIMER is set, the air conditioner automatically adjusts the temperature setting (0.5°C up in COOL, 2.0°C down in HEAT) to prevent excessive cooling (heating) during sleeping hours.

TIMER Operation



ON/OFF TIMER Operation

[ON/OFF TIMER with an hours remaining display]
(COUNT UP-DOWN ON/OFF TIMER)

- Check that the clock is correct. If not, set the clock to the present time. ►Page 10

1. Press and hold  or  for about 5 seconds, when signal indicator (▲) blink, release your finger from the button.



For ON TIMER
"1hr" appears on the LCD.
"ON" blinks.



For OFF TIMER
"1hr" appears on the LCD.
"OFF" blinks.

- "▲" and day of the week disappear from the LCD.

2. Press  until the time setting reaches the point you like.

- Each pressing of the button increases or decreases the time setting by 1 hour.
- The time can be set between 1 and 12 hours.

3. Press  or  again.

- "ON" and "OFF", and setting time appear on the LCD.
- The TIMER lamp lights orange.
- To change the display mode from COUNT UP-DOWN ON/OFF TIMER to 24 HOUR ON/OFF TIMER, repeat **STEP 1**, and continue from **STEP 2** of "24 HOUR ON/OFF TIMER".
►Page 19, 20
- Timer display mode cannot be changed while "ON" or "OFF" are blinking.
Press  or  to stop the blinking.



Display

■ Combining display modes

- Different display modes (24 HOUR ON/OFF TIMER and COUNT UP-DOWN ON/OFF TIMER) can be used for the ON TIMER and OFF TIMER simultaneously.



NOTE

Notes on COUNT UP-DOWN ON/OFF TIMER

- After setting, the time displayed on the LCD will automatically count down by 1 hour for every 1 hour that passes.
For example, if the OFF TIMER is set to 3 hours, the display will change as follows: 3hr → 2hr → 1hr → blank (OFF)

Switching between ON/OFF TIMER display modes (24 hour clock and hours remaining display)

- When changing timer display modes (from 24 HOUR ON/OFF TIMER to COUNT UP-DOWN ON/OFF TIMER or vice versa), the timer cancels. The timer must be set again to activate it.

TIMER Operation



WEEKLY TIMER Operation

Up to 4 timer settings can be saved for each day of the week. This is convenient to adapt the WEEKLY TIMER to your family's life style.

■ Setting example of the WEEKLY TIMER

The same timer settings are used from Monday through Friday, while different timer settings are used for the weekend.

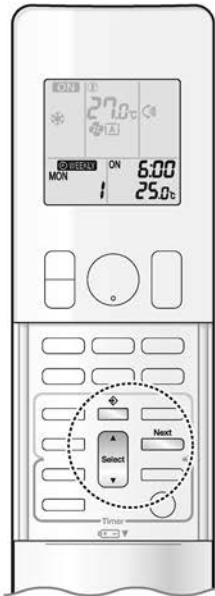
[Monday]	Make timer settings for programmes 1-4. ▶ Page 23 Programme 1 ON 25.0°C 6:00 Programme 2 OFF 8:30 Programme 3 ON 27.0°C 17:30 Programme 4 OFF 22:00
[Tuesday] to [Friday]	Use the copy mode to make settings for Tuesday to Friday, because these settings are the same as those for Monday. ▶ Page 25 Programme 1 ON 25.0°C 6:00 Programme 2 OFF 8:30 Programme 3 ON 27.0°C 17:30 Programme 4 OFF 22:00
[Saturday]	No timer settings
[Sunday]	Make timer settings for programmes 1-4. ▶ Page 23 Programme 1 ON 25.0°C 8:00 Programme 2 OFF 10:00 Programme 3 OFF 19:00 Programme 4 ON 27.0°C 21:00

- Up to 4 reservations per day and 28 reservations per week can be set using the WEEKLY TIMER. The effective use of the copy mode simplifies timer programming.
- The use of ON-ON-ON-ON settings, for example, makes it possible to schedule operating mode and set temperature changes. Furthermore, by using OFF-OFF-OFF-OFF settings, only the turn off time of each day can be set. This will turn off the air conditioner automatically if you forget to turn it off.

TIMER Operation



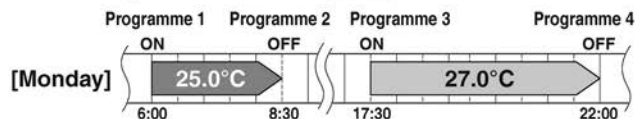
WEEKLY TIMER Operation



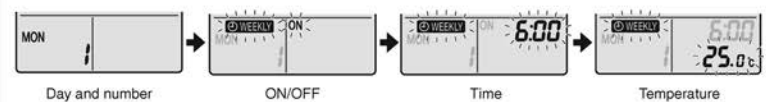
■ To use WEEKLY TIMER operation

Setting mode

- Make sure the day of the week and time are set.
If not, set the day of the week and time. ▶Page 10



Setting Displays

1. Press .

- The day of the week and the reservation number of the current day will be displayed.
- 1 to 4 settings can be made per day.

2. Press to select the desired day of the week and reservation number.

- Pressing  changes the reservation number and the day of the week.

3. Press .

- The day of the week and reservation number will be set.
- "WEEKLY" and "ON" blink.

4. Press to select the desired mode.

- Pressing  changes the "ON" or "OFF" setting in sequence.

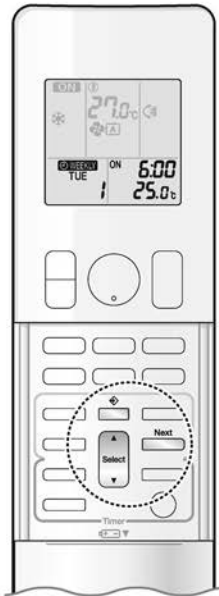


- In case the reservation has already been set, selecting "blank" deletes the reservation.
- Proceed to **STEP 9** if "blank" is selected.
- To return to the day of the week and reservation number setting, press .

5. Press .

- The ON/OFF TIMER mode will be set.
- "WEEKLY" and the time blink.

TIMER Operation



6. Press to select the desired time.

- The time can be set between 0:00 and 23:50 in 10-minute intervals.
- To return to the ON/OFF TIMER mode setting, press .
- Proceed to **STEP 9** when setting the OFF TIMER.

7. Press .

- The time will be set.
- "WEEKLY" and the temperature blink.

8. Press to select the desired temperature.

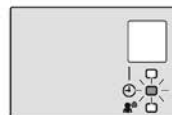
- The temperature can be set between 10.0°C and 32.0°C.
COOL or AUTO: The unit operates at 18.0°C even if it is set to 10.0°C to 17.0°C. [▶Page 11](#)
HEAT or AUTO: The unit operates at 30.0°C even if it is set to 31.0°C to 32.0°C. [▶Page 11](#)
- To return to the time setting, press .
- The set temperature is only displayed when the mode setting is on.

9. Press .

- The temperature will be set and go to the next reservation setting.
- The temperature is set while in ON TIMER operation, and the time is set while in OFF TIMER operation.
- The next reservation screen will appear.
- To continue further settings, repeat the procedure from **STEP 4**.

10. Press to complete the setting.

- Be sure to direct the remote controller toward the indoor unit and check for a receiving tone and blinking of the OPERATION lamp.
- "WEEKLY" appears on the LCD and WEEKLY TIMER operation is activated.
- The TIMER lamp lights orange.



Display

- A reservation made once can be easily copied and the same settings used for another day of the week. Refer to [Copy mode](#). [▶Page 25](#)

NOTE

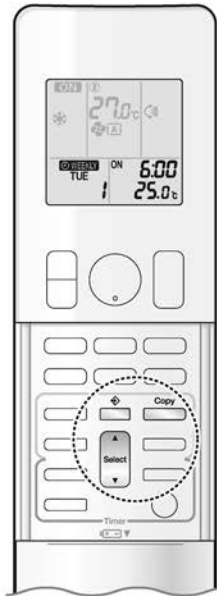
Notes on WEEKLY TIMER operation

- Do not forget to set the clock on the remote controller first. [▶Page 10](#)
- The day of the week, ON/OFF TIMER mode, time and set temperature (only for ON TIMER mode) can be set with the WEEKLY TIMER. Other settings for the ON TIMER are based on the settings just before the operation.
- WEEKLY TIMER and ON/OFF TIMER operation cannot be used at the same time. The ON/OFF TIMER operation has priority if it is set while WEEKLY TIMER is still active. The WEEKLY TIMER will enter the standby state, and "WEEKLY" will disappear from the LCD. When the ON/OFF TIMER is up, the WEEKLY TIMER will automatically become active.
- Only the time and set temperature with the WEEKLY TIMER are sent with the . Set the WEEKLY TIMER only after setting the operation mode, the airflow rate and the airflow direction ahead of time.
- Turning off the circuit breaker, power failure, and other similar events will render operation of the indoor unit's internal clock inaccurate. Reset the clock. [▶Page 10](#)
-  can be used only for the time and temperature settings. It cannot be used to go back to the reservation number.

TIMER Operation

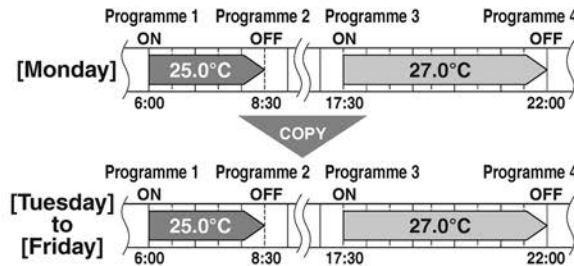


WEEKLY TIMER Operation

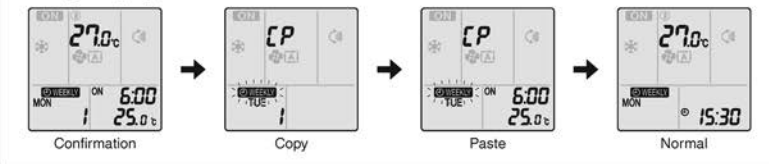


Copy mode

- A reservation made once can be copied to another day of the week. The whole reservation of the selected day of the week will be copied.



Setting Displays



- Press .
- Press to confirm the day of the week to be copied.
- Press .
 - The whole reservation of the selected day of the week will be copied.
- Press to select the destination day of the week.
- Press .
 - The reservation will be copied to the selected day of the week. The whole reservation of the selected day of the week will be copied.
 - To continue copying the settings to other days of the week, repeat **STEP 4** and **STEP 5**.
- Press to complete the setting.
 - " WEEKLY" appears on the LCD and WEEKLY TIMER operation is activated.

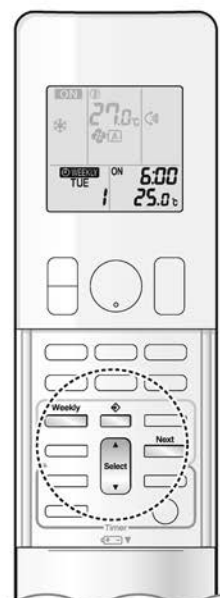
NOTE

Note on COPY MODE

- The entire reservation of the source day of the week is copied in the copy mode.

In the case of making a reservation change for any day of the week individually after copying the content of weekly reservations, press and change the settings in the steps of **Setting mode** **Page 23**

TIMER Operation



■ Confirming a reservation

- The reservation can be confirmed.



1. Press .

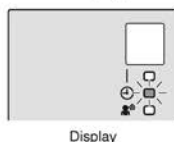
- The day of the week and the reservation number of the current day will be displayed.

2. Press to select the day of the week and the reservation number to be confirmed.

- Pressing  displays the reservation details.
- To change the confirmed reserved settings, select the reservation number and press . The mode is switched to setting mode. Proceed to **Setting mode STEP 4.** [▶ Page 23](#)

3. Press to exit the confirmation mode.

- “ WEEKLY” appears on the LCD and WEEKLY TIMER operation is activated.
- The TIMER lamp lights orange.



■ To deactivate WEEKLY TIMER operation

▶ Press while “ WEEKLY” is displayed on the LCD.

- “ WEEKLY” disappears from the LCD.
- The TIMER lamp goes off.
- To reactivate the WEEKLY TIMER operation, press  again.
- If a reservation deactivated with  is activated once again, the last reservation mode will be used.

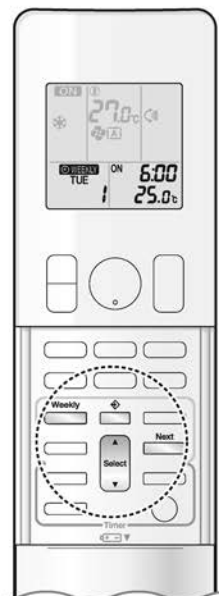
NOTE

- If not all the reservation settings are reflected, deactivate the WEEKLY TIMER operation once. Then press  again to reactivate the WEEKLY TIMER operation.

TIMER Operation



WEEKLY TIMER Operation



■ To delete a reservation

An individual reservation

1. Press .

- The day of the week and the reservation number will be displayed.

2. Press to select the day of the week and the reservation number to be deleted.3. Press .

- " WEEKLY" and "ON" or "OFF" blink.

4. Press until no icon is displayed.

- Pressing  changes the ON/OFF TIMER mode in sequence.
- Selecting "blank" will cancel any reservation you may have.

5. Press .

- The selected reservation will be deleted.

6. Press .

- If there are still other reservations, WEEKLY TIMER operation will be activated.

Reservations for each day of the week

- This function can be used for deleting reservations for each day of the week.
- It can be used while confirming or setting reservations.

1. Press .

- The day of the week and the reservation number will be displayed.

2. Press to select the day of the week to be deleted.3. Hold for about 5 seconds.

- The reservation of the selected day of the week will be deleted.

4. Press .

- If there are still other reservations, WEEKLY TIMER operation will be activated.

All reservations

▶ Hold for about 5 seconds with the normal display.

- Be sure to direct the remote controller toward the indoor unit and check for a receiving tone.
- This operation cannot be used for the WEEKLY TIMER setting display.
- All reservations will be deleted.

Care

Care and Cleaning

⚠ CAUTION

- Before cleaning, be sure to stop the operation and turn off the circuit breaker.
- Do not touch the aluminum fins of the indoor unit. If you touch those parts, this may cause an injury.

■ Quick reference

Cleaning parts

Front panel

- Wipe it with a soft damp cloth.
- Only neutral detergent may be used.

If dirty



Air filter

- Vacuum dust or wash the filter.

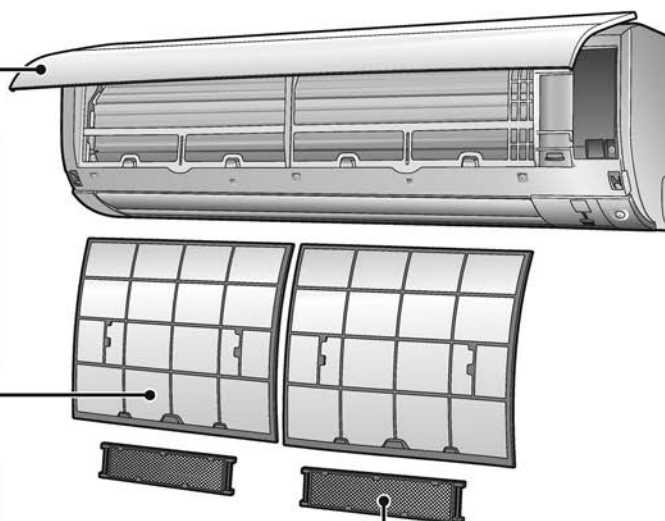
Once every 2 weeks

▶ Page 29

Indoor unit, outdoor unit and remote controller

- Wipe them with a soft cloth.

If dirty



Titanium apatite deodorizing filter (included with the air conditioner) and dust collection filter (PM2.5) (sold separately)

To purchase, contact the dealer where you bought the air conditioner.

Item	Part No.
Titanium apatite deodorizing filter - without frame (set of 2)	KAF970A46
Dust collection filter (PM2.5) - without frame (set of 2)	BAFP046A41

Titanium apatite deodorizing filter

- Vacuum dust or replace the filter.

[Cleaning]

Once every 6 months

▶ Page 30

[Replacement]

Once every 3 years

▶ Page 30

Dust collection filter (PM2.5)

- No cleaning is required. Please replace the filter every 6 months. ▶ Page 30

Notes on cleaning

For cleaning, do not use any of the following:

- Water hotter than 40°C
- Volatile liquid such as benzene, petrol and thinner
- Polishing compounds
- Rough materials such as a scrubbing brush



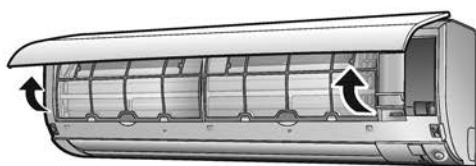
Care

Care and Cleaning

■ Air filter

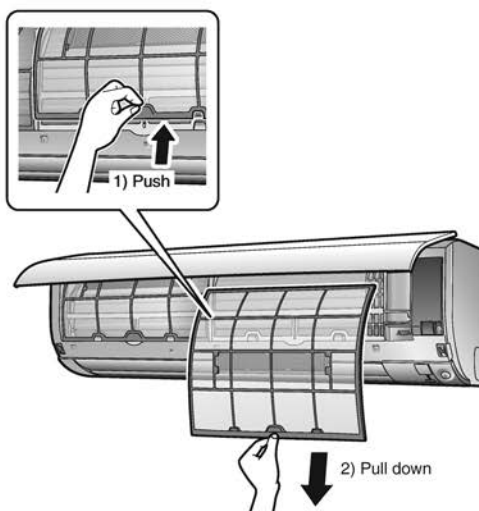
1. Open the front panel.

- Hold the front panel by the sides and open it.



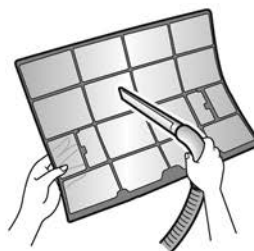
2. Pull out the air filters.

- Push the filter tab at the centre of each air filter a little upwards, then pull it down.



3. Wash the air filters with water or clean them with vacuum cleaner.

- It is recommended to clean the air filters every 2 weeks.



If the dust does not come off easily

- Wash the air filters with neutral detergent thinned with lukewarm water, then dry them up in the shade.



4. Reattach the filters.

5. Close the front panel slowly.

- Press the panel at both sides and the centre.



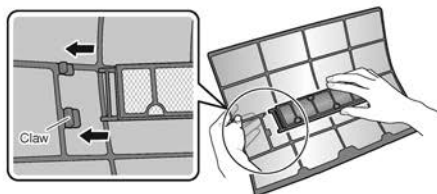
- Make sure that the front panel is securely fixed.

Care

■ Titanium apatite deodorizing filter (included with the air conditioner) / dust collection filter (PM2.5) (sold separately)

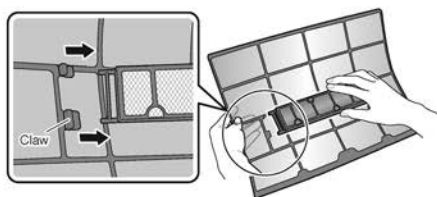
Attaching

1. Open the front panel and pull out the air filters. **▶ Page 29**
2. Insert the titanium apatite deodorizing filters or dust collection filters (PM2.5).
 - Check that the filters are inserted securely under the tabs.



Cleaning and Replacement

1. Open the front panel and pull out the air filters. **▶ Page 29**
2. Take off the titanium apatite deodorizing filters or dust collection filters (PM2.5).
 - Hold the recessed parts of the frame and unhook the 4 claws.
 - Remove the filters from the tabs.



3. Clean or replace the titanium apatite deodorizing filters or dust collection filters (PM2.5).

[Cleaning for titanium apatite deodorizing filter (Blue-Black)]

3-1 Vacuum dust, and soak in lukewarm water or water for about 10 to 15 minutes if very dirty.

3-2 After washing, shake off remaining water and let them dry in the shade.

- This filter can be renewed by washing it with water once **every 6 months** and we recommend replacing it once **every 3 years**.
- Do not wring out the filter when removing water from it.

[Cleaning for dust collection filter (PM2.5) (White)]

- Do not wash the filters as this will reduce their efficiency.
- It is recommended to replace it once **every 6 months**.

[Replacement]

Remove the filter from the filter frame and prepare a new one.

- Do not throw away the filter frame. Reuse the filter frame when replacing titanium apatite deodorizing filter or dust collection filter (PM2.5).
- When attaching the filter, check that the filter is properly set in the tabs.



4. Set the filters as they were and close the front panel.

- Press the front panel at both sides and the center.

NOTE

- Operation with dirty filters:
 - cannot deodorise the air,
 - cannot clean the air,
 - results in poor cooling,
 - may cause odour.
- Dispose of old filters as non-flammable waste.

Care

Care and Cleaning

■ Prior to a long period of non-use

1. Operate the FAN mode for several hours on a fine day to dry out the inside.

- 1) Press  and select "  ".
- 2) Press  and start the operation.

2. After operation stops, turn off the circuit breaker for the room air conditioner.

3. Clean the air filters and reattach them.

4. To prevent battery leakage, take out the batteries from the remote controller.

■ We recommend periodical maintenance

- In certain operating conditions, the inside of the air conditioner may get foul after several seasons of use, resulting in poor performance. It is recommended to have periodical maintenance by a specialist in addition to regular cleaning by the user.
- For specialist maintenance, please contact the dealer where you bought the air conditioner.
- The maintenance cost must be borne by the user.

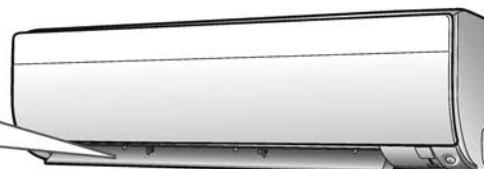
When the Need Arises

FAQ

Indoor unit

The flaps do not start swinging immediately.

- The air conditioner is adjusting the position of the flaps. The flaps will start moving soon.

**The air conditioner stops generating airflow during HEAT operation.**

- Once the set temperature is reached, the airflow rate is reduced and operation stopped in order to avoid generating a cool airflow. Operation will resume automatically when the indoor temperature falls.

HEAT operation stops suddenly and a flowing sound is heard.

- The outdoor unit is defrosting. HEAT operation starts after the frost on the outdoor unit has been removed. This can take about 4 to 12 minutes.

Operation does not start soon.

- When  was pressed soon after operation was stopped.
- When the mode was reselected.
 - This is to protect the air conditioner. You should wait for about 3 minutes.

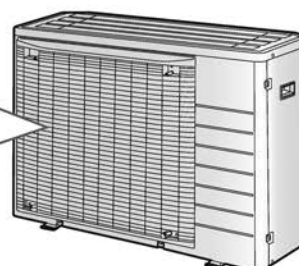
Different sounds are heard.

- **A sound like flowing water**
 - This sound is generated because the refrigerant in the air conditioner is flowing.
 - This is a pumping sound of the water in the air conditioner and can be heard when the water is pumped out from the air conditioner during COOL or DRY operation.
- **Blowing sound**
 - This sound is generated when the flow of the refrigerant in the air conditioner is switched over.
- **Ticking sound**
 - This sound is generated when the cabinet and frame of the air conditioner slightly expand or shrink as a result of temperature changes.
- **Whistling sound**
 - This sound is generated when refrigerant flows during defrosting operation.
- **Clicking sound during operation or idle time**
 - This sound is generated when the refrigerant control valves or the electrical parts operate.
- **Clopping sound**
 - This sound is heard from the inside of the air conditioner when the exhaust fan is activated while the room doors are closed. Open the window or turn off the exhaust fan.

Outdoor unit

The outdoor unit emits water or steam.

- **In HEAT operation**
 - The frost on the outdoor unit melts into water or steam when the air conditioner is in defrosting operation.
- **In COOL or DRY operation**
 - Moisture in the air condenses into water on the cool surface of the outdoor unit piping and drips.



When the Need Arises

Troubleshooting

Before making an inquiry or a request for repair, please check the following.
If the problem persists, consult your dealer.



Not a problem

This case is not a problem.



Check

Please check again before requesting repairs.

The air conditioner does not operate

Case	Description / what to check
OPERATION lamp is off.	- Has the circuit breaker been tripped or the fuse blown? - Is there a power failure? - Are batteries set in the remote controller?
OPERATION lamp is blinking.	Turn off the power with the circuit breaker and restart operation with the remote controller. If the OPERATION lamp is still blinking, check the error code and consult your dealer. ▶ Page 36

The air conditioner suddenly stops operating

Case	Description / what to check
OPERATION lamp is on.	To protect the system, the air conditioner may stop operating after sudden large voltage fluctuations. It automatically resumes operation in about 3 minutes.
OPERATION lamp is blinking.	Is there anything blocking the air inlet or air outlet of the indoor unit or outdoor unit? Stop operation and after turning off the circuit breaker, remove the obstruction. Then restart operation with the remote controller. If the OPERATION lamp is still blinking, check the error code and consult your dealer. ▶ Page 36

The air conditioner does not stop operating

Case	Description / what to check
The air conditioner continues operating even after operation is stopped.	Immediately after the air conditioner is stopped - The outdoor unit fan continues rotating for about another 1 minute to protect the system. While the air conditioner is not in operation - When the outdoor temperature is high, the outdoor unit fan may start rotating to protect the system.

The room does not cool down / warm up

Case	Description / what to check
Air does not come out.	In HEAT operation - The air conditioner is warming up. Wait for about 1 to 4 minutes. - During defrosting operation, hot air does not flow out of the indoor unit. In COOL or DRY operation - If COOL/DRY operation is started with airflow settings on "AUTO", the indoor unit will initially deodorise, which functions to suppress any internal odours. As a result, the indoor unit will not emit air immediately. (Wait for about 1 minute.) When the air conditioner operates immediately after the circuit breaker is turned on - The air conditioner is preparing to operate. Wait for about 3 to 15 minutes.
Air does not come out / Air comes out.	Is the airflow rate setting appropriate? - Is the airflow rate setting low, such as "Indoor unit quiet" or "Airflow rate 1"? Increase the airflow rate setting. Is the set temperature appropriate? Is the adjustment of the airflow direction appropriate?
Air comes out.	- Is there any furniture directly under or beside the indoor unit? - Is the air conditioner in ECONO operation or OUTDOOR UNIT QUIET operation? ▶ Page 18 - Are the air filters dirty? - Is there anything blocking the air inlet or air outlet of the indoor unit or outdoor unit? - Is a window or door open? - Is an exhaust fan turning?

When the Need Arises

Mist comes out

Case	Description / what to check
Mist comes out of the indoor unit.	☒ • This happens when the air in the room is cooled into mist by the cold airflow during COOL or other operation.

Remote controller

Case	Description / what to check
The unit does not receive signals from the remote controller or has a limited operating range.	☒ • The batteries may be exhausted. Replace both batteries with new dry batteries AAA.LR03 (alkaline). For details, refer to "Preparation Before Operation". ▶Page 9 ☐ • Signal communication may be disabled if an electronic-starter-type fluorescent lamp (such as inverter-type lamps) is in the room. Consult your dealer if that is the case. ☐ • The remote controller may not function correctly if the transmitter is exposed to direct sunlight.
LCD is faint, is not working, or the display is erratic.	☒ • The batteries may be exhausted. Replace both batteries with new dry batteries AAA.LR03 (alkaline). For details, refer to "Preparation Before Operation". ▶Page 9
The LCD is blinking and the remote controller cannot be operated.	☒ • Battery power has run out. Replace both batteries at the same time with new size AAA.LR03 (alkaline) batteries. Leaving exhausted batteries in the remote controller can result in injury due to battery leakage, rupturing or heating, or lead to equipment failure. (Even when the LCD is blinking, the OFF button remains functional.)
Other electric devices start operating.	☒ • If the remote controller activates other electric devices, move them away or consult your dealer.

Air has an odour

Case	Description / what to check
The air conditioner gives off an odour.	☒ • The room odour absorbed in the unit is discharged with the airflow. We recommend you to have the indoor unit cleaned. Please consult your dealer.

Others

Case	Description / what to check
The air conditioner suddenly starts behaving strangely during operation.	☐ • The air conditioner may malfunction due to lightning or radio. If the air conditioner malfunctions, turn off the power with the circuit breaker and restart the operation with the remote controller.
The ON/OFF TIMER does not operate according to the settings.	☒ • Check if the ON/OFF TIMER and the WEEKLY TIMER are set to the same time. Change or deactivate the settings in the WEEKLY TIMER. ▶Page 22

NOTE

Notes on the operating conditions

- If operation continues under any conditions other than those listed in the table,
 - A safety device may activate to stop the operation.
 - Dew may form on the indoor unit and drip from it when COOL or DRY operation is selected.

Mode	Operating conditions
COOL / DRY	Outdoor temperature: -10.0-46.0°C Indoor temperature: 18.0-32.0°C Indoor humidity: 80% max.
HEAT	Outdoor temperature: -15.0-24.0°C Indoor temperature: 10.0-30.0°C

When the Need Arises

Troubleshooting

■ Call your dealer immediately

WARNING

When an abnormality (such as a burning smell) occurs, stop operation and turn off the circuit breaker.

- Continued operation in an abnormal condition may result in problems, electric shock or fire.
- Consult the dealer where you bought the air conditioner.

Do not attempt to repair or modify the air conditioner by yourself.

- Incorrect work may result in electric shock or fire.
- Consult the dealer where you bought the air conditioner.

If one of the following symptoms takes place, call your dealer immediately.

- The power cord is abnormally hot or damaged.
- An abnormal sound is heard during operation.
- The circuit breaker, a fuse, or the earth leakage circuit breaker cuts off the operation frequently.
- A switch or a button often fails to work properly.
- There is a burning smell.
- Water leaks from the indoor unit.

Turn off the circuit breaker and call your dealer.



■ After a power failure

- The air conditioner automatically resumes operation in about 3 minutes. Please wait for a while.

■ Lightning

- If there is a risk lightning could strike in the neighbourhood, stop operation and turn off the circuit breaker to protect the system.

■ Disposal requirements

- Dismantling of the unit, handling of the refrigerant, oil and other parts, should be done in accordance with the relevant local and national regulations.

When the Need Arises



■ Fault diagnosis by remote controller

- In case the air conditioner is malfunctioning, you can check the error code via the remote controller before consulting with service personnel and informing them of the malfunction code.

1. When  is held down for about 5 seconds, “00” blinks in the temperature display section.

2. Press  repeatedly until a long beep is produced.

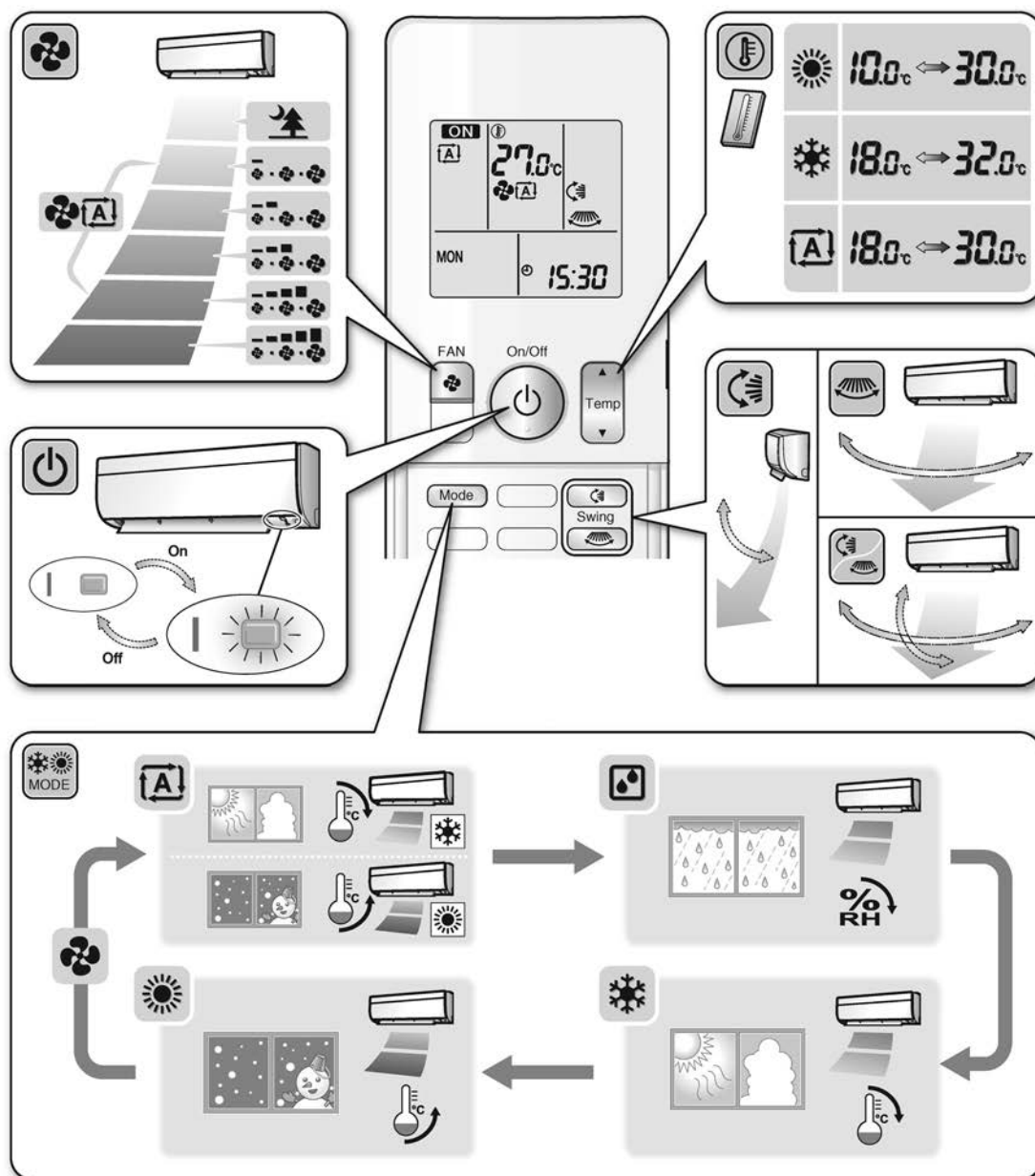
- The code indication changes as shown below, and notifies you with a long beep.

	CODE	MEANING
SYSTEM	00	NORMAL
	UA	INDOOR-OUTDOOR UNIT COMBINATION FAULT
	U0	REFRIGERANT SHORTAGE
	U2	DROP VOLTAGE OR MAIN CIRCUIT OVERVOLTAGE
	U4	FAILURE OF TRANSMISSION (BETWEEN INDOOR UNIT AND OUTDOOR UNIT)
INDOOR UNIT	A1	INDOOR PCB DEFECTIVENESS
	A5	HIGH PRESSURE CONTROL OR FREEZE-UP PROTECTOR
	A6	FAN MOTOR FAULT
	C4	FAULTY HEAT EXCHANGER TEMPERATURE SENSOR
	C9	FAULTY SUCTION AIR TEMPERATURE SENSOR
OUTDOOR UNIT	EA	COOLING-HEATING SWITCHING ERROR
	E1	CIRCUIT BOARD FAULT
	E5	OL STARTED
	E6	FAULTY COMPRESSOR START UP
	E7	DC FAN MOTOR FAULT
	E8	OVERCURRENT INPUT
	F3	HIGH TEMPERATURE DISCHARGE PIPE CONTROL
	F6	HIGH PRESSURE CONTROL (IN COOLING)
	H0	SENSOR FAULT
	H6	OPERATION HALT DUE TO FAULTY POSITION DETECTION SENSOR
	H8	DC CURRENT SENSOR FAULT
	H9	FAULTY SUCTION AIR TEMPERATURE SENSOR
	J3	FAULTY DISCHARGE PIPE TEMPERATURE SENSOR
	J6	FAULTY HEAT EXCHANGER TEMPERATURE SENSOR
	L3	ELECTRICAL PARTS HEAT FAULT
	L4	HIGH TEMPERATURE AT INVERTER CIRCUIT HEATSINK
	L5	OUTPUT OVERCURRENT
	P4	FAULTY INVERTER CIRCUIT HEATSINK TEMPERATURE SENSOR

NOTE

- A short beep indicate non-corresponding codes.
- To cancel the code display, hold  down for about 5 seconds. The code display also clears if no button is pressed for 1 minute.

Quick Reference



3P552666-6B

13. Optional Accessories

13.1 Indoor Unit

	Option Name		20/25/35/46 Class	50/60/71 Class	Manual No.	
					Installation Manual	Operation Manual
1	Wireless LAN connecting adaptor		BRP072C42 ★1	BRP072C42 ★2	4P518835-2G	—
2	Remote control PC-board set		BRP067A42	—	1P518834-1	—
			—	BRP980B42	1P518836-1	—
3	Wired remote controller ★5		BRC944B2 ★1	BRC944B2 ★2	3P202923-2B	3P202922-2B
			BRC073A4 ★1	BRC073A4 ★2	3P422721-1	3P422724-1
4	Wired remote controller cord (shielded wire)	Length 3 m	BRCW901A03		3P201487-1	—
		Length 8 m	BRCW901A08			
5	Wiring adaptor for timer clock / remote controller ★3 (normal open pulse contact / normal open contact)		KRP413BB1S ★1	KRP413BB1S ★2	3P248024-2G	—
6	Central remote controller ★4		DCS302CA61		2P162816-1	3P124623-8D
7	Unified ON/OFF controller ★4		DCS301BA61		1P162827-1A	2P167410-1
8	Schedule timer ★4		DST301BA61		3P162015-1A	3P124623-5C
9	Interface adaptor for DIII-NET (residential air conditioner)		KRP928BB2S ★1	KRP928BB2S ★2	3P248024-1F 3P248024-3D	—
10	Titanium apatite deodorizing filter (without frame) ★6		KAF970A46		—	—
11	Dust collection filter (PM2.5) (without frame)		BAFP046A41		—	—
12	Remote controller loss prevention with chain		KKF910A4		—	—

- Notes:**
- ★1 Remote control PC-board set (BRP067A42) is also required for each indoor unit.
 - ★2 Remote control PC-board set (BRP980B42) is also required for each indoor unit.
 - ★3 Timer clock and other devices ; obtained locally.
 - ★4 An interface adaptor (KRP928BB2S) is also required for each indoor unit.
 - ★5 3 m (BRCW901A03) or 8 m (BRCW901A08) length wired remote controller cord is necessary.
 - ★6 Standard accessory

13.2 Outdoor Unit

	Option Name	20/25/35 Class	46/50/60/71 Class	Manual No.	
				Installation Manual	Operation Manual
1	Air direction adjustment grille	KPW937E4	—	2P286387-1C	—
		—	KPW5E112	3P243204-1	—
2	Drain plug ★	KKP937A4		—	—
3	Demand adaptor	BRP070A43	—	3P376835-5E	—
		—	BRP070A44	3P376835-6E	—

- Note:** ★ Standard accessory



Warning



- Ask a qualified installer or contractor to install this product. Do not try to install the product yourself. Improper installation can result in water or refrigerant leakage, electrical shock, fire or explosion.
- Use only those parts and accessories supplied or specified by Daikin. Ask a qualified installer or contractor to install those parts and accessories. Use of unauthorised parts and accessories or improper installation of parts and accessories can result in water or refrigerant leakage, electrical shock, fire or explosion.
- Read the user's manual carefully before using this product. The user's manual provides important safety instructions and warnings. Be sure to follow these instructions and warnings.

If you have any enquiries, please contact your local importer, distributor and/or retailer.

Cautions on product corrosion

1. Air conditioners should not be installed in areas where corrosive gases, such as acid gas or alkaline gas, are produced.
2. If the outdoor unit is to be installed close to the sea shore, direct exposure to the sea breeze should be avoided. If you need to install the outdoor unit close to the sea shore, contact your local distributor.